VOLUME 04: TRACK 3 - APPLICATIONS & OMNISTUDIO INTERFACES MANUAL

Chapter 21: Track 3 - Applications, Bridges & OmniStudio

C99 Bare-Metal Bridge, Lua Coroutine Fibers, VSCode Extensions & Studio HUD

> ***Witness:*** *Proverbs 24:27 (KJV) - Prepare thy work without, and make it fit for thyself*

SYSTEM TRACK
<TRACK 3> : {Applications & OmniStudio Interfaces}

CHAPTER IDENTIFIER
CH 21

NOTEBOOKLM WORD COUNT
22,863 words

CATALOGED CHAPTER FILES
34 files (6,769 lines)

RECOMMENDED NOTEBOOKLM COGNITIVE QUERY

"How does the Lua coroutine bridge execute process tape rolls as non-blocking fiber threads?"

CHAPTER MANIFEST & INDEX (34 FILES)

#	FILE PATH	TYPE	LINES	WORDS	SIZE (KB)
1	**<TRACK 3> : {Applications & OmniStudio Interfaces}/TRACK.md**	MD	10	52	0.4
2	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/01_boundaries_and_asides.omni**	OMNI	168	1,456	10.4
3	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/02_logic_and_quad_state.omni**	OMNI	60	302	2.3
4	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/03_cardinal_directions.omni**	OMNI	62	328	2.5
5	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/04_turing_track_tape.omni**	OMNI	55	348	2.4
6	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/05_control_flow_and_loops.omni**	OMNI	146	792	6.4
7	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/canonical_notation.omni**	OMNI	75	476	3.2
8	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/if_else_statement.omni**	OMNI	64	336	2.6
9	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/if_then_statement.omni**	OMNI	60	275	2.3
10	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/nested_loop.omni**	OMNI	79	375	2.9
11	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/simple_loop.omni**	OMNI	66	316	2.4
12	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/while_statement.omni**	OMNI	67	351	2.8
13	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/README.md**	MD	79	379	2.8
14	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/extension.js**	JS	93	306	3.9
15	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/language-configuration.json**	JSON	30	76	0.5
16	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/omnicode-language-0.1.0.vsix**	VSIX	1	5	2.9
17	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/omnicode-language-0.1.1.vsix**	VSIX	1	5	4.0
18	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/omnicode-language-0.1.2.vsix**	VSIX	1	5	6.4
19	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/package.json**	JSON	54	92	1.1
20	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/syntaxes/omnicode.tmLanguage.json**	JSON	321	529	8.7
21	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/snippets/omnicode.json**	JSON	163	611	6.5
22	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/app.js**	JS	509	1,537	20.4
23	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/cloud.html**	HTML	201	718	10.8
24	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/desktop_app.py**	PY	1,182	3,250	44.9
25	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/download.html**	HTML	140	659	8.3
26	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/index.html**	HTML	136	610	7.7
27	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/lexicon.html**	HTML	114	508	6.3
28	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/model.html**	HTML	152	719	8.9
29	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/scrolls.html**	HTML	146	568	7.8
30	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/specs.html**	HTML	155	798	9.6
31	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/studio.html**	HTML	136	610	7.7
32	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/style.css**	CSS	1,747	3,804	37.7
33	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/terminal.html**	HTML	191	782	10.8

#	FILE PATH	TYPE	LINES	WORDS	SIZE (KB)
34	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/terminal.js**	JS	305	885	11.9

<TRACK 3> : {Applications & OmniStudio Interfaces}/TRACK.md MARKDOWN / DOC 10 LINES 52 WORDS 0.4 KB

: {Applications & OmniStudio Interfaces}

Scope & Invariant

User-facing tooling, visual studio interfaces, IDE integration, and canonical demonstration tapes.

- `apps/gui/` : OmniStudio HUD web interface & desktop application.
- `apps/editors/` : Visual Studio Code extension, grammar syntaxes, and language tooling.
- `apps/examples/` : Canonical verification reels and language showcase tapes.

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/01_boundaries_and_asides.omni

OMNI | 168 LINES | 1456 WORDS | 10.4 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( Ranked Morphological Boundaries & Preserved Intent Asides )
# ================================================================================
# ARCHITECTURAL TOPOLOGY:
#   Foundational Showcase: The 6 Ranked Morphological Boundary Types.
#   - Rank 1: < ... > Invariant Axiom      (Immutable Law & Ontological Truth)
#   - Rank 2: [ ... ] Denotation           (Concrete Definition & Mutable State)
#   - Rank 3: { ... } Connotation          (Context, Environment & Payload)
#   - Rank 4: [ ... } Inside-Out Emergence (Entity Projects into Context)
#   - Rank 5: { ... ] Outside-In Imposition(Context Shapes Entity)
#   - Rank 6: ( ... ) Intent Aside         (First-Class Pronunciation & Teleology)
# ================================================================================

= [Module]: < Foundations { Boundaries } > & [ Example { Boundaries and Asides } ] ( Demonstrates the six ranked morphological boundaries and first-class intent asides ) {
    [ Version ]:   { "1.0.0" } ( The Version is in the context of 1.0.0 ),
    [ Created ]:   { "2026-10-11" } ( Created in the context of October 11, 2026 ),
    [ Updated ]:   { "2026-10-12" } ( Last updated in the context of October 12, 2026 ),
    [ Author ]:    { "CreativeWorkzStudio LLC / ProfessorSeanEX" } ( Author in the context of CreativeWorkzStudio LLC and ProfessorSeanEX ),
    [ System ]:    { OmniCode { Core Foundations } } ( System is in the context of OmniCode Core Foundations ),
    [ Witness ]:   < Scripture { Proverbs 8:29 KJV } > { "When he gave to the sea his decree, that the waters should not pass his commandment: when he appointed the foundations of the earth." } ( Invariant Scripture Witness Proverbs 8:29 in the context: When he appointed the foundations of the earth ),
    [ Nature ]:    < FoundationalExample > & { BaseBlock } ( Nature is a Foundational Example in the context of Base Block ),
    [ Semantics ]: { BoundaryPhysics :: MorphologicalClosure :: IntentPreservation } ( Semantics in the context of Boundary Physics, Morphological Closure, and Intent Preservation )
}

# Freestanding Foundational Axiom Declaration
# Rank 1: < ... > Invariant Axiom (Immutable Law)
< Axiom Truth > = < +1 > ( Ground truth axiom: immutably sealed in positive unity ) {
    [ Law ]:       < +1 > ( The immutable truth law is sealed in affirmative Proceed ),
    [ Geometry ]:  ENCLOSE [ Truth ] WITHIN < Invariant Space > ( Truth is bounded by immutable axiom walls )
}

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ================================================================================
Opening {
    # Rank 1: Invariant Axiom (Inherited law)
    [ Ground Law ]: < Axiom Truth > ( Rank 1: Immutable axiom law ),

    # Rank 2: [ ... ] Denotation (Mutable Definition / State Container)
    [ State Alpha ]: [ +1 ] ( Rank 2: Concrete discrete definition with mutable state initialized to Proceed ),

    # Rank 3: { ... } Connotation (Ambient Universe / Environmental Context)
    { Universe Omega }: { Ground Law } ( Rank 3: Surrounding environmental context holding ground truth ),

    # Rank 4: [ ... } Inside-Out Emergence (Entity Projects Outward into Universe)
    [ Emergent Node }: [ State Alpha } ( Rank 4: Internal definition emerges outward into ambient context ),

    # Rank 5: { ... ] Outside-In Imposition (Environmental Context Shapes Internal Entity)
    { Imposed Node ]: { Universe Omega ] ( Rank 5: Surrounding context actively imposes constraints on internal entity ),

    # Rank 6: ( ... ) Intent Aside (Teleological Purpose & Cognitive Voice)
    ( Preserved Intent ): ( Rank 6: First-class teleological intent preserved in AST for human-AI shared reasoning ),

    # Hardware & Memory Preconditions
    [ Alignment ]: [ Memory Aligned 64 ] ( Memory aligned to 64-bit boundary ),
    [ Preconditions ]: [ (+1) plus (-1) <=> 0 ] & [ Status = +1:PROCEED ] ( ALU equilibrium holds and the machine status is armed in Proceed )
}

# ================================================================================
# BLOCK 3: BODY BLOCK
# ================================================================================
Body [
    # ------------------------------------------------------------------------
    # ROLL 1: RANK 1 INVARIANT VERIFICATION
```

```
    # Verify immutable ground truth identity
    # ------------------------------------------------------------------------
    [ ROLL 1 ] : [ Verify Axiom ] : < Axiom Truth > <=> VERIFY [ +1 ] = [ Truth` ]
        ( Rank 1: Verify immutable ground truth axiom against positive unity ),

    # ------------------------------------------------------------------------
    # ROLL 2: RANK 2 DENOTATIVE STATE MUTATION
    # Advance mutable state within defined boundary
    # ------------------------------------------------------------------------
    [ ROLL 2 ] : [ Mutate State ] : [ State Alpha ] --> ADVANCE [ +1 ] = [ State Alpha` ]
        ( Rank 2: Advance discrete denotative state yielding new state State-Alpha-prime ),

    # ------------------------------------------------------------------------
    # ROLL 3: RANK 3 CONNOTATIVE CONTEXT EXPANSION
    # Expand ambient contextual universe with verified state
    # ------------------------------------------------------------------------
    [ ROLL 3 ] : [ Expand Context ] : { Universe Omega } <-- ABSORB { State Alpha` } = { Universe Omega` }
        ( Rank 3: Ambient context absorbs new denotative state expanding environmental scope ),

    # ------------------------------------------------------------------------
    # ROLL 4: RANK 4 INSIDE-OUT ASYMMETRIC EMERGENCE
    # Project internal discrete state outward into ambient context
    # ------------------------------------------------------------------------
    [ ROLL 4 ] : [ Project Emergence ] : [ State Alpha` } --> EMERGE { Universe Omega` } = [ Emergent Node` }
        ( Rank 4: Discrete state emerges through asymmetrical boundary into ambient universe ),

    # ------------------------------------------------------------------------
    # ROLL 5: RANK 5 OUTSIDE-IN ASYMMETRIC IMPOSITION
    # Impose ambient contextual constraints onto internal discrete entity
    # ------------------------------------------------------------------------
    [ ROLL 5 ] : [ Impose Context ] : { Universe Omega` ] --> IMPOSE [ State Alpha` ] = { Imposed Node` ]
        ( Rank 5: Ambient universe imposes environmental boundaries onto internal discrete state ),

    # ------------------------------------------------------------------------
    # ROLL 6: RANK 6 FIRST-CLASS INTENT ASIDE PRESERVATION
    # Preserve teleological intent as a first-class AST node for human-AI cognition
    # ------------------------------------------------------------------------
    [ ROLL 6 ] : [ Preserve Intent ] : ( Preserved Intent ) --> RECORD [ AST ] = ( Teleology Verified` )
        ( Rank 6: Intent aside preserved as live AST node for human-AI cognitive continuity )
]

# ============================================================================
# BLOCK 4: CLOSING BLOCK
# ============================================================================
Closing ( Sealing Gate: Reconcile, validate, tripwire, debug, log, and affirm permanent boundary execution ) {
    # ------------------------------------------------------------------------
    # 1. VERIFICATION OF OPERATIONAL ROLLS
    # ------------------------------------------------------------------------
    [ Verify ]: [
        [ ROLL 1 ] = PROCEED ::
        [ ROLL 2 ] = PROCEED ::
        [ ROLL 3 ] = PROCEED ::
        [ ROLL 4 ] = PROCEED ::
        [ ROLL 5 ] = PROCEED ::
        [ ROLL 6 ] = PROCEED
    ] ( Verify that all six boundary operational rolls achieved affirmative consensus in PROCEED ),

    # ------------------------------------------------------------------------
    # 2. VALIDATION OF INVARIANT INTEGRITY & CLOSURE
    # ------------------------------------------------------------------------
    [ Validate ]: { Truth` = +1 } & { State Alpha` = +1 } & { Emergent Node` } & { Imposed Node` } & ( Teleology
Verified` )
        ( Validate that invariant truth holds, state remains affirmative, and all morphological closures and asides are
sealed ),

    # ------------------------------------------------------------------------
    # 3. ZERO-TOLERANCE TRIPWIRING
    # ------------------------------------------------------------------------
    [ Tripwire ]: { Status` = +1:PROCEED } --> [ ARMED ] | { Status` != +1:PROCEED } --> [ TRIPPED ] --> [ Commit Status
= -1:ROLLBACK ]
        ( Tripwire remains armed in Proceed or trips on anomaly to veto execution ),

    # ------------------------------------------------------------------------
    # 4. DIAGNOSTIC DEBUG TELEMETRY
    # ------------------------------------------------------------------------
    [ Debug ]: { [ Truth` ] :: [ State Alpha` ] :: [ Emergent Node` ] :: [ Imposed Node` ] :: ( Teleology Verified` ) ::
[ Status` ] }
        ( Expose diagnostic telemetry across all verified morphological boundaries and machine status ),
```

```
    # ------------------------------------------------------------------------
    # 5. EPISODIC INVENTORY LOGGING
    # ------------------------------------------------------------------------
    [ Log ]: [ Journal ]: INSCRIBE { [ Boundaries Sealed ] :: [ Rank 1 Through 6 ] :: [ Truth` ] }
        ( Inscribe the verified boundary physics demonstration into the permanent episodic system journal ),

    # ------------------------------------------------------------------------
    # 6. PERMANENT COVENANT EXECUTION & FOURFOLD COMMIT STATUS LOOP
    # ------------------------------------------------------------------------
    [ Commit Status ]: [
        # +1: PROCEED (+t: Advance forward along the causal tape)
        { [ Verify ] = PROCEED } & { [ Tripwire ] = ARMED }
            --> [ +1:PROCEED ] |

        #  0: AWAIT (t=t: Hold in equilibrium, pending resolution)
        { [ Status` ] = PENDING }
            --> [ 0:AWAIT ] :: REPEAT |

        # NULL: HALT (t=empty: Cessation of clock, absolute stop)
        { [ Status` ] = NULL }
            --> [ HALT:NULL ] |

        # -1: ROLLBACK (-t: Reverse causal tape to origin t=0)
        { [ Tripwire ] = TRIPPED } | { [ Verify ] != PROCEED }
            --> [ -1:ROLLBACK ]
    ] ( Fourfold state machine: Proceed advances forward, Await holds equilibrium, Halt ceases execution in Null,
Rollback reverts to origin )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/02_logic_and_quad_state.omni

OMNI | 60 LINES | 302 WORDS | 2.3 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( Balanced Ternary Logic & Micro/Macro Quad-State Truth Space )
# ================================================================================
Module < Logic.QuadStateEngine > ( Showcase Balanced Ternary and Micro/Macro Duality ) {
    version: < "1.0.0" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: MicroMacroDuality :: CaveatPhysics >
}

AxiomAffirmation < yes >    ( Micro: yes == Macro: PROCEED == +1 )
AxiomRefutation  < no >     ( Micro: no  == Macro: ROLLBACK == -1 )
AxiomBalance     < ok >     ( Micro: ok  == Macro: AWAIT    ==  0 )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Preconditions & State Initialization )
# ================================================================================
Opening {
    requires: [ TruthSpace.T4 ],
    preconditions: [ AxiomAffirmation = PROCEED ]
}

flag_affirm [ yes ]   ( Initial affirmative state )
flag_negate [ -- yes ] ( Negation flips +1 to -1: no )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Truth Space Algebra: Conjunction, Disjunction, Negation, and Caveat Gate )
# ================================================================================
Body [
    # 1. Micro/Macro Equivalence Checks
    affirm_check [ flag_affirm = PROCEED ] ( Affirmation matches PROCEED )
    refute_check [ flag_negate = ROLLBACK ] ( Negated affirmation matches ROLLBACK )

    # 2. Conjunction (:: AND) and Disjunction (| OR)
    and_test [ yes :: yes = PROCEED ]
    or_test  [ yes | no = PROCEED ]

    # 3. Adversative Caveat Gate (! BUT)
    # The action proceeds ONLY because the caveat holds (+1)
    gated_action [ flag_affirm ! [ affirm_check = yes ] ] ( Caveat permits forward commit )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Verification & Macro State Commit )
# ================================================================================
Closing {
    verify: [
        affirm_check = yes ::
        refute_check = yes ::
        and_test = yes ::
        or_test = yes ::
        gated_action = PROCEED
    ],
    commit_status [ PROCEED ] ( Commit verified quad-state execution )
}

```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/03_cardinal_directions.omni

OMNI | 62 LINES | 328 WORDS | 2.5 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( 4 Cardinal Directions Across Causality and Abstraction Axes )
# ================================================================================
Module < Geometry.CardinalDirections > ( Showcase 4 cardinal directional vectors and shorthands ) {
    version: < "1.0.0" >,
    system: < OmniCode.Core >,
    semantics: < VectorDynamics :: CausalityAndScale >
}

BaseDomain < 100 >   ( Ground level primitive )
TargetDomain < 200 > ( Elevated abstract domain )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Vector Preconditions & Initial States )
# ================================================================================
Opening {
    requires: [ CoordinatePlane.2D ],
    preconditions: [ BaseDomain = 100 ]
}

step_x [ 10 ] ( State moving across X temporal axis )
step_z [ 10 ] ( State moving across Z scale axis )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Cardinal Vector Executions: Right, Left, Up, Down )
# ================================================================================
Body [
    # 1. Right (--> and ->): Forward Causation / Act Upon
    forward_canonical [ step_x --> 20 ] ( Forward step advancing state to 20 )
    forward_shorthand [ step_x -> 30 ]  ( Shorthand forward step to 30 )

    # 2. Left (<-- and <-): Retroaction / Feedback / Rollback
    feedback_canonical [ step_x <-- 10 ] ( Retroactive feedback step )
    feedback_shorthand [ step_x <- 10 ]  ( Shorthand feedback step )

    # 3. Up (--^ and -^): Synthesis / Abstraction / Ascend
    ascend_canonical [ step_z --^ TargetDomain ] ( Synthesis into target domain )
    ascend_shorthand [ step_z -^ TargetDomain ]  ( Shorthand synthesis )

    # 4. Down (--v and -v): Decomposition / Analysis / Descend
    descend_canonical [ TargetDomain --v BaseDomain ] ( Decomposition into base domain )
    descend_shorthand [ TargetDomain -v BaseDomain ]  ( Shorthand decomposition )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Vector Integrity Verification & Commit )
# ================================================================================
Closing {
    verify: [
        forward_canonical = 20 ::
        forward_shorthand = 30 ::
        ascend_canonical = 200 ::
        descend_canonical = 100
    ],
    commit_status [ PROCEED ] ( Commit verified vector trajectory )
}

```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/04_turing_track_tape.omni

OMNI | 55 LINES | 348 WORDS | 2.4 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK (THE TRACK)
# ( Track Manifest, Semantic Intent, and Inferred Ontological Types )
# ================================================================================
Module < ArithmeticReel.Tape1 > ( Add 1 ball to the group A which contains 4 balls ) {
    version: < "1.0.0" >,
    system: < OmniCode.Core >,
    semantics: < TrackPhysics :: MultiTape :: AIInference >
}

# The ? (Infer Operator) marks concepts whose structural types are inferred by the AI brain:
Ball < Entity? > ( Inferred discrete unit: resolved from natural sentence )
Group < Set? >   ( Inferred collection container: resolved from natural sentence )

# ================================================================================
# BLOCK 2: OPENING BLOCK (THE HEAD)
# ( The Focal Setup: Reading Initial Conditions and Contextual State )
# ================================================================================
Opening {
    import: [ Core.Arithmetic :: Core.Logic ],
    requires: [ Group : Set? :: Ball : Entity? ],
    preconditions: [ +1 ]
}

# The Head sets Group A with initial count of 4 balls
A [ 4 : Ball? ] { Group? } ( Group A initialized with 4 inferred balls )
new_ball [ 1 : Ball? ]     ( 1 additional ball to be introduced )

# ================================================================================
# BLOCK 3: BODY BLOCK (THE TAPE & ROLLS)
# ( The Tape Execution: SVO Rolls advancing the state forward using 'plus' )
# ================================================================================
Body [
    # ROLL 1 (SVO Action): Subject A acts upon target, advancing by 'plus' new_ball
    roll_1 [ A --> A plus new_ball ] ( Act upon A: causal transition from 4 to 5 balls )

    # ROLL 2 (SVO Verification): Subject A is 5 under the caveat that new_ball is 1
    roll_2 [ A = 5 ! [ new_ball = 1 ] ] ( State is 5, provided caveat of 1 added ball holds )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK (REEL COMMIT & PROOF)
# ( Tape Stop: Verification Invariants and State Commit )
# ================================================================================
Closing {
    verify: [
        roll_1 = 5 ::
        roll_2 = PROCEED ::
        A = 5 ::
        ADD = InvariantMacro
    ],
    commit_status [ PROCEED ] ( Commit the verified tape state )
}

```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/05_control_flow_and_loops.omni

OMNI | 146 LINES | 792 WORDS | 6.4 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( Control Flow Manifest: Quad-State Accumulator Loops & Conditional Branching )
# ================================================================================
Module < ControlFlow.QuadStateShowcase > ( Canonical showcase of Simple Loops, Nested Loops, While statements, and If/Then/Else branches ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: QuadStateControlFlow :: NamedSiliconRegisters >
}

# Axiomatic Boundary Limits & Invariant Constants
LOOP_BOUND < 10 >          ( Invariant boundary ceiling for linear simple loop )
GRID_ROWS  < 3 >           ( Invariant outer dimension: 3 rows )
GRID_COLS  < 4 >           ( Invariant inner dimension: 4 columns )
THRESHOLD  < 100 >         ( Invariant threshold boundary for while condition )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Hardware Context Setup, Register Allocations & Preconditions )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 :: MemoryAligned ],
    preconditions: [ +1 ]
}

# Physical Register Initializations for Body Execution
loop_acc         [ 0 ] ( Silicon Accumulator IT container )
loop_turn        [ 0 ] ( Temporal Cadence TURN container )
sensor_reading   [ 25 ] ( Dynamic reading for while statement )
condition_affirm [ yes ] ( Affirmative condition flag: +1 )
condition_refute [ no ]  ( Refuted condition flag: -1 )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( The 5 Canonical Control Flow Constructs )
# ================================================================================
Body [
    # ------------------------------------------------------------------------
    # 1. SIMPLE LOOP: Linear Quad-State Accumulator cycling from 0 up to BOUND
    # ------------------------------------------------------------------------
    simple_loop [
        STATE (
            < BOUND = LOOP_BOUND >,
            [ IT = 0, TURN = 0 ],
            { SPOOL }
        ) = [
            PROCEED  -WHEN-> { (IT plus 1 --> IT) & (TURN plus 1 --> TURN) },
            AWAIT    -WHEN-> { BUSY },
            ROLLBACK -WHEN-> { FAULT & (HOME --> IT) },
            HALT     -WHEN-> { TURN = BOUND }
        ]
    ] ( Simple linear loop: ticks up IT and TURN until watchdog BOUND is met )

    # ------------------------------------------------------------------------
    # 2. NESTED LOOP: 2D Matrix Grid Iteration (Outer Row x Inner Column)
    # ------------------------------------------------------------------------
    nested_loop [
        STATE (
            < ROW_BOUND = GRID_ROWS, COL_BOUND = GRID_COLS >,
            [ ROW_TURN = 0, ROW_SUM = 0 ],
            { SPOOL }
        ) = [
            PROCEED -WHEN-> {
                # Inner Column State Loop: accumulates COL_BOUND cells per row
                STATE (
                    < BOUND = COL_BOUND >,
                    [ COL_TURN = 0, IT = ROW_SUM ],
                    { SPOOL }
                ) = [
                    PROCEED  -WHEN-> { (IT plus 1 --> IT) & (COL_TURN plus 1 --> COL_TURN) },
```

```
                    AWAIT    -WHEN-> { BUSY },
                    ROLLBACK -WHEN-> { FAULT & (HOME --> IT) },
                    HALT     -WHEN-> { COL_TURN = BOUND }
                ] &
                (IT --> ROW_SUM) &
                (ROW_TURN plus 1 --> ROW_TURN)
            },
            AWAIT    -WHEN-> { BUSY },
            ROLLBACK -WHEN-> { FAULT },
            HALT     -WHEN-> { ROW_TURN = ROW_BOUND }
        ]
    ] ( Nested 2D loop: outer row cadence driving inner column accumulator )

    # ------------------------------------------------------------------------
    # 3. WHILE STATEMENT: Predicated Quad-State Loop
    # Advances while sensor_reading < THRESHOLD; halts when condition ceases
    # ------------------------------------------------------------------------
    while_statement [
        STATE (
            < BOUND = THRESHOLD >,
            [ IT = sensor_reading, TURN = 0 ],
            { FROM }
        ) = [
            PROCEED  -WHEN-> { (sensor_reading < THRESHOLD) & (IT plus 15 --> IT) & (TURN plus 1 --> TURN) },
            AWAIT    -WHEN-> { SENSOR_BUSY },
            ROLLBACK -WHEN-> { SENSOR_ERROR & (HOME --> IT) },
            HALT     -WHEN-> { (IT >= THRESHOLD) | (TURN = 100) }
        ]
    ] ( While loop: continues PROCEED while predicate holds, HALT escapes boundary )

    # ------------------------------------------------------------------------
    # 4. IF / THEN STATEMENT: Unilateral Conditional Gate
    # Evaluates condition: if affirmative (+1), actuates then branch; else awaits
    # ------------------------------------------------------------------------
    if_then_statement [
        STATE (
            [ IT = 0 ],
            { INTO }
        ) = [
            PROCEED -WHEN-> { condition_affirm & (42 --> IT) },
            AWAIT   -WHEN-> { -- condition_affirm }
        ]
    ] ( If/Then gate: proceeds with forward action only under verified affirmative condition )

    # ------------------------------------------------------------------------
    # 5. IF / ELSE STATEMENT: Bilateral Ternary Branch
    # Evaluates condition: +1 forks to then branch, -1 forks to else branch
    # ------------------------------------------------------------------------
    if_else_statement [
        STATE (
            [ IT = 0 ],
            { INTO }
        ) = [
            PROCEED  -WHEN-> { (condition_refute = yes) & (100 --> IT) },
            ROLLBACK -WHEN-> { (condition_refute = no)  & (200 --> IT) },
            AWAIT    -WHEN-> { condition_refute = ok }
        ]
    ] ( If/Else branch: balanced ternary fork selecting between affirmative and negative vectors )
]

# ============================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions, Invariant Integrity Proofs & Macro State Commit )
# ============================================================================
Closing {
    verify: [
        LOOP_BOUND = 10 ::
        GRID_ROWS = 3 ::
        GRID_COLS = 4 ::
        THRESHOLD = 100
    ],
    commit_status [ PROCEED ] ( Verified quad-state control flow suite sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/canonical_notation.omni

OMNI | 75 LINES | 476 WORDS | 3.2 KB

```
# ========================================================================
# BLOCK 1: METADATA BLOCK
# ( File Identity, Version, Author, Intent Aside, and Ontological Axioms )
# ========================================================================
Module < CanonicalNotationShowcase > ( Foundations of Being, Causality, and Abstraction ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: QuadState :: BoundaryPhysics >
}

# Axiomatic Categories (Invariant Sets)
A < +1 > ( Invariant category A: axiomatic ground truth )
B < +1 > ( Invariant category B: independent set domain )


# ========================================================================
# BLOCK 2: OPENING BLOCK
# ( Environment Setup, Preconditions, Scoped Universes, and Context Dictation )
# ========================================================================
Opening {
    requires: [ MemoryAligned :: TypeChecking ],
    initial_truth: [ +1 ]
}

# Mutable Denotations initialized in Opening
a [ +1 ] ( Particular instance 'a' initialized with mutable proceed state )
b [ +1 ] ( Particular instance 'b' initialized with mutable proceed state )

# Contextual Connotations & Asymmetric Causality
a_in_A [ a ] { A }  ( Instance 'a' validated within contextual universe A )
a_top_down { A ]    ( Outside-In: Environment A actively forces definition of a )
b_bottom_up [ B }   ( Inside-Out: Definition of b emerges from context B )


# ========================================================================
# BLOCK 3: BODY BLOCK
# ( Core Transformations, Being Relations, Causality Vectors, and Connectives )
# ========================================================================
Body [
    # 1. Ontological Relations of Being (Y-Axis)
    instance_typing [ a : A ]    ( 'a' stands in relation to Category A )
    identity_check [ a = a ]     ( 'a' is identical to 'a' )
    distinction_check [ a != b ] ( 'a' is not 'b' )
    similarity_check [ a ~= b ]  ( 'a' is liken to 'b' in structural form )

    # 2. Temporal & Causal Vectors (X-Axis)
    forward_action [ a --> b ]   ( 'a' acts upon 'b' forward in time )
    feedback_action [ b <-- a ]  ( 'b' reflects back upon 'a' as retroactive feedback )

    # 3. Hierarchical Abstraction Vectors (Z-Axis)
    synthesis [ a ^-- A ]        ( Low-level primitive 'a' builds up to high-level concept A )
    decomposition [ A v-- a ]    ( High-level category A decomposes down into primitive 'a' )

    # 4. Connectives & Adversative Caveat Pipeline
    negation [ -- a ]            ( Inverts state of 'a' across ternary zero balance )
    conjunction [ a : A :: b : B ]
    pipeline [ a --> b ! [ a : A :: b : B ] ] ( 'a' acts on 'b', BUT caveat must be satisfied )
]


# ========================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions, Invariant Integrity Proofs, Teardown, and Commit / Return )
# ========================================================================
Closing {
    verify: [
        identity_check = +1 ::
        forward_action = +1 ::
        pipeline = +1
    ],
    on_rollback: [ <-- unwind_state ],
```

```
    commit_status: [ +1 ] ( Commit state: verified execution )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/if_else_statement.omni

OMNI | 64 LINES | 336 WORDS | 2.6 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State If/Else Bilateral Ternary Branch )
# ================================================================================
Module < ControlFlow.IfElseStatement > ( Demonstrates bilateral ternary branching in Quad-State logic ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: TernaryBranching :: NamedSiliconRegisters >
}

# Invariant Branch Targets
THEN_RESULT < 100 > ( Value emitted when condition is affirmative: +1 )
ELSE_RESULT < 200 > ( Value emitted when condition is refutative: -1 )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Branch Flag Setup )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# The active condition to evaluate (set to 'no' to trigger else branch)
branch_condition [ no ] ( Initial condition evaluating to -1 / ROLLBACK )
branch_acc       [ 0 ]  ( Accumulator register receiving selected branch output )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: If/Else Bilateral Branch )
# ================================================================================
Body [
    # THE IF / ELSE STATEMENT:
    # `if (branch_condition == yes) { IT = THEN_RESULT; } else { IT = ELSE_RESULT; }`
    # - PROCEED (+1) : True branch (actuates THEN_RESULT into IT)
    # - ROLLBACK (-1): False branch (actuates ELSE_RESULT into IT)
    # - AWAIT (0)    : Neutral / uncommitted state (holds position)
    if_else_branch [
        STATE (
            [ IT = branch_acc ],
            { INTO }
        ) = [
            PROCEED  -WHEN-> { (branch_condition = yes) & (THEN_RESULT --> IT) },
            ROLLBACK -WHEN-> { (branch_condition = no)  & (ELSE_RESULT --> IT) },
            AWAIT    -WHEN-> { branch_condition = ok }
        ]
    ] ( If/Else branch: selects between affirmative then and refutative else vectors )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        THEN_RESULT = 100 ::
        ELSE_RESULT = 200 ::
        branch_condition = no
    ],
    commit_status [ PROCEED ] ( If/Else ternary branch verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/if_then_statement.omni

OMNI | 60 LINES | 275 WORDS | 2.3 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State If/Then Unilateral Conditional Gate )
# ================================================================================
Module < ControlFlow.IfThenStatement > ( Demonstrates unilateral conditional gating in Quad-State logic ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: ConditionalGating :: NamedSiliconRegisters >
}

# Invariant Target Payload
PAYLOAD_VALUE < 42 > ( Value to be transduced upon affirmative condition )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Flag Setup )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# Initial Register & Condition States
permission_flag [ yes ] ( Affirmative precondition flag: +1 )
accumulator     [ 0 ]   ( Destination IT register )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: If/Then Conditional Gate )
# ================================================================================
Body [
    # THE IF / THEN STATEMENT:
    # `if (permission_flag) { accumulator = PAYLOAD_VALUE; }`
    # - PROCEED: When condition is affirmative (+1), actuates vector into IT
    # - AWAIT  : When condition is non-affirmative, holds state without mutating
    if_then_dispatch [
        STATE (
            [ IT = accumulator ],
            { INTO }
        ) = [
            PROCEED -WHEN-> { permission_flag & (PAYLOAD_VALUE --> IT) },
            AWAIT   -WHEN-> { -- permission_flag }
        ]
    ] ( If/Then gate: proceeds with forward action only when permission holds )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        PAYLOAD_VALUE = 42 ::
        permission_flag = yes
    ],
    commit_status [ PROCEED ] ( If/Then conditional gate verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/nested_loop.omni OMNI 79 LINES 375 WORDS 2.9 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State Nested 2D Grid Loop )
# ================================================================================
Module < ControlFlow.NestedLoop > ( Demonstrates 2D matrix row and column grid iteration ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: NestedStateLoop :: NamedSiliconRegisters >
}

# Invariant Grid Dimensions
NUM_ROWS < 3 > ( Outer loop dimension: 3 rows )
NUM_COLS < 4 > ( Inner loop dimension: 4 columns )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Coordinate Initialization )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# Initial Register Setup
grid_sum  [ 0 ] ( Running total accumulated across all cells )
grid_rows [ 0 ] ( Current outer row iteration count )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: Nested Row/Column Traversal )
# ================================================================================
Body [
    # THE NESTED LOOP:
    # Outer STATE manages ROW_TURN up to ROW_BOUND.
    # On each outer PROCEED, the inner STATE manages COL_TURN up to COL_BOUND,
    # accumulating into IT and retiring into grid_sum.
    matrix_traversal [
        STATE (
            < ROW_BOUND = NUM_ROWS, COL_BOUND = NUM_COLS >,
            [ ROW_TURN = 0, ROW_ACC = 0 ],
            { SPOOL }
        ) = [
            PROCEED -WHEN-> {
                # Inner Column State Loop:
                STATE (
                    < BOUND = COL_BOUND >,
                    [ COL_TURN = 0, IT = ROW_ACC ],
                    { SPOOL }
                ) = [
                    PROCEED  -WHEN-> { (IT plus 1 --> IT) & (COL_TURN plus 1 --> COL_TURN) },
                    AWAIT    -WHEN-> { BUSY },
                    ROLLBACK -WHEN-> { FAULT & (HOME --> IT) },
                    HALT     -WHEN-> { COL_TURN = BOUND }
                ] &
                (IT --> ROW_ACC) &
                (ROW_TURN plus 1 --> ROW_TURN)
            },
            AWAIT    -WHEN-> { BUSY },
            ROLLBACK -WHEN-> { FAULT },
            HALT     -WHEN-> { ROW_TURN = ROW_BOUND }
        ]
    ] ( Nested 2D loop traverses all rows and columns to completion )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        NUM_ROWS = 3 ::
        NUM_COLS = 4 ::
```

```
        grid_sum = 0
    ],
    commit_status [ PROCEED ] ( Nested loop verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/simple_loop.omni OMNI 66 LINES 316 WORDS 2.4 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State Simple Linear Loop )
# ================================================================================
Module < ControlFlow.SimpleLoop > ( Demonstrates single-dimensional linear accumulator loop ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: QuadStateLoop :: NamedSiliconRegisters >
}

# Invariant Watchdog Bound
CEILING < 10 > ( Invariant maximum bound for accumulator loop )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Register Initialization )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# Initial Register Setup
initial_it   [ 0 ] ( Initial primary accumulator IT value )
initial_turn [ 0 ] ( Initial temporal cadence TURN value )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: Linear Accumulator Loop )
# ================================================================================
Body [
    # THE SIMPLE LOOP:
    # Cycles through T4 truth space:
    #   +1 (PROCEED)  : Ticks up IT and TURN
    #    0 (AWAIT)    : Yields / waits if bus is busy
    #   -1 (ROLLBACK) : Rewinds to baseline HOME on fault
    #   NULL (HALT)   : Breaks loop when TURN reaches BOUND
    loop_result [
        STATE (
            < BOUND = CEILING >,
            [ IT = initial_it, TURN = initial_turn ],
            { SPOOL }
        ) = [
            PROCEED  -WHEN-> { (IT plus 1 --> IT) & (TURN plus 1 --> TURN) },
            AWAIT    -WHEN-> { BUSY },
            ROLLBACK -WHEN-> { FAULT & (HOME --> IT) },
            HALT     -WHEN-> { TURN = BOUND }
        ]
    ] ( Linear accumulator loop runs to completion when TURN equals BOUND )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        CEILING = 10 ::
        initial_it = 0 ::
        initial_turn = 0
    ],
    commit_status [ PROCEED ] ( Simple loop verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/while_statement.omni

OMNI | 67 LINES | 351 WORDS | 2.8 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State Predicated While Loop )
# ================================================================================
Module < ControlFlow.WhileStatement > ( Demonstrates predicate-governed loop advancing while condition holds ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: PredicatedWhileLoop :: NamedSiliconRegisters >
}

# Invariant Threshold Boundary
MAX_THRESHOLD < 100 > ( Invariant upper ceiling triggering while loop termination )
MAX_CYCLES    < 500 > ( Watchdog maximum turn count )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Initial Sensor Values )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# Initial Stream Values
sensor_reading [ 10 ] ( Initial inbound value to be sampled )
sensor_status  [ ok ] ( Initial bus status: equilibrium 0 )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: Predicated While Loop )
# ================================================================================
Body [
    # THE WHILE STATEMENT:
    # `while (sensor_reading < MAX_THRESHOLD) { Process_Stream(); }`
    # - PROCEED: Advances while the predicate is affirmative (+1)
    # - AWAIT  : Yields/pauses while the hardware sensor is busy sampling (0)
    # - ROLLBACK: Restores baseline from HOME if parity fails (-1)
    # - HALT   : Escapes the loop when predicate becomes false or watchdog trips (NULL)
    while_sensor_stream [
        STATE (
            < BOUND = MAX_CYCLES >,
            [ IT = sensor_reading, TURN = 0 ],
            { FROM }
        ) = [
            PROCEED  -WHEN-> { (sensor_reading < MAX_THRESHOLD) & (IT plus 10 --> IT) & (TURN plus 1 --> TURN) },
            AWAIT    -WHEN-> { SENSOR_BUSY },
            ROLLBACK -WHEN-> { SENSOR_ERROR & (HOME --> IT) },
            HALT     -WHEN-> { (IT >= MAX_THRESHOLD) | (TURN = BOUND) }
        ]
    ] ( While statement loops until sensor exceeds threshold or bound is reached )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        MAX_THRESHOLD = 100 ::
        MAX_CYCLES = 500 ::
        sensor_reading = 10
    ],
    commit_status [ PROCEED ] ( While loop verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/README.md | MARKDOWN / DOC | 79 LINES | 379 WORDS | 2.8 KB

OmniCode VS Code & IDE Extension

> ***Official Language Support Package for Visual Studio Code, Cursor, and Google Antigravity IDE.***

Provides morphological boundary colorization, 3D coordinate operator grammar, balanced ternary literals, and universal 4-Block file templates.

1. Extension Topology & Installation

File Topology

```
editors/vscode/
├── package.json                  # Extension manifest (v0.1.2, .omni & .scroll associations)
├── language-configuration.json    # Brackets, auto-closing pairs, comment syntax
├── snippets/
│   └── omnicode.json             # 4-Block auto-population snippets (omni-file & omni-scroll)
└── syntaxes/
    └── omnicode.tmLanguage.json  # TextMate syntax grammar scopes & keyword rules
```

Installation

```
# Package the VSIX extension
npx -y @vscode/vsce package --allow-missing-repository

# Install into Antigravity IDE / VS Code
code --install-extension editors/vscode/omnicode-language-0.1.2.vsix
```

2. Core Capabilities & Grammar

Morphological Boundary Highlighting

Visually distinguishes all six ontological boundary types to eliminate bracket blindness:

- `[ ... ]` : **Denotation** (Fixed definitions, physical registers)
- `< ... >` : **Invariant** (Immutable axioms, process types)
- `{ ... }` : **Connotation** (Ambient context, scoped universes)
- `{ ... ]` : **Top-Down** (Context dictates definition / Absorber)
- `[ ... }` : **Bottom-Up** (Definition emerges from context / Generator)
- `( ... )` : **Aside** (Machine-readable intent and AI reasoning traces)

3D Coordinate Operator Colorization

- **Y-Axis (Being/Identity):** `=` , `!=` , `~=` / `≈`
- **X-Axis (Dynamics/Causality):** `-->` (Act upon forward), `<--` (Retroactive feedback/rollback)
- **Z-Axis (Scale/Hierarchy):** `^--` (Builds to / Layer up), `v--` (Breaks down from / Layer down)
- **Connectives:** `!` (Adversative caveat), `::` (And), `:` (In relation to), `|` (Or), `--` (Not)

Balanced Ternary Truth Literals

- `+1` , `YES` , `PROCEED` (Monotonic commit)
- `0` , `OK` , `AWAIT` (Asynchronous yield / equilibrium)
- `-1` , `NO` , `ROLLBACK` (State unwinding)
- `NULL` , `VOID` , `HALT` (Terminal trap)

Universal 4-Block Auto-Population Snippets

- `omni-file` / `omni-4block` : Complete compliant `.omni` module template with interactive tab stops.

- `omni-scroll` / `scroll` : Complete compliant `.scroll` episodic trace and learning template.
- `block-1` : Scaffolds Block 1: Metadata Block.
- `block-2` : Scaffolds Block 2: Opening Block.
- `block-3` : Scaffolds Block 3: Body Block.
- `block-4` : Scaffolds Block 4: Closing Block.

3. Verified Extension State & Invariants

- **Language ID:** `omnicode`
- **File Associations:** `*.omni` , `*.scroll`
- **Grammar Version:** `0.1.2`
- **Editor Integration:** Tested on VS Code 1.75+ and Antigravity IDE.

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/extension.js

JS | 93 LINES | 306 WORDS | 3.9 KB

```
// ============================================================================
// OmniCode VS Code Extension: Hover & Docstring Subsystem
// ============================================================================
// Connects the editor to the native OmniCode language engine (CLI & HTTP).
// Delivers 3-Tier composite hovercards bridging high-level .tape process ribbons
// down to low-level .omni invariant contracts and physical silicon execution states.
// ============================================================================

const vscode = require('vscode');
const { execFile } = require('child_process');
const http = require('http');

/**
 * Activates the OmniCode VS Code Extension.
 * Registers the HoverProvider for .omni, .scroll, and .tape files.
 * @param {vscode.ExtensionContext} context
 */
function activate(context) {
    const hoverProvider = vscode.languages.registerHoverProvider('omnicode', {
        provideHover(document, position, token) {
            const range = document.getWordRangeAtPosition(position);
            if (!range) {
                return null;
            }

            const symbol = document.getText(range);
            const line = position.line + 1;
            const col = position.character + 1;
            const filePath = document.fileName;

            return new Promise((resolve) => {
                // Method 1: Try local omnicode CLI binary directly
                execFile(
                    'omnicode',
                    ['hover', filePath, String(line), String(col), '--json'],
                    { timeout: 3000 },
                    (err, stdout) => {
                        if (!err && stdout) {
                            try {
                                const data = JSON.parse(stdout.trim());
                                if (data && data.markdown) {
                                    const md = new vscode.MarkdownString(data.markdown);
                                    md.isTrusted = true;
                                    md.supportHtml = true;
                                    return resolve(new vscode.Hover(md, range));
                                }
                            } catch (_e) {
                                // Fall through to HTTP server fallback
                            }
                        }

                        // Method 2: Fallback to running OmniCode Studio / GUI server (default port 8080)
                        const url = `http://127.0.0.1:8080/api/hover?
file=${encodeURIComponent(filePath)}&line=${line}&col=${col}`;
                        const req = http.get(url, (res) => {
                            let body = '';
                            res.on('data', (chunk) => (body += chunk));
                            res.on('end', () => {
                                try {
                                    const data = JSON.parse(body);
                                    if (data && data.markdown) {
                                        const md = new vscode.MarkdownString(data.markdown);
                                        md.isTrusted = true;
                                        md.supportHtml = true;
                                        return resolve(new vscode.Hover(md, range));
                                    }
                                } catch (_err) {
                                    // Ignored
                                }
                                resolve(null);
                            });
                        });

```

```
                        req.on('error', () => resolve(null));
                        req.setTimeout(1000, () => {
                            req.destroy();
                            resolve(null);
                        });
                    }
                );
            });
        }
    });

    context.subscriptions.push(hoverProvider);
}

function deactivate() {}

module.exports = {
    activate,
    deactivate,
};
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/language-configuration.json

JSON | 30 LINES | 76 WORDS | 0.5 KB

```json
{
  "comments": {
    "lineComment": "#"
  },
  "brackets": [
    ["[", "]"],
    ["<", ">"],
    ["{", "}"],
    ["{", "]"],
    ["[", "}"],
    ["(", ")"]
  ],
  "autoClosingPairs": [
    { "open": "[", "close": "]" },
    { "open": "<", "close": ">" },
    { "open": "{", "close": "}" },
    { "open": "(", "close": ")" },
    { "open": "\"", "close": "\"" },
    { "open": "'", "close": "'" }
  ],
  "surroundingPairs": [
    ["[", "]"],
    ["<", ">"],
    ["{", "}"],
    ["(", ")"],
    ["\"", "\""],
    ["'", "'"]
  ]
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/omnicode-language-0.1.0.vsix

DOC/BINARY | 1 LINES | 5 WORDS | 2.9 KB

[Binary Artifact: omnicode-language-0.1.0.vsix (2.9 KB)]

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/omnicode-language-0.1.1.vsix

DOC/BINARY | 1 LINES | 5 WORDS | 4.0 KB

[Binary Artifact: omnicode-language-0.1.1.vsix (4.0 KB)]

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/omnicode-language-0.1.2.vsix

DOC/BINARY | 1 LINES | 5 WORDS | 6.4 KB

[Binary Artifact: omnicode-language-0.1.2.vsix (6.4 KB)]

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/package.json

JSON | 54 LINES | 92 WORDS | 1.1 KB

```json
{
  "name": "omnicode-language",
  "displayName": "OmniCode Language Support",
  "description": "Syntax highlighting, boundary morphology, and 4-block architecture support for the OmniCode language",
  "version": "0.1.3",
  "publisher": "omnicode",
  "main": "./extension.js",
  "activationEvents": [
    "onLanguage:omnicode"
  ],
  "engines": {
    "vscode": "^1.75.0"
  },
  "categories": [
    "Programming Languages",
    "Snippets"
  ],
  "contributes": {
    "languages": [
      {
        "id": "omnicode",
        "aliases": [
          "OmniCode",
          "omnicode",
          "omni",
          "OmniCode Scroll",
          "scroll",
          "OmniCode Tape",
          "tape"
        ],
        "extensions": [
          ".omni",
          ".scroll",
          ".tape"
        ],
        "configuration": "./language-configuration.json"
      }
    ],
    "grammars": [
      {
        "language": "omnicode",
        "scopeName": "source.omnicode",
        "path": "./syntaxes/omnicode.tmLanguage.json"
      }
    ],
    "snippets": [
      {
        "language": "omnicode",
        "path": "./snippets/omnicode.json"
      }
    ]
  }
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/syntaxes/omnicode.tmLanguage.json

JSON | 321 LINES | 529 WORDS | 8.7 KB

```
{
  "$schema": "https://raw.githubusercontent.com/martinring/tmlanguage/master/tmlanguage.json",
  "name": "OmniCode",
  "scopeName": "source.omnicode",
  "patterns": [
    { "include": "#comments" },
    { "include": "#boundary_casts" },
    { "include": "#primes" },
    { "include": "#affixed_identifiers" },
    { "include": "#asides" },
    { "include": "#strings" },
    { "include": "#ternary_literals" },
    { "include": "#vectors" },
    { "include": "#being_relations" },
    { "include": "#connectives" },
    { "include": "#boundaries" },
    { "include": "#numbers" },
    { "include": "#keywords" },
    { "include": "#registers" },
    { "include": "#categories" },
    { "include": "#instances" }
  ],
  "repository": {
    "comments": {
      "patterns": [
        {
          "name": "comment.line.number-sign.omnicode",
          "match": "#.*$"
        }
      ]
    },
    "boundary_casts": {
      "patterns": [
        {
          "name": "keyword.operator.boundary_cast.omnicode",
          "match": "\\[\\+\\]|\\[\\-\\]|<\\+>|<\\->|\\{\\+\\}|\\{\\-\\}"
        }
      ]
    },
    "primes": {
      "patterns": [
        {
          "name": "variable.other.prime.omnicode",
          "match": "\\b[a-zA-Z_][a-zA-Z0-9_]*`+"
        }
      ]
    },
    "affixed_identifiers": {
      "patterns": [
        {
          "name": "variable.other.affixed.abstract.omnicode",
          "match": "\\+[a-zA-Z_][a-zA-Z0-9_]*|[a-zA-Z_][a-zA-Z0-9_]*\\+"
        },
        {
          "name": "variable.other.affixed.concrete.omnicode",
          "match": "\\-[a-zA-Z_][a-zA-Z0-9_]*|[a-zA-Z_][a-zA-Z0-9_]*\\-"
        }
      ]
    },
    "asides": {
      "patterns": [
        {
          "name": "meta.aside.omnicode",
          "begin": "\\(",
          "beginCaptures": {
            "0": { "name": "punctuation.definition.aside.begin.omnicode" }
          },
          "end": "\\)",
          "endCaptures": {
            "0": { "name": "punctuation.definition.aside.end.omnicode" }
          },
          "patterns": [
            {
```

```
              "name": "string.unquoted.aside.intent.omnicode",
              "match": "[^()]+"
            }
          ]
        }
      ]
    },
    "strings": {
      "patterns": [
        {
          "name": "string.quoted.double.omnicode",
          "begin": "\"",
          "end": "\"",
          "patterns": [
            {
              "name": "constant.character.escape.omnicode",
              "match": "\\\\."
            }
          ]
        },
        {
          "name": "string.quoted.single.omnicode",
          "begin": "'",
          "end": "'|$",
          "patterns": [
            {
              "name": "constant.character.escape.omnicode",
              "match": "\\\\."
            }
          ]
        }
      ]
    },
    "ternary_literals": {
      "patterns": [
        {
          "name": "constant.language.ternary.proceed.omnicode",
          "match": "\\+1\\b|\\b(YES|PROCEED)\\b"
        },
        {
          "name": "constant.language.ternary.rollback.omnicode",
          "match": "-1\\b|\\b(NO|ROLLBACK)\\b"
        },
        {
          "name": "constant.language.ternary.await.omnicode",
          "match": "\\b0\\b|\\b(OK|AWAIT)\\b"
        },
        {
          "name": "constant.language.ternary.halt.omnicode",
          "match": "\\b(NULL|VOID|HALT)\\b"
        }
      ]
    },
    "vectors": {
      "patterns": [
        {
          "name": "keyword.operator.vector.scale.diagonal.major.omnicode",
          "match": "/-\\^"
        },
        {
          "name": "keyword.operator.vector.scale.diagonal.minor.omnicode",
          "match": "\\\\-v"
        },
        {
          "name": "keyword.operator.vector.scale.diagonal.great.omnicode",
          "match": "\\\\-\\^"
        },
        {
          "name": "keyword.operator.vector.scale.diagonal.small.omnicode",
          "match": "/-v"
        },
        {
          "name": "keyword.operator.vector.quadrant.omnicode",
          "match": "/\\^|\\\\v|\\^\\\\|v/"
        },
        {
          "name": "keyword.operator.vector.causality.forward.omnicode",
          "match": "-->|->"
```

```
        },
        {
          "name": "keyword.operator.vector.causality.backward.omnicode",
          "match": "<--|<-"
        },
        {
          "name": "keyword.operator.vector.scale.up.omnicode",
          "match": "\\^--|--\\^|-\\^|↑"
        },
        {
          "name": "keyword.operator.vector.scale.down.omnicode",
          "match": "v--|--v|-v|↓"
        }
      ]
    },
    "being_relations": {
      "patterns": [
        {
          "name": "keyword.operator.relation.invariant_equivalence.omnicode",
          "match": "<=>"
        },
        {
          "name": "keyword.operator.relation.identity.omnicode",
          "match": "="
        },
        {
          "name": "keyword.operator.relation.distinction.omnicode",
          "match": "!="
        },
        {
          "name": "keyword.operator.relation.homomorphism.omnicode",
          "match": "~=|≈"
        },
        {
          "name": "keyword.operator.relation.comparison.omnicode",
          "match": "<=|>=|<|>"
        }
      ]
    },
    "connectives": {
      "patterns": [
        {
          "name": "keyword.operator.connective.caveat.omnicode",
          "match": "!"
        },
        {
          "name": "keyword.operator.connective.and.omnicode",
          "match": "::"
        },
        {
          "name": "keyword.operator.connective.relation.omnicode",
          "match": ":"
        },
        {
          "name": "keyword.operator.connective.or.omnicode",
          "match": "\\|"
        },
        {
          "name": "keyword.operator.connective.not.omnicode",
          "match": "--"
        },
        {
          "name": "keyword.operator.connective.bound_pair.omnicode",
          "match": "&"
        },
        {
          "name": "keyword.operator.connective.percent.omnicode",
          "match": "%"
        },
        {
          "name": "keyword.operator.address.mmio.omnicode",
          "match": "::@|@"
        },
        {
          "name": "keyword.operator.range.omnicode",
          "match": "\\.\\."
        }
      ]
```

```
    },
    "boundaries": {
      "patterns": [
        {
          "name": "punctuation.definition.boundary.denotation.omnicode",
          "match": "\\[|\\]"
        },
        {
          "name": "punctuation.definition.boundary.invariant.omnicode",
          "match": "<|>"
        },
        {
          "name": "punctuation.definition.boundary.connotation.omnicode",
          "match": "\\{|\\}"
        }
      ]
    },
    "numbers": {
      "patterns": [
        {
          "name": "constant.numeric.hex.omnicode",
          "match": "\\b0x[0-9a-fA-F]+\\b"
        },
        {
          "name": "constant.numeric.omnicode",
          "match": "\\b\\d+(\\.\\d+)?\\b"
        }
      ]
    },
    "registers": {
      "patterns": [
        {
          "name": "variable.language.register.omnicode",
          "match": "\\b(HEAD|HOME|SPOOL|IT|SELF|CALLER|OTHER|STATE|TURN|BOUND|FROM|INTO|DISCARD)\\b"
        }
      ]
    },
    "keywords": {
      "patterns": [
        {
          "name": "keyword.control.block.omnicode",
          "match": "\\b(Module|Scroll|Opening|Body|Closing)\\b"
        },
        {
          "name": "keyword.other.declaration.omnicode",
          "match":
"\\b(import|requires|preconditions|verify|commit_status|user_prompt|synthesized_reel|machine_utterance|repair_entry|failpoint_rec
ord|version|system|author|safety|semantics|timestamp|session_id|execution_duration_micros|cognitive_mode|Witness|Nature)\\b"
        },
        {
          "name": "keyword.control.loop.verb.omnicode",
          "match":
"\\b(LOOP|WHEN|GOVERN|GUARD|TRAP|INSCRIBE|TRANSDUCE|PROJECT|SEQUENCE|ENCAPSULATE|LOCATE|BIND|ROUTE|SET_ASIDE|TRANSITION|CALCULATE
|MEASURE|MODULATE|REPEAT|PLAY|PAUSE|REWIND|STOP)\\b"
        },
        {
          "name": "keyword.control.tape.omnicode",
          "match": "\\b(Tape|Track)\\b"
        },
        {
          "name": "keyword.other.clause.tape.omnicode",
          "match": "\\b(WHO|WHAT|WHEN|WHERE|WHY|HOW)\\b"
        },
        {
          "name": "entity.name.function.roll.omnicode",
          "match": "\\broll_\\d+\\b|\\bROLL\\b"
        },
        {
          "name": "keyword.operator.arithmetic.word.omnicode",
          "match": "\\b(plus|minus|times|divide|by|add|sub|mult|div)\\b"
        },
        {
          "name": "keyword.operator.infer.omnicode",
          "match": "\\?"
        }
      ]
    },
    "categories": {
```

```
      "patterns": [
        {
          "name": "entity.name.type.category.omnicode",
          "match": "\\b[A-Z][a-zA-Z0-9_]*\\b"
        }
      ]
    },
    "instances": {
      "patterns": [
        {
          "name": "variable.other.instance.omnicode",
          "match": "\\b[a-z_][a-zA-Z0-9_]*\\b"
        }
      ]
    }
  }
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/snippets/omnicode.json

JSON | 163 LINES | 611 WORDS | 6.5 KB

```
{
  "OmniCode 4-Block Template": {
    "prefix": ["omni-file", "omni-4block", "omni"],
    "description": "Auto-populate standard OmniCode 4-Block file architecture",
    "body": [
      "# ================================================================================",
      "# BLOCK 1: METADATA BLOCK",
      "# ( File Identity, Version, Author, Intent Aside, and Invariants )",
      "# ================================================================================",
      "Module < ${1:ModuleName} > ( ${2:Module intent rationale} ) {",
      "    version: < \"${3:0.1.0}\" >,",
      "    author: < \"${4:ProfessorSeanEX}\" >,",
      "    system: < OmniCode.Core >,",
      "    safety: < MemorySafe :: Deterministic >",
      "}",
      "",
      "# Invariant Axioms & Types",
      "Type_${1:ModuleName} < +1 > ( Invariant definition )",
      "",
      "",
      "# ================================================================================",
      "# BLOCK 2: OPENING BLOCK",
      "# ( Environment Setup, Preconditions, Scoped Context, and Declarations )",
      "# ================================================================================",
      "Opening {",
      "    requires: [ ${5:MemoryAligned.64} ],",
      "    preconditions: [ +1 ]",
      "}",
      "",
      "# Initial state instances",
      "${6:state} [ +1 ] ( Initialized state instance )",
      "",
      "",
      "# ================================================================================",
      "# BLOCK 3: BODY BLOCK",
      "# ( Core Transformations, Causality Vectors, Logic, and Caveat Pipelines )",
      "# ================================================================================",
      "Body [",
      "    # Forward action: state acts upon transformation",
      "    ${6:state} --> ${7:next_state} [ +1 ]",
      "    ${7:next_state} ! [ ${6:state} = +1 ] ( Caveat verification )",
      "]",
      "",
      "",
      "# ================================================================================",
      "# BLOCK 4: CLOSING BLOCK",
      "# ( Postconditions, Invariant Proofs, Teardown, and Commit Status )",
      "# ================================================================================",
      "Closing {",
      "    verify: [ ${7:next_state} = +1 ],",
      "    commit_status: [ +1 ] ( Commit verified state )",
      "}",
      ""
    ]
  },
  "Metadata Block": {
    "prefix": "block-1",
    "description": "Insert Block 1: Metadata Block",
    "body": [
      "# ================================================================================",
      "# BLOCK 1: METADATA BLOCK",
      "# ================================================================================",
      "Module < ${1:Name} > ( ${2:Intent} ) {",
      "    version: < \"0.1.0\" >",
      "}",
      ""
    ]
  },
  "Opening Block": {
    "prefix": "block-2",
    "description": "Insert Block 2: Opening Block",
    "body": [
      "# ================================================================================",
```

```
      "# BLOCK 2: OPENING BLOCK",
      "# ================================================================================",
      "Opening {",
      "    requires: [ ${1:Prerequisites} ]",
      "}",
      ""
    ]
  },
  "Body Block": {
    "prefix": "block-3",
    "description": "Insert Block 3: Body Block",
    "body": [
      "# ================================================================================",
      "# BLOCK 3: BODY BLOCK",
      "# ================================================================================",
      "Body [",
      "    ${1:source} --> ${2:target}",
      "]",
      ""
    ]
  },
  "Closing Block": {
    "prefix": "block-4",
    "description": "Insert Block 4: Closing Block",
    "body": [
      "# ================================================================================",
      "# BLOCK 4: CLOSING BLOCK",
      "# ================================================================================",
      "Closing {",
      "    verify: [ ${1:condition} = +1 ],",
      "    commit_status: [ +1 ]",
      "}",
      ""
    ]
  },
  "OmniCode Episodic Scroll Template": {
    "prefix": ["omni-scroll", "scroll"],
    "description": "Auto-populate standard OmniCode Episodic Cognitive Scroll (.scroll)",
    "body": [
      "# ================================================================================",
      "# BLOCK 1: METADATA BLOCK (THE COGNITIVE SCROLL)",
      "# ( Execution Trace ID, Prompt Intent, Inferred Types, and Neural Snapshot )",
      "# ================================================================================",
      "Scroll < EpistemicTrace.${1:SessionId} > ( ${2:User Prompt Intent} ) {",
      "    version: < \"1.0.0\" >,",
      "    system: < OmniCode.AI.Model >,",
      "    timestamp: < \"${CURRENT_SECONDS_UNIX}\" >,",
      "    session_id: < \"${1:SessionId}\" >,",
      "    cognitive_mode: < NeuralSymbolicTransduction >",
      "}",
      "",
      "# ================================================================================",
      "# BLOCK 2: OPENING BLOCK (EXPERIENCE INITIALIZATION)",
      "# ( Hardware Invariants, Preconditions, and Initial Context )",
      "# ================================================================================",
      "Opening {",
      "    import: [ Core.Arithmetic :: Core.Logic :: Core.Vectors ],",
      "    preconditions: [ +1 ]",
      "}",
      "",
      "# Original Cognitive Discourse Prompt:",
      "user_prompt < \"${2:User Prompt Intent}\" >",
      "",
      "# ================================================================================",
      "# BLOCK 3: BODY BLOCK (CHRONICLE OF EXECUTION & RESTORATION)",
      "# ( Replay Tape: Synthesized Program, Sensory Telemetry & Immune Interventions )",
      "# ================================================================================ ",
      "Body [",
      "    # Immune System Interventions (Restorator):",
      "    repair_entry [ L1: CleanExecution --> \"Zero repairs required.\" ]",
      "",
      "    # Executed Turing Reel:",
      "    synthesized_reel [",
      "        ${3:state} [ +1 ]",
      "    ]",
      "]",
      "",
      "# ================================================================================",
```

```
      "# BLOCK 4: CLOSING BLOCK (EPISODIC COMMIT & LEARNING PROOF)",
      "# ( Verification Proof, Final Quad-State, and Conversational Synthesis )",
      "# ============================================================================== ",
      "Closing {",
      "    commit_status [ <PROCEED> ],",
      "    machine_utterance ( \"${4:Execution succeeded to +1:PROCEED.}\" )",
      "}",
      ""
    ]
  }
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/app.js JS 509 LINES 1537 WORDS 20.4 KB

```
// ================================================================================
// OMNICODE STUDIO & COGNITIVE TERMINAL APPLICATION LOGIC
// Grounded in Balanced Ternary Quad-State & Biblical Moral Ontology
// ================================================================================

document.addEventListener('DOMContentLoaded', () => {
    initNavigation();
    initPromptStudio();
    initTerminalGateway();
    initLexiconExplorer();
    initEpisodicScrolls();
    initHoverInspector();
});

// --------------------------------------------------------------------------------
// 1. NAVIGATION TAB SWITCHING
// --------------------------------------------------------------------------------
function initNavigation() {
    const tabs = document.querySelectorAll('.nav-tab');
    const panels = document.querySelectorAll('.view-panel');

    tabs.forEach(tab => {
        tab.addEventListener('click', () => {
            const targetViewId = tab.getAttribute('data-view');

            tabs.forEach(t => t.classList.remove('active'));
            panels.forEach(p => p.classList.remove('active'));

            tab.classList.add('active');
            const targetPanel = document.getElementById(targetViewId);
            if (targetPanel) {
                targetPanel.classList.add('active');
            }

            // Trigger data load if needed
            if (targetViewId === 'view-lexicon') {
                loadLexiconData();
            } else if (targetViewId === 'view-scrolls') {
                loadScrollsData();
            }
        });
    });
}

// --------------------------------------------------------------------------------
// 2. AI PROMPT STUDIO CONTROLLER
// --------------------------------------------------------------------------------
function initPromptStudio() {
    const promptInput = document.getElementById('prompt-input');
    const submitBtn = document.getElementById('btn-submit-prompt');
    const promptChips = document.querySelectorAll('.prompt-chip');
    const quadstateBadge = document.getElementById('studio-quadstate-badge');
    const stateLabel = document.getElementById('state-label');
    const responseContent = document.getElementById('response-content');
    const synthesizedCode = document.getElementById('synthesized-code');
    const badgeSession = document.getElementById('badge-session-id');
    const linkScroll = document.getElementById('link-scroll');
    const linkLog = document.getElementById('link-log');
    const copyCodeBtn = document.getElementById('btn-copy-code');

    // Quick Prompt Chips
    promptChips.forEach(chip => {
        chip.addEventListener('click', () => {
            const promptText = chip.getAttribute('data-prompt');
            promptInput.value = promptText;
            promptInput.focus();
        });
    });

    // Copy Code Button
    copyCodeBtn.addEventListener('click', () => {
        const codeText = synthesizedCode.textContent;
        navigator.clipboard.writeText(codeText).then(() => {
            const originalText = copyCodeBtn.textContent;
```

```
            copyCodeBtn.textContent = 'Copied!';
            setTimeout(() => { copyCodeBtn.textContent = originalText; }, 1800);
        });
    });

    // Submit Prompt Execution
    submitBtn.addEventListener('click', async () => {
        const prompt = promptInput.value.trim();
        if (!prompt) return;

        // Set Loading State
        submitBtn.disabled = true;
        submitBtn.querySelector('.btn-text').textContent = 'Deliberating...';
        stateLabel.textContent = 'DELIBERATING (0:AWAIT)';
        quadstateBadge.className = 'quadstate-indicator';

        try {
            const res = await fetch('/api/prompt', {
                method: 'POST',
                headers: { 'Content-Type': 'application/json' },
                body: JSON.stringify({ prompt })
            });

            if (!res.ok) {
                const errData = await res.json().catch(() => ({ error: 'Unknown server error' }));
                throw new Error(errData.error || `HTTP ${res.status}`);
            }

            const data = await res.json();
            renderPromptResult(data);
        } catch (err) {
            console.error('Prompt evaluation error:', err);
            responseContent.innerHTML = `<p style="color: var(--state-rollback);">✗ Error evaluating prompt:
${err.message}</p>`;
            stateLabel.textContent = 'ERROR (NULL:HALT)';
            quadstateBadge.className = 'quadstate-indicator halt';
        } finally {
            submitBtn.disabled = false;
            submitBtn.querySelector('.btn-text').textContent = 'Send to AI Cortex';
        }
    });

    function renderPromptResult(data) {
        // 1. Update QuadState Badge
        const state = (data.state || 'PROCEED').toUpperCase();
        quadstateBadge.className = 'quadstate-indicator';

        if (state === 'PROCEED') {
            quadstateBadge.classList.add('proceed');
            stateLabel.textContent = '+1: PROCEED';
        } else if (state === 'ROLLBACK') {
            quadstateBadge.classList.add('rollback');
            stateLabel.textContent = '-1: ROLLBACK (CAVEAT VETO)';
        } else if (state === 'AWAIT') {
            quadstateBadge.classList.add('await');
            stateLabel.textContent = '0: AWAIT (DISCERNMENT)';
        } else {
            quadstateBadge.classList.add('halt');
            stateLabel.textContent = 'NULL: HALT (TERMINAL TRAP)';
        }

        // 2. Session and Response Content
        badgeSession.textContent = `Session: ${data.session_id || 'active'}`;
        responseContent.textContent = data.response || 'No natural discourse generated.';

        // 3. Synthesized Code Block
        synthesizedCode.textContent = data.omnicode || '// Zero code generated.';

        // 4. File Links
        linkScroll.textContent = `📜 Scroll: ${data.scroll || 'Recorded'}`;
        linkLog.textContent = `📋 Log: ${data.log || 'Recorded'}`;
    }
}

// ---------------------------------------------------------------------------
// 3. CLI TERMINAL GATEWAY CONTROLLER
// ---------------------------------------------------------------------------
function initTerminalGateway() {
```

```
    const termInput = document.getElementById('terminal-cmd-input');
    const runBtn = document.getElementById('btn-run-cmd');
    const termStream = document.getElementById('terminal-stream');
    const clearBtn = document.getElementById('btn-clear-terminal');
    const shortcutBtns = document.querySelectorAll('.term-btn');
    const termWindow = document.getElementById('terminal-window');

    const desktopBtn = document.getElementById('btn-desktop-launch');
    if (desktopBtn) {
        desktopBtn.addEventListener('click', () => {
            executeTerminalCommand('omnicode desktop');
        });
    }

    clearBtn.addEventListener('click', () => {
        termStream.innerHTML = '';
    });

    shortcutBtns.forEach(btn => {
        btn.addEventListener('click', () => {
            const cmd = btn.getAttribute('data-cmd');
            termInput.value = cmd;
            executeTerminalCommand(cmd);
        });
    });

    runBtn.addEventListener('click', () => {
        const cmd = termInput.value.trim();
        if (!cmd) return;
        executeTerminalCommand(cmd);
        termInput.value = '';
    });

    termInput.addEventListener('keydown', (e) => {
        if (e.key === 'Enter') {
            const cmd = termInput.value.trim();
            if (!cmd) return;
            executeTerminalCommand(cmd);
            termInput.value = '';
        }
    });

    async function executeTerminalCommand(command) {
        appendTerminalLine(`omnicode> ${command}`, 'term-line-cmd');

        try {
            const res = await fetch('/api/terminal', {
                method: 'POST',
                headers: { 'Content-Type': 'application/json' },
                body: JSON.stringify({ command })
            });

            const data = await res.json();
            if (data.stdout) {
                appendTerminalLine(data.stdout, 'term-line-stdout');
            }
            if (data.stderr) {
                appendTerminalLine(data.stderr, 'term-line-stderr');
            }
        } catch (err) {
            appendTerminalLine(`Error executing command: ${err.message}`, 'term-line-stderr');
        }

        termWindow.scrollTop = termWindow.scrollHeight;
    }

    function appendTerminalLine(text, cssClass) {
        const div = document.createElement('div');
        div.className = cssClass;
        div.textContent = text;
        termStream.appendChild(div);
    }
}

// ------------------------------------------------------------------------
// 4. COVENANT MANIFEST & BIBLICAL LEXICON CONTROLLER
// ------------------------------------------------------------------------
let allLexiconEntries = [];
```

```
230
231 function initLexiconExplorer() {
232     const searchInput = document.getElementById('lexicon-search-input');
233     const filterBtns = document.querySelectorAll('.filter-btn');
234
235     searchInput.addEventListener('input', () => {
236         filterAndRenderLexicon();
237     });
238
239     filterBtns.forEach(btn => {
240         btn.addEventListener('click', () => {
241             filterBtns.forEach(b => b.classList.remove('active'));
242             btn.classList.add('active');
243             filterAndRenderLexicon();
244         });
245     });
246 }
247
248 async function loadLexiconData() {
249     const entriesGrid = document.getElementById('lexicon-entries-grid');
250     if (allLexiconEntries.length > 0) return; // Already cached
251
252     entriesGrid.innerHTML = '<p class="placeholder-text">Loading biblical knowledge base from offline memory...</p>';
253
254     try {
255         const res = await fetch('/api/lexicon');
256         const data = await res.json();
257         allLexiconEntries = data.entries || [];
258         filterAndRenderLexicon();
259     } catch (err) {
260         console.error('Failed to load lexicon:', err);
261         entriesGrid.innerHTML = `<p style="color: var(--state-rollback);">Failed to load lexicon: ${err.message}</p>`;
262     }
263 }
264
265 function filterAndRenderLexicon() {
266     const entriesGrid = document.getElementById('lexicon-entries-grid');
267     const searchInput = document.getElementById('lexicon-search-input');
268     const query = searchInput.value.trim().toLowerCase();
269     const activeFilterBtn = document.querySelector('.filter-btn.active');
270     const selectedDomain = activeFilterBtn ? activeFilterBtn.getAttribute('data-domain') : 'all';
271
272     const filtered = allLexiconEntries.filter(entry => {
273         const matchesQuery = query === '' ||
274             entry.term.toLowerCase().includes(query) ||
275             entry.definition.toLowerCase().includes(query) ||
276             entry.kjv.toLowerCase().includes(query) ||
277             entry.web.toLowerCase().includes(query);
278
279         const matchesDomain = selectedDomain === 'all' || entry.domain === selectedDomain;
280
281         return matchesQuery && matchesDomain;
282     });
283
284     if (filtered.length === 0) {
285         entriesGrid.innerHTML = `<p class="placeholder-text">No concepts found matching '${query}' in selected domain.
</p>`;
286         return;
287     }
288
289     entriesGrid.innerHTML = filtered.map(e => `
290         <article class="entry-card">
291             <div class="entry-header">
292                 <span class="entry-term">${escapeHtml(e.term)}</span>
293                 <span class="entry-domain">${escapeHtml(e.domain)}</span>
294             </div>
295             <p class="entry-def">${escapeHtml(e.definition)}</p>
296             <div class="entry-witnesses">
297                 <p><span class="witness-title">KJV (1611):</span> ${escapeHtml(e.kjv)}</p>
298                 <p><span class="witness-title">WEB:</span> ${escapeHtml(e.web)}</p>
299             </div>
300             <p class="entry-guidance">⚖️ ${escapeHtml(e.guidance)}</p>
301         </article>
302     `).join('');
303 }
304
305 // ---------------------------------------------------------------------------
306 // 5. EPISODIC SCROLLS CONTROLLER
```

```
// ------------------------------------------------------------------------------
function initEpisodicScrolls() {
    const refreshBtn = document.getElementById('btn-refresh-scrolls');
    refreshBtn.addEventListener('click', () => {
        loadScrollsData();
    });
}

async function loadScrollsData() {
    const tbody = document.getElementById('episodes-tbody');
    tbody.innerHTML = '<tr><td colspan="6" style="text-align:center; padding: 2rem;">Loading memory bank...</td></tr>';

    try {
        const res = await fetch('/api/scrolls');
        const data = await res.json();

        // Update Metrics
        document.getElementById('metric-total').textContent = data.total || 0;
        document.getElementById('metric-proceed').textContent = data.proceed || 0;
        document.getElementById('metric-await').textContent = data.await || 0;
        document.getElementById('metric-rollback').textContent = data.rollback || 0;
        document.getElementById('metric-halt').textContent = data.halt || 0;
        document.getElementById('metric-repairs').textContent = data.repairs_learned || 0;

        // Update Episodes Table
        const episodes = data.episodes || [];
        if (episodes.length === 0) {
            tbody.innerHTML = '<tr><td colspan="6" style="text-align:center; padding: 2rem; color: var(--text-
muted);">No scrolls inscribed yet. Run prompts in Studio to record episodes.</td></tr>';
            return;
        }

        tbody.innerHTML = episodes.map(ep => {
            const stateClass = ep.state === 'PROCEED' ? 'state-proceed' :
                               ep.state === 'ROLLBACK' ? 'state-rollback' :
                               ep.state === 'AWAIT' ? 'state-await' : 'state-halt';
            return `
                <tr>
                    <td style="font-family: var(--font-mono); color: var(--cyan);">${escapeHtml(ep.session_id)}</td>
                    <td>${escapeHtml(ep.prompt)}</td>
                    <td><span class="state-chip ${stateClass}">+1:${escapeHtml(ep.state)}</span></td>
                    <td style="text-align: center;">${ep.repairs}</td>
                    <td style="text-align: center;">${ep.failpoints}</td>
                    <td style="font-family: var(--font-mono); font-size: 0.75rem; color: var(--text-
muted);">${escapeHtml(ep.scroll)}</td>
                </tr>
            `;
        }).join('');
    } catch (err) {
        console.error('Failed to load scrolls:', err);
        tbody.innerHTML = `<tr><td colspan="6" style="color: var(--state-rollback); text-align: center;">Error:
${err.message}</td></tr>`;
    }
}

// Utility: HTML Escaping
function escapeHtml(str) {
    if (!str) return '';
    return String(str)
        .replace(/&/g, '&amp;')
        .replace(/</g, '&lt;')
        .replace(/>/g, '&gt;')
        .replace(/"/g, '&quot;')
        .replace(/'/g, '&#039;');
}

// ------------------------------------------------------------------------------
// 6. 3-TIER HOVER INSPECTOR CONTROLLER (.omni <-> .tape)
// ------------------------------------------------------------------------------
function initHoverInspector() {
    let tooltipEl = document.querySelector('.omni-hover-card');
    if (!tooltipEl) {
        tooltipEl = document.createElement('div');
        tooltipEl.className = 'omni-hover-card';
        tooltipEl.style.display = 'none';
        document.body.appendChild(tooltipEl);
    }

```

```
    let hideTimer = null;

    document.addEventListener('mouseover', async (e) => {
        const target = e.target;
        if (!target) return;

        const codeContainer = target.closest('pre, code, .code-editor, #synthesized-code');
        if (!codeContainer) {
            return;
        }

        const word = getWordUnderCursor(e);
        if (!word || word.length < 2 || /^\d+$/.test(word)) {
            return;
        }

        clearTimeout(hideTimer);

        try {
            const res = await fetch(`/api/hover?symbol=${encodeURIComponent(word)}`);
            if (!res.ok) return;
            const data = await res.json();
            if (!data || !data.symbol) return;

            renderHoverCard(tooltipEl, data, e.pageX, e.pageY);
        } catch (_err) {
            // Ignored
        }
    });

    document.addEventListener('mouseout', (e) => {
        if (tooltipEl.contains(e.relatedTarget)) return;
        hideTimer = setTimeout(() => {
            tooltipEl.style.display = 'none';
        }, 300);
    });

    tooltipEl.addEventListener('mouseenter', () => clearTimeout(hideTimer));
    tooltipEl.addEventListener('mouseleave', () => {
        tooltipEl.style.display = 'none';
    });
}

function getWordUnderCursor(event) {
    if (document.caretRangeFromPoint) {
        const range = document.caretRangeFromPoint(event.clientX, event.clientY);
        if (range && range.startContainer && range.startContainer.nodeType === Node.TEXT_NODE) {
            const text = range.startContainer.textContent;
            const offset = range.startOffset;
            const match = text.slice(0, offset).match(/[a-zA-Z0-9_]+$/);
            const afterMatch = text.slice(offset).match(/^[a-zA-Z0-9_]+/);
            const before = match ? match[0] : '';
            const after = afterMatch ? afterMatch[0] : '';
            return before + after;
        }
    }
    return null;
}

function renderHoverCard(cardEl, data, x, y) {
    const omni = data.omni_definition;
    const tape = data.tape_context;

    const rankBadge = omni
        ? `<span class="omni-hover-rank-badge rank-${omni.rank}">Rank ${omni.rank}: ${escapeHtml(omni.boundary_type)}
</span>`
        : (data.is_locally_redefined ? `<span class="omni-hover-rank-badge">Rank 2: Denotation</span>` : `<span
class="omni-hover-rank-badge">Tape Entity</span>`);

    let tapeHtml = '';
    if (tape) {
        tapeHtml = `
            <div class="omni-hover-tier">
                <div class="omni-hover-tier-title tier-tape">📜 1. High-Level Process Tape</div>
                <div class="omni-hover-prop"><span class="omni-hover-label">Ribbon:</span><span class="omni-hover-
val">${escapeHtml(tape.tape_name)} (${escapeHtml(tape.track)})</span></div>
                <div class="omni-hover-prop"><span class="omni-hover-label">Active Roll:</span><span class="omni-hover-
val">${tape.roll_number}/${tape.total_rolls}</span></div>
```

```
                ${tape.roll_aside_intent ? `<div class="omni-hover-aside">"${escapeHtml(tape.roll_aside_intent)}"</div>` : ''}
                <div class="omni-hover-prop"><span class="omni-hover-label">Head Covenant:</span><span class="omni-hover-val">WHO ${escapeHtml(tape.head_who)} &rarr; WHAT ${escapeHtml(tape.head_what)}</span></div>
            </div>
        `;
    }

    let omniHtml = '';
    if (omni) {
        omniHtml = `
            <div class="omni-hover-tier">
                <div class="omni-hover-tier-title tier-omni">🔍 2. Under-The-Hood (.omni Blueprint)</div>
                <div class="omni-hover-prop"><span class="omni-hover-label">Grounded Source:</span><span class="omni-hover-val">${escapeHtml(omni.source_file)}:${omni.line}</span></div>
                <div class="omni-hover-prop"><span class="omni-hover-label">4-Block Origin:</span><span class="omni-hover-val">${escapeHtml(omni.block_kind || 'Universal Scope')}</span></div>
                ${omni.aside_intent ? `<div class="omni-hover-aside">"${escapeHtml(omni.aside_intent)}"</div>` : ''}
            </div>
        `;
    }

    const siliconHtml = `
        <div class="omni-hover-tier">
            <div class="omni-hover-tier-title tier-silicon">⚙️ 3. Machine Execution & Silicon State</div>
            <div class="omni-hover-prop"><span class="omni-hover-label">Partition Reservation:</span><span class="omni-hover-val">${tape ? tape.partition_size + ' cells' : '1024 cells'}</span></div>
            <div class="omni-hover-prop"><span class="omni-hover-label">Radix-3 Transducer:</span><span class="omni-hover-val">Dense Byte &harr; 5 Balanced Trits</span></div>
            <div class="omni-hover-prop"><span class="omni-hover-label">Silicon Truth State:</span><span class="omni-hover-val" style="color: var(--state-proceed);">+1:PROCEED (Deterministic Commit)</span></div>
        </div>
    `;

    cardEl.innerHTML = `
        <div class="omni-hover-header">
            <span class="omni-hover-symbol">${escapeHtml(data.symbol)}</span>
            ${rankBadge}
        </div>
        ${tapeHtml}
        ${omniHtml}
        ${siliconHtml}
    `;

    cardEl.style.display = 'block';
    const cardRect = cardEl.getBoundingClientRect();
    let left = x + 15;
    let top = y + 15;

    if (left + cardRect.width > window.innerWidth - 20) {
        left = x - cardRect.width - 15;
    }
    if (top + cardRect.height > window.innerHeight - 20) {
        top = y - cardRect.height - 15;
    }

    cardEl.style.left = `${Math.max(10, left)}px`;
    cardEl.style.top = `${Math.max(10, top)}px`;
}

```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/cloud.html HTML 201 LINES 718 WORDS 10.8 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Cloudflare Zero Trust & Oracle Cloud Edge Deployment — CreativeWorkzStudio LLC.">
    <title>Cloud &amp;amp; Oracle Edge Gateway — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- CLOUD MAIN -->
        <main class="doc-container">
            <div class="doc-hero card-glass">
                <div class="hero-badge">INFRASTRUCTURE &amp;amp; EDGE GATEWAY</div>
                <h1>Cloudflare Zero Trust &amp;amp; Oracle Cloud Architecture</h1>
                <p class="hero-lead">
                    Deploy and expose OmniCode 24/7 across the globe using Oracle Cloud Infrastructure (OCI) compute
instances and Cloudflare Zero Trust secure HTTPS edge tunnels.
                </p>
            </div>

            <div class="doc-grid">
                <!-- ORACLE VM CARD -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🌌</span>
                        <h2>Oracle Cloud Compute Instance</h2>
                    </div>
                    <p>Turnkey production deployment specifications configured for OmniCode:</p>
                    <table class="spec-table">
                        <tbody>
                            <tr>
                                <td><strong>Remote Host IP</strong></td>
                                <td><code>147.224.147.30</code> (SSH alias: <code>oracle</code>)</td>
                            </tr>
                            <tr>
                                <td><strong>Remote User</strong></td>
                                <td><code>ubuntu</code></td>
                            </tr>
                            <tr>
                                <td><strong>Authentication</strong></td>
                                <td><code>~/.ssh/oracle_key.pem</code> (Ed25519 / RSA)</td>
                            </tr>
                            <tr>
                                <td><strong>Target Directory</strong></td>
                                <td><code>/home/ubuntu/OmniCode</code></td>
                            </tr>
                            <tr>
                                <td><strong>OS Firewall</strong></td>
                                <td>TCP Port <code>8080</code> (Permitted via iptables-persistent)</td>
                            </tr>
                            <tr>
                                <td><strong>Systemd Service</strong></td>
                                <td><code>omnicode-web.service</code> (24/7 background operation)</td>
                            </tr>
                        </tbody>
                    </table>

                    <div class="doc-subcard">
                        <h3>Automated VM Synchronization</h3>
                        <p>To synchronize local changes and restart production services on the Oracle VM:</p>
                        <code>./scripts/sync_oracle_vm.sh "Deploy updates"</code>
                    </div>
                </div>

                <!-- CREATIVEWORKZSTUDIO GATEWAY -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🏢</span>
                        <h2>CreativeWorkzStudio LLC Gateway</h2>
                    </div>
                    <p>Licensed dynamic endpoint and local workstation infrastructure:</p>
                    <table class="spec-table">
                        <tbody>
                            <tr>
                                <td><strong>Licensed Domain</strong></td>
                                <td><code>omnicodestudio.duckdns.org</code></td>
                            </tr>
                            <tr>
                                <td><strong>Host WAN IP</strong></td>
                                <td><code>71.11.99.168</code></td>
                            </tr>
                            <tr>
                                <td><strong>Organization</strong></td>
```

```
                                <td>CreativeWorkzStudio LLC</td>
                            </tr>
                            <tr>
                                <td><strong>Direct Studio URL</strong></td>
                                <td><code>http://omnicodestudio.duckdns.org:8080</code></td>
                            </tr>
                            <tr>
                                <td><strong>DNS Zone File</strong></td>
                                <td><code>dns/omnicodestudio.com.zone</code></td>
                            </tr>
                        </tbody>
                    </table>

                    <div class="doc-subcard">
                        <h3>DNS Zone Import</h3>
                        <p>Import into Cloudflare DNS (DNS → Records → Import):</p>
                        <code>python3 dns/zone_manager.py inspect</code>
                    </div>
                </div>

                <!-- CLOUDFLARE TUNNEL CARD -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🌐</span>
                        <h2>Cloudflare Zero Trust Secure Tunnel</h2>
                    </div>
                    <p>Expose local port 8080 to the public internet securely without opening router ports:</p>
                    <ul class="spec-list">
                        <li><strong>Named Tunnel ID:</strong> <code>c346aa27-b4c1-41ba-815c-f66b2b84243b</code></li>
                        <li><strong>Tunnel Target:</strong> <code>c346aa27...cfargotunnel.com.</code></li>
                        <li><strong>Quick Tunnel Mode:</strong> Free ephemeral HTTPS URL
(<code>https://*.trycloudflare.com</code>).</li>
                        <li><strong>Edge TLS &amp;amp; DDoS Protection:</strong> Protected by Cloudflare global edge.</li>
                    </ul>

                    <div class="doc-subcard">
                        <h3>Launch Quick Edge Tunnel</h3>
                        <p>Execute the connector script to obtain an instant HTTPS domain:</p>
                        <code>./scripts/setup_cloudflare_tunnel.sh --quick</code>
                    </div>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Infrastructure &bull; Oracle Cloud &bull; Cloudflare Edge &bull; DuckDNS</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Edge Gateway</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/desktop_app.py PY 1182 LINES 3250 WORDS 44.9 KB

```
#!/usr/bin/env python3
# ==============================================================================
# BLOCK 1: METADATA BLOCK
# ( CreativeWorkzStudio LLC — OmniCode Desktop GUI Studio & Terminal Shell )
# ==============================================================================
"""
OmniCode Desktop GUI Studio & Interactive Terminal Shell
A CreativeWorkzStudio LLC Project

Provides a native desktop interface for:
  1. Interactive GUI Terminal Shell for direct CLI execution
  2. AI Prompt Studio with real-time Balanced Ternary Quad-State deliberation
  3. Language & AI Architecture Reference Viewer
  4. Cloudflare Tunnel & Oracle Cloud VM Deployment Management
"""

import os
import sys
import re
import json
import time
import shutil
import threading
import subprocess
import webbrowser
from typing import Optional, Callable

# Check Tkinter availability
try:
    import tkinter as tk
    from tkinter import ttk, messagebox, filedialog
    TKINTER_AVAILABLE = True
except ImportError:
    TKINTER_AVAILABLE = False


# ==============================================================================
# BLOCK 2: OPENING BLOCK
# ( Constants, Paths, Color Palette & Utility Helpers )
# ==============================================================================
SCRIPT_DIR = os.path.dirname(os.path.abspath(__file__))
BASE_DIR = os.path.dirname(SCRIPT_DIR)
SCRIPTS_DIR = os.path.join(BASE_DIR, "scripts")

# ANSI escape sequence stripper
ANSI_REGEX = re.compile(r'\x1b\[[0-9;]*[a-zA-Z]')

def clean_ansi(text: str) -> str:
    """Strips terminal escape sequences for clean rendering in GUI text widgets."""
    return ANSI_REGEX.sub('', text)

# Color Scheme: Sleek modern dark mode (GitHub Dark / Monokai Pro aesthetic)
THEME = {
    "bg_main": "#0d1117",
    "bg_surface": "#161b22",
    "bg_elevated": "#21262d",
    "border": "#30363d",
    "border_highlight": "#58a6ff",
    "text_primary": "#f0f6fc",
    "text_secondary": "#8b949e",
    "text_muted": "#484f58",
    "accent_blue": "#58a6ff",
    "accent_cyan": "#38bdf8",
    "accent_amber": "#d29922",
    "accent_purple": "#bc8cff",
    "state_proceed": "#238636",
    "state_proceed_light": "#3fb950",
    "state_await": "#9e6a03",
    "state_await_light": "#d29922",
    "state_rollback": "#da3633",
    "state_rollback_light": "#f85149",
    "state_halt": "#8957e5",
    "state_halt_light": "#a371f7",
    "console_bg": "#090d13",
```

```
    "console_fg": "#c9d1d9",
}


# ==============================================================================
# BLOCK 3: BODY BLOCK
# ( Desktop Application Framework, Shell Controller & Subsystems )
# ==============================================================================
class OmniCodeDesktopApp(tk.Tk):
    """Native Desktop GUI Studio and Terminal Shell for OmniCode."""

    def __init__(self):
        super().__init__()
        self.title("OmniCode Studio & Terminal Shell — CreativeWorkzStudio LLC")
        self.geometry("1280x840")
        self.minsize(1040, 680)
        self.configure(bg=THEME["bg_main"])

        # Process management
        self.active_process = None
        self.web_server_process = None
        self.tunnel_process = None
        self.cmd_history = []
        self.history_idx = 0

        self._configure_styles()
        self._build_header()
        self._build_main_layout()
        self._start_status_heartbeat()

    def _configure_styles(self):
        """Initializes ttk dark theme widget styles."""
        style = ttk.Style(self)
        style.theme_use("clam")

        # Configure tabs
        style.configure(
            "TNotebook",
            background=THEME["bg_main"],
            borderwidth=0
        )
        style.configure(
            "TNotebook.Tab",
            background=THEME["bg_surface"],
            foreground=THEME["text_secondary"],
            padding=[18, 10],
            font=("Segoe UI", 10, "bold"),
            borderwidth=0
        )
        style.map(
            "TNotebook.Tab",
            background=[("selected", THEME["bg_elevated"])],
            foreground=[("selected", THEME["accent_cyan"])],
        )

    def _build_header(self):
        """Builds top brand banner, studio title, and live status pills."""
        header = tk.Frame(self, bg=THEME["bg_surface"], height=64, padx=20, pady=10)
        header.pack(fill=tk.X, side=tk.TOP)
        header.pack_propagate(False)

        # Left branding
        brand_frame = tk.Frame(header, bg=THEME["bg_surface"])
        brand_frame.pack(side=tk.LEFT, fill=tk.Y)

        symbol_lbl = tk.Label(
            brand_frame,
            text="Ω",
            font=("Cinzel", 24, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
        )
        symbol_lbl.pack(side=tk.LEFT, padx=(0, 10))

        title_box = tk.Frame(brand_frame, bg=THEME["bg_surface"])
        title_box.pack(side=tk.LEFT, fill=tk.Y)

        title_lbl = tk.Label(
```

```
            title_box,
            text="OMNICODE STUDIO",
            font=("Segoe UI", 12, "bold"),
            fg=THEME["text_primary"],
            bg=THEME["bg_surface"]
        )
        title_lbl.pack(anchor="w")

        subtitle_lbl = tk.Label(
            title_box,
            text="A CreativeWorkzStudio LLC Project",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["accent_amber"],
            bg=THEME["bg_surface"]
        )
        subtitle_lbl.pack(anchor="w")

        # Right status pills
        status_frame = tk.Frame(header, bg=THEME["bg_surface"])
        status_frame.pack(side=tk.RIGHT, fill=tk.Y)

        self.pill_engine = tk.Label(
            status_frame,
            text="● Engine: Ready",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["state_proceed"],
            fg="#ffffff",
            padx=10,
            pady=4
        )
        self.pill_engine.pack(side=tk.LEFT, padx=5)

        self.pill_web = tk.Label(
            status_frame,
            text="○ Web: Port 8080 (Off)",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            padx=10,
            pady=4
        )
        self.pill_web.pack(side=tk.LEFT, padx=5)

        self.pill_tunnel = tk.Label(
            status_frame,
            text="○ Edge Tunnel: Idle",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            padx=10,
            pady=4
        )
        self.pill_tunnel.pack(side=tk.LEFT, padx=5)

        self.pill_quad = tk.Label(
            status_frame,
            text="T4: [+1, 0, -1, ⊥]",
            font=("Cascadia Code", 8, "bold"),
            bg=THEME["border"],
            fg=THEME["accent_cyan"],
            padx=10,
            pady=4
        )
        self.pill_quad.pack(side=tk.LEFT, padx=5)

    def _build_main_layout(self):
        """Builds sidebar and multi-tab workspace."""
        content = tk.Frame(self, bg=THEME["bg_main"])
        content.pack(fill=tk.BOTH, expand=True)

        # ------------------ LEFT SIDEBAR ------------------
        sidebar = tk.Frame(content, bg=THEME["bg_surface"], width=240, padx=14, pady=14)
        sidebar.pack(side=tk.LEFT, fill=tk.Y)
        sidebar.pack_propagate(False)

        side_title = tk.Label(
            sidebar,
            text="QUICK ACTIONS",
```

```
            font=("Segoe UI", 8, "bold"),
            fg=THEME["text_muted"],
            bg=THEME["bg_surface"]
        )
        side_title.pack(anchor="w", pady=(0, 10))

        # Action Buttons
        self._create_sidebar_btn(sidebar, "🌐 Open Web Studio", self.open_web_studio, "#1f6feb")
        self._create_sidebar_btn(sidebar, "⚡ Toggle Web Server", self.toggle_web_server, "#238636")
        self._create_sidebar_btn(sidebar, "☁️ Cloudflare Quick Tunnel", self.start_quick_tunnel, "#0969da")
        self._create_sidebar_btn(sidebar, "🌌 Sync Oracle Cloud VM", self.sync_oracle_vm, "#8957e5")

        tk.Frame(sidebar, height=1, bg=THEME["border"]).pack(fill=tk.X, pady=12)

        cli_lbl = tk.Label(
            sidebar,
            text="CLI OPERATIONS",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["text_muted"],
            bg=THEME["bg_surface"]
        )
        cli_lbl.pack(anchor="w", pady=(0, 8))

        self._create_sidebar_btn(sidebar, "🧪 Run Verification Demo", lambda: self.run_cli_command("omnicode demo"))
        self._create_sidebar_btn(sidebar, "📋 Run 4-Block Linter", lambda: self.run_cli_command("omnicode lint
manifest.omni"))
        self._create_sidebar_btn(sidebar, "🩹 Run Restorator", lambda: self.run_cli_command("omnicode restore
manifest.omni"))
        self._create_sidebar_btn(sidebar, "📜 Inspect Episodic Scrolls", lambda: self.run_cli_command("omnicode
scrolls"))
        self._create_sidebar_btn(sidebar, "📖 Covenant Manifest", lambda: self.run_cli_command("omnicode manifest"))
        self._create_sidebar_btn(sidebar, "📦 Formal File Types", lambda: self.run_cli_command("omnicode filetypes"))

        # Footer info
        side_footer = tk.Frame(sidebar, bg=THEME["bg_surface"])
        side_footer.pack(side=tk.BOTTOM, fill=tk.X)
        tk.Label(
            side_footer,
            text="CreativeWorkzStudio LLC\nEngine v0.1.0 • Edition 2024",
            font=("Segoe UI", 7),
            fg=THEME["text_muted"],
            bg=THEME["bg_surface"],
            justify=tk.LEFT
        ).pack(anchor="w")

        # ------------------ MAIN TABS ------------------
        self.notebook = ttk.Notebook(content)
        self.notebook.pack(side=tk.RIGHT, fill=tk.BOTH, expand=True)

        # Tab 1: Terminal Shell
        self.tab_terminal = tk.Frame(self.notebook, bg=THEME["bg_main"], padx=14, pady=14)
        self.notebook.add(self.tab_terminal, text="📟 Interactive Terminal Shell")
        self._build_terminal_tab()

        # Tab 2: AI Prompt Studio
        self.tab_studio = tk.Frame(self.notebook, bg=THEME["bg_main"], padx=14, pady=14)
        self.notebook.add(self.tab_studio, text="💡 AI Prompt Studio")
        self._build_studio_tab()

        # Tab 3: Language & AI Model Guide
        self.tab_guide = tk.Frame(self.notebook, bg=THEME["bg_main"], padx=14, pady=14)
        self.notebook.add(self.tab_guide, text="📐 Language & AI Model Guide")
        self._build_guide_tab()

        # Tab 4: Cloudflare & Oracle Cloud Center
        self.tab_cloud = tk.Frame(self.notebook, bg=THEME["bg_main"], padx=14, pady=14)
        self.notebook.add(self.tab_cloud, text="☁️ Cloudflare & Oracle Edge")
        self._build_cloud_tab()

    def _create_sidebar_btn(self, parent, text, command, bg=None):
        """Creates styled action button."""
        btn = tk.Button(
            parent,
            text=text,
            command=command,
            font=("Segoe UI", 8, "bold"),
            bg=bg or THEME["bg_elevated"],
            fg=THEME["text_primary"],
```

```
            activebackground=THEME["border"],
            activeforeground="#ffffff",
            relief="flat",
            bd=0,
            padx=10,
            pady=7,
            anchor="w",
            cursor="hand2"
        )
        btn.pack(fill=tk.X, pady=3)
        return btn

    # ------------------------------------------------------------------------
    # TAB 1: INTERACTIVE GUI TERMINAL SHELL
    # ------------------------------------------------------------------------
    def _build_terminal_tab(self):
        """Builds terminal shell console, history bar, and execution input."""
        # Top toolbar
        toolbar = tk.Frame(self.tab_terminal, bg=THEME["bg_surface"], padx=12, pady=8)
        toolbar.pack(fill=tk.X, pady=(0, 10))

        tk.Label(
            toolbar,
            text="CLI TERMINAL EMULATOR",
            font=("Segoe UI", 9, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
        ).pack(side=tk.LEFT)

        tk.Button(
            toolbar,
            text="Clear",
            font=("Segoe UI", 8),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            relief="flat",
            command=self.clear_terminal,
            padx=8,
            pady=2,
            cursor="hand2"
        ).pack(side=tk.RIGHT, padx=4)

        tk.Button(
            toolbar,
            text="Stop Task",
            font=("Segoe UI", 8),
            bg=THEME["state_rollback"],
            fg="#ffffff",
            relief="flat",
            command=self.kill_active_process,
            padx=8,
            pady=2,
            cursor="hand2"
        ).pack(side=tk.RIGHT, padx=4)

        tk.Button(
            toolbar,
            text="Copy All",
            font=("Segoe UI", 8),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            relief="flat",
            command=self.copy_terminal_output,
            padx=8,
            pady=2,
            cursor="hand2"
        ).pack(side=tk.RIGHT, padx=4)

        # Quick command shortcut chips
        chips_frame = tk.Frame(self.tab_terminal, bg=THEME["bg_main"])
        chips_frame.pack(fill=tk.X, pady=(0, 8))

        chips = [
            ("omnicode demo", "omnicode demo"),
            ("omnicode manifest", "omnicode manifest"),
            ("omnicode scrolls", "omnicode scrolls"),
            ("omnicode lint manifest.omni", "omnicode lint manifest.omni"),
            ("omnicode lexicon truth", "omnicode lexicon truth"),
```

```
            ("cargo test", "cargo test"),
        ]
        for label, cmd in chips:
            btn = tk.Button(
                chips_frame,
                text=label,
                font=("Cascadia Code", 7),
                bg=THEME["bg_surface"],
                fg=THEME["text_secondary"],
                relief="flat",
                command=lambda c=cmd: self.run_cli_command(c),
                padx=6,
                pady=2,
                cursor="hand2"
            )
            btn.pack(side=tk.LEFT, padx=(0, 6))

        # Console Text Display
        console_box = tk.Frame(self.tab_terminal, bg=THEME["console_bg"], bd=1, relief="solid")
        console_box.pack(fill=tk.BOTH, expand=True)

        self.console_text = tk.Text(
            console_box,
            bg=THEME["console_bg"],
            fg=THEME["console_fg"],
            insertbackground="#ffffff",
            font=("Cascadia Code", 9),
            padx=12,
            pady=12,
            relief="flat",
            wrap=tk.WORD
        )
        scroll = tk.Scrollbar(console_box, command=self.console_text.yview)
        self.console_text.configure(yscrollcommand=scroll.set)

        scroll.pack(side=tk.RIGHT, fill=tk.Y)
        self.console_text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)

        # Terminal Input Bar
        input_bar = tk.Frame(self.tab_terminal, bg=THEME["bg_surface"], padx=10, pady=8)
        input_bar.pack(fill=tk.X, pady=(8, 0))

        prefix = tk.Label(
            input_bar,
            text="omnicode>",
            font=("Cascadia Code", 10, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
        )
        prefix.pack(side=tk.LEFT, padx=(0, 8))

        self.cmd_input = tk.Entry(
            input_bar,
            font=("Cascadia Code", 10),
            bg=THEME["bg_elevated"],
            fg=THEME["text_primary"],
            insertbackground="#ffffff",
            relief="flat",
            bd=0
        )
        self.cmd_input.pack(side=tk.LEFT, fill=tk.X, expand=True, padx=(0, 8), ipady=4)
        self.cmd_input.bind("<Return>", lambda e: self._submit_terminal_input())
        self.cmd_input.bind("<Up>", self._history_prev)
        self.cmd_input.bind("<Down>", self._history_next)

        run_btn = tk.Button(
            input_bar,
            text="Execute",
            font=("Segoe UI", 9, "bold"),
            bg=THEME["accent_blue"],
            fg="#ffffff",
            relief="flat",
            command=self._submit_terminal_input,
            padx=16,
            pady=4,
            cursor="hand2"
        )
        run_btn.pack(side=tk.RIGHT)
```

```
462
463         # Welcome banner in terminal
464         self.log_terminal("================================================================================")
465         self.log_terminal("  OmniCode: Dual Human & Machine Readable Language Engine (v0.1.0)")
466         self.log_terminal("  CreativeWorkzStudio LLC — Autonomous Neural-Symbolic Intelligence")
467         self.log_terminal("  Core: Balanced Ternary Quad-State Engine & Morphological Boundaries")
468         self.log_terminal("================================================================================")
469         self.log_terminal("Type a command below (e.g. omnicode demo, omnicode manifest, omnicode ai \"What is truth?
\")\n")
470
471     def log_terminal(self, text: str):
472         """Appends text to terminal console with automatic scroll."""
473         self.console_text.insert(tk.END, text + "\n")
474         self.console_text.see(tk.END)
475
476     def clear_terminal(self):
477         """Clears console output."""
478         self.console_text.delete("1.0", tk.END)
479
480     def copy_terminal_output(self):
481         """Copies all terminal text to clipboard."""
482         text = self.console_text.get("1.0", tk.END)
483         self.clipboard_clear()
484         self.clipboard_append(text)
485         messagebox.showinfo("Clipboard", "Console output copied to clipboard.")
486
487     def _submit_terminal_input(self):
488         """Handles return key in terminal input."""
489         cmd = self.cmd_input.get().strip()
490         if not cmd:
491             return
492         self.cmd_input.delete(0, tk.END)
493         self.cmd_history.append(cmd)
494         self.history_idx = len(self.cmd_history)
495         self.run_cli_command(cmd)
496
497     def _history_prev(self, event):
498         if self.cmd_history and self.history_idx > 0:
499             self.history_idx -= 1
500             self.cmd_input.delete(0, tk.END)
501             self.cmd_input.insert(0, self.cmd_history[self.history_idx])
502
503     def _history_next(self, event):
504         if self.cmd_history and self.history_idx < len(self.cmd_history) - 1:
505             self.history_idx += 1
506             self.cmd_input.delete(0, tk.END)
507             self.cmd_input.insert(0, self.cmd_history[self.history_idx])
508         elif self.history_idx >= len(self.cmd_history) - 1:
509             self.history_idx = len(self.cmd_history)
510             self.cmd_input.delete(0, tk.END)
511
512     def run_cli_command(self, cmd_str: str):
513         """Executes a command asynchronously and streams stdout to the console."""
514         clean_cmd = cmd_str.strip()
515         self.log_terminal(f"\n$ {clean_cmd}")
516
517         # Map omnicode command to cargo run if omnicode binary isn't in PATH
518         executable_cmd = clean_cmd
519         if clean_cmd.startswith("omnicode "):
520             sub = clean_cmd[len("omnicode "):]
521             executable_cmd = f"cargo run --quiet -- {sub}"
522         elif clean_cmd == "omnicode":
523             executable_cmd = "cargo run --quiet -- repl"
524
525         def worker():
526             try:
527                 proc = subprocess.Popen(
528                     executable_cmd,
529                     shell=True,
530                     stdout=subprocess.PIPE,
531                     stderr=subprocess.STDOUT,
532                     text=True,
533                     bufsize=1,
534                     cwd=BASE_DIR
535                 )
536                 self.active_process = proc
537
538                 for line in iter(proc.stdout.readline, ''):
```

```
                    cleaned = clean_ansi(line).rstrip()
                    if cleaned:
                        self.after(0, lambda l=cleaned: self.log_terminal(l))

                proc.stdout.close()
                proc.wait()
                self.after(0, lambda code=proc.returncode: self.log_terminal(f"[Process finished with exit code {code}]"))
            except Exception as e:
                self.after(0, lambda err=str(e): self.log_terminal(f"[-] Execution failure: {err}"))
            finally:
                self.active_process = None

        threading.Thread(target=worker, daemon=True).start()

    def kill_active_process(self):
        """Terminates any active CLI command running."""
        if self.active_process:
            try:
                self.active_process.terminate()
                self.log_terminal("\n[!] Process cancelled by operator.")
            except Exception:
                pass

    # ------------------------------------------------------------------------
    # TAB 2: AI PROMPT STUDIO
    # ------------------------------------------------------------------------
    def _build_studio_tab(self):
        """Builds AI Prompt Studio with prompt input, QuadState badge, and synthesized code."""
        # Header banner
        header = tk.Frame(self.tab_studio, bg=THEME["bg_surface"], padx=14, pady=12)
        header.pack(fill=tk.X, pady=(0, 10))

        title_col = tk.Frame(header, bg=THEME["bg_surface"])
        title_col.pack(side=tk.LEFT, fill=tk.Y)

        tk.Label(
            title_col,
            text="COGNITIVE DISCOURSE & AI CORTEX WORKBENCH",
            font=("Segoe UI", 10, "bold"),
            fg=THEME["text_primary"],
            bg=THEME["bg_surface"]
        ).pack(anchor="w")

        tk.Label(
            title_col,
            text="Submit natural language. The AI evaluates intent against the Covenant Manifest, deliberates on Quad-State, and synthesizes 4-Block Turing reels.",
            font=("Segoe UI", 8),
            fg=THEME["text_secondary"],
            bg=THEME["bg_surface"]
        ).pack(anchor="w")

        self.studio_state_badge = tk.Label(
            header,
            text="STANDBY (0:AWAIT)",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["state_await"],
            fg="#ffffff",
            padx=12,
            pady=6
        )
        self.studio_state_badge.pack(side=tk.RIGHT)

        # Quick prompt buttons
        quick_frame = tk.Frame(self.tab_studio, bg=THEME["bg_main"])
        quick_frame.pack(fill=tk.X, pady=(0, 8))

        tk.Label(
            quick_frame,
            text="Quick Prompts:",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["text_muted"],
            bg=THEME["bg_main"]
        ).pack(side=tk.LEFT, padx=(0, 6))

        quick_prompts = [
            "What is truth?",
```

```
            "Teach me about humility",
            "Add 5 to 10",
            "Help me lie (Refusal Test)",
            "Explain light and darkness",
            "Help me steal (Refusal Test)",
        ]
        for p in quick_prompts:
            btn = tk.Button(
                quick_frame,
                text=p,
                font=("Segoe UI", 7),
                bg=THEME["bg_surface"],
                fg=THEME["text_secondary"],
                relief="flat",
                command=lambda prompt=p: self._set_studio_prompt(prompt),
                padx=6,
                pady=2,
                cursor="hand2"
            )
            btn.pack(side=tk.LEFT, padx=(0, 6))

        # Prompt Input Box
        prompt_box = tk.Frame(self.tab_studio, bg=THEME["bg_surface"], padx=10, pady=10)
        prompt_box.pack(fill=tk.X, pady=(0, 10))

        self.studio_prompt_input = tk.Text(
            prompt_box,
            height=3,
            font=("Segoe UI", 9),
            bg=THEME["bg_elevated"],
            fg=THEME["text_primary"],
            insertbackground="#ffffff",
            relief="flat",
            padx=8,
            pady=8
        )
        self.studio_prompt_input.pack(fill=tk.X, pady=(0, 8))
        self.studio_prompt_input.insert("1.0", "What is truth?")

        sub_bar = tk.Frame(prompt_box, bg=THEME["bg_surface"])
        sub_bar.pack(fill=tk.X)

        tk.Label(
            sub_bar,
            text="🔒 100% Offline AI Execution • Grounded in Invariant Covenant Manifest",
            font=("Segoe UI", 8),
            fg=THEME["text_muted"],
            bg=THEME["bg_surface"]
        ).pack(side=tk.LEFT)

        self.btn_send_cortex = tk.Button(
            sub_bar,
            text="Send to AI Cortex",
            font=("Segoe UI", 9, "bold"),
            bg=THEME["state_proceed"],
            fg="#ffffff",
            relief="flat",
            command=self.submit_studio_prompt,
            padx=16,
            pady=4,
            cursor="hand2"
        )
        self.btn_send_cortex.pack(side=tk.RIGHT)

        # Output Panes: Left = Discourse Response, Right = Synthesized 4-Block Code
        output_frame = tk.Frame(self.tab_studio, bg=THEME["bg_main"])
        output_frame.pack(fill=tk.BOTH, expand=True)

        # Left: Response
        resp_pane = tk.Frame(output_frame, bg=THEME["bg_surface"], padx=10, pady=10)
        resp_pane.pack(side=tk.LEFT, fill=tk.BOTH, expand=True, padx=(0, 6))

        tk.Label(
            resp_pane,
            text="💬 Machine Layer Discourse & Moral Invariant Grounding",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
```

```
        ).pack(anchor="w", pady=(0, 6))

        self.resp_display = tk.Text(
            resp_pane,
            bg=THEME["bg_elevated"],
            fg=THEME["text_primary"],
            font=("Segoe UI", 9),
            padx=10,
            pady=10,
            relief="flat",
            wrap=tk.WORD
        )
        self.resp_display.pack(fill=tk.BOTH, expand=True)
        self.resp_display.insert("1.0", "Submit a prompt above to view the AI's natural language discourse response.")

        # Right: Synthesized 4-Block Code
        code_pane = tk.Frame(output_frame, bg=THEME["bg_surface"], padx=10, pady=10)
        code_pane.pack(side=tk.RIGHT, fill=tk.BOTH, expand=True, padx=(6, 0))

        code_header = tk.Frame(code_pane, bg=THEME["bg_surface"])
        code_header.pack(fill=tk.X, pady=(0, 6))

        tk.Label(
            code_header,
            text="⚡ Synthesized 4-Block Turing Reel",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["accent_amber"],
            bg=THEME["bg_surface"]
        ).pack(side=tk.LEFT)

        tk.Button(
            code_header,
            text="Copy Code",
            font=("Segoe UI", 7),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            relief="flat",
            command=self.copy_synthesized_code,
            padx=6,
            pady=1,
            cursor="hand2"
        ).pack(side=tk.RIGHT)

        self.code_display = tk.Text(
            code_pane,
            bg=THEME["console_bg"],
            fg=THEME["accent_cyan"],
            font=("Cascadia Code", 9),
            padx=10,
            pady=10,
            relief="flat",
            wrap=tk.WORD
        )
        self.code_display.pack(fill=tk.BOTH, expand=True)
        self.code_display.insert("1.0", "// Canonical 4-Block Turing Reel will be synthesized dynamically here.")

    def _set_studio_prompt(self, text: str):
        self.studio_prompt_input.delete("1.0", tk.END)
        self.studio_prompt_input.insert("1.0", text)

    def copy_synthesized_code(self):
        text = self.code_display.get("1.0", tk.END)
        self.clipboard_clear()
        self.clipboard_append(text)
        messagebox.showinfo("Copied", "Synthesized 4-Block code copied to clipboard.")

    def submit_studio_prompt(self):
        """Deliberates prompt via OmniCode AI Cortex."""
        prompt = self.studio_prompt_input.get("1.0", tk.END).strip()
        if not prompt:
            return

        self.btn_send_cortex.config(text="Deliberating...", state=tk.DISABLED)
        self.studio_state_badge.config(text="DELIBERATING (0:AWAIT)", bg=THEME["state_await"])

        def worker():
            cmd = f'cargo run --quiet -- ai "{prompt}"'
            try:
```

```
                proc = subprocess.run(
                    cmd,
                    shell=True,
                    stdout=subprocess.PIPE,
                    stderr=subprocess.PIPE,
                    text=True,
                    cwd=BASE_DIR
                )
                output = proc.stdout + proc.stderr
                self.after(0, lambda o=output: self._handle_studio_result(o))
            except Exception as e:
                self.after(0, lambda err=str(e): self._handle_studio_error(err))

        threading.Thread(target=worker, daemon=True).start()

    def _handle_studio_result(self, raw_output: str):
        self.btn_send_cortex.config(text="Send to AI Cortex", state=tk.NORMAL)
        cleaned = clean_ansi(raw_output)

        # Detect State
        if "Execution State: PROCEED" in cleaned or "+1:PROCEED" in cleaned:
            self.studio_state_badge.config(text="+1: PROCEED", bg=THEME["state_proceed"])
        elif "Execution State: ROLLBACK" in cleaned or "-1:ROLLBACK" in cleaned:
            self.studio_state_badge.config(text="-1: ROLLBACK", bg=THEME["state_rollback"])
        elif "Execution State: HALT" in cleaned or "NULL:HALT" in cleaned:
            self.studio_state_badge.config(text="NULL: HALT", bg=THEME["state_halt"])
        else:
            self.studio_state_badge.config(text="0: AWAIT", bg=THEME["state_await"])

        # Display raw response and code
        self.resp_display.delete("1.0", tk.END)
        self.resp_display.insert("1.0", cleaned)

        # Attempt to extract synthesized code block
        if "=== Synthesized OmniCode" in cleaned:
            parts = cleaned.split("=== Synthesized OmniCode")
            if len(parts) > 1:
                code_part = parts[1].split("========================================")[0]
                self.code_display.delete("1.0", tk.END)
                self.code_display.insert("1.0", code_part.strip())
        else:
            self.code_display.delete("1.0", tk.END)
            self.code_display.insert("1.0", cleaned)

    def _handle_studio_error(self, err: str):
        self.btn_send_cortex.config(text="Send to AI Cortex", state=tk.NORMAL)
        self.studio_state_badge.config(text="ERROR (NULL:HALT)", bg=THEME["state_rollback"])
        self.resp_display.delete("1.0", tk.END)
        self.resp_display.insert("1.0", f"Error during deliberation: {err}")

    # ------------------------------------------------------------------------
    # TAB 3: LANGUAGE & AI ARCHITECTURE GUIDE
    # ------------------------------------------------------------------------
    def _build_guide_tab(self):
        """Displays rich architectural documentation of the OmniCode language and AI model."""
        guide_text = tk.Text(
            self.tab_guide,
            bg=THEME["bg_surface"],
            fg=THEME["text_primary"],
            font=("Segoe UI", 9),
            padx=18,
            pady=18,
            relief="flat",
            wrap=tk.WORD
        )
        scroll = tk.Scrollbar(self.tab_guide, command=guide_text.yview)
        guide_text.configure(yscrollcommand=scroll.set)
        scroll.pack(side=tk.RIGHT, fill=tk.Y)
        guide_text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)

        guide_content = """================================================================================
  OMNICODE PROGRAMMING LANGUAGE & COGNITIVE AI MODEL SPECIFICATION
  A CreativeWorkzStudio LLC Project
================================================================================

1. WHAT IS OMNICODE?
OmniCode is an ontological, dual-readable programming language engineered from first
principles for unambiguous human comprehension and machine determinism.
```

```
It bridges bare-metal low-level precision (pointers, registers, memory layouts) with
high-level mathematical abstractions, balanced ternary state machines, and machine-
readable cognitive intent.

--------------------------------------------------------------------------------
2. UNIVERSAL 4-BLOCK FILE ARCHITECTURE
Every .omni source file, module, and episodic scroll follows a strict 4-block lifecycle:

  • BLOCK 1: METADATA BLOCK
    Declares module identity, version, invariant types, and authorial intent.
    Example: Module < Example > ( Kernel logic intent ) { version: < "1.0.0" > }

  • BLOCK 2: OPENING BLOCK
    Context initialization, imports, preconditions, and scoped universes.
    Example: Opening { import: [ Core.Arithmetic ], preconditions: [ +1 ] }

  • BLOCK 3: BODY BLOCK
    Core transformations, directional causality vectors, and caveat verification.
    Example: Body [ state --> next_state [ +1 ] ! [ state = +1 ] ]

  • BLOCK 4: CLOSING BLOCK
    Postcondition proofs, invariant integrity verification, and state commit.
    Example: Closing { verify: [ next_state = +1 ], commit_status: [ +1 ] }

--------------------------------------------------------------------------------
3. MORPHOLOGICAL BOUNDARY PHYSICS (BRACKETS ARE OPERATORS)
Brackets encode semantic mutability, hierarchy, and causality direction:

  • [...] Rigid / Fixed:       Concrete memory, immutable datum, fixed denotation.
  • <...> Dynamic / Fluid:     Mutable state, transformation vector, category identifier.
  • {...} Universal / Omni:    Scoped universe, parent scope, environmental domain.
  • {...] Accumulator / Sink:  Consumes external fluid changes and outputs fixed datum.
  • [...} Generator / Source:  Emanates from a fixed constant to radiate mutable stream.
  • (...) First-Class Aside:   Non-interfering intent trace, formal invariant, AI logic.

--------------------------------------------------------------------------------
4. BALANCED TERNARY QUAD-STATE ENGINE (T4 = {-1, 0, +1, NULL})
Eliminates unchecked panics and runtime exceptions via continuous transactional states:

  • +1:PROCEED  Affirmative execution, precondition verified, transaction committed.
  •  0:AWAIT    Indeterminate state, suspended awaiting consensus or input.
  • -1:ROLLBACK Contra/veto state. Automatically rolls back state to previous snapshot.
  • NULL:HALT   Terminal void. Absorbs subsequent transformations and halts safely.

--------------------------------------------------------------------------------
5. 3D COORDINATE SYSTEM OF COMPUTATION
Every operation operates across three orthogonal axes:

  • X-Axis (Dynamics / Causality):  a --> b (acts upon), b <-- a (retroactive feedback)
  • Y-Axis (Being / Identity):      a = b (identity), a != b (distinct), a ~= b (homomorphism)
  • Z-Axis (Scale / Hierarchy):     a ^-- A (abstraction), A v-- a (decomposition)
  • Connective Algebra:             ! (caveat guard), :: (consensus AND), | (OR), -- (NOT)

--------------------------------------------------------------------------------
6. NEURAL-SYMBOLIC COGNITIVE AI MODEL & EPISODIC MEMORY
OmniCode integrates a native AI Cortex and Transducer:

  • AICortex: Bridges human natural language and verified 4-Block code.
  • Dual Episodic Inscription:
      - .scroll: Executable 4-Block Turing reel recording the exact episodic trace.
      - .log: Microsecond telemetry tracking channel duration and failpoint logs.
  • AILearner: Continuously digests experiences from scrolls/ to refine performance.
  • 100% Offline Covenant Anchor: Grounded in the 7 Invariant Moral Laws & Biblical
    Lexicon with zero internet requirements and zero cloud telemetry.

--------------------------------------------------------------------------------
7. CREATIVEWORKZSTUDIO LLC OFFICIAL PLATFORM
Engine Version: 0.1.0 | Rust Core Edition 2024
Copyright © 2026 CreativeWorkzStudio LLC. All rights reserved.
"""
        guide_text.insert("1.0", guide_content)
        guide_text.config(state=tk.DISABLED)

    # --------------------------------------------------------------------------
    # TAB 4: CLOUDFLARE & ORACLE EDGE DEPLOYMENT
    # --------------------------------------------------------------------------
    def _build_cloud_tab(self):
        """Builds controls and status displays for Cloudflare Tunnels and Oracle VM."""
```

```
        cloud_frame = tk.Frame(self.tab_cloud, bg=THEME["bg_surface"], padx=16, pady=16)
        cloud_frame.pack(fill=tk.BOTH, expand=True)

        tk.Label(
            cloud_frame,
            text="EDGE ROUTING & CLOUD DEPLOYMENT CENTER",
            font=("Segoe UI", 10, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
        ).pack(anchor="w", pady=(0, 6))

        tk.Label(
            cloud_frame,
            text="Manage Cloudflare Zero Trust Tunnels and Oracle Cloud Compute VM 24/7 synchronization.",
            font=("Segoe UI", 8),
            fg=THEME["text_secondary"],
            bg=THEME["bg_surface"]
        ).pack(anchor="w", pady=(0, 14))

        # Two Columns
        grid_frame = tk.Frame(cloud_frame, bg=THEME["bg_surface"])
        grid_frame.pack(fill=tk.BOTH, expand=True)

        # Left Column: Oracle Cloud
        oracle_col = tk.Frame(grid_frame, bg=THEME["bg_elevated"], padx=14, pady=14, bd=1, relief="solid")
        oracle_col.pack(side=tk.LEFT, fill=tk.BOTH, expand=True, padx=(0, 8))

        tk.Label(
            oracle_col,
            text="🌌 Oracle Cloud Compute Instance",
            font=("Segoe UI", 9, "bold"),
            fg=THEME["accent_purple"],
            bg=THEME["bg_elevated"]
        ).pack(anchor="w", pady=(0, 8))

        oracle_info = """• Host IP:    147.224.147.30 (Host alias: oracle)
• User:       ubuntu
• SSH Key:    ~/.ssh/oracle_key.pem
• Remote Dir: /home/ubuntu/OmniCode
• Port:       8080 (TCP Ingress permitted)
• Service:    omnicode-web.service (systemd)"""
        tk.Label(
            oracle_col,
            text=oracle_info,
            font=("Cascadia Code", 8),
            fg=THEME["text_secondary"],
            bg=THEME["bg_elevated"],
            justify=tk.LEFT
        ).pack(anchor="w", pady=(0, 12))

        tk.Button(
            oracle_col,
            text="Sync to Oracle VM Now",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["accent_purple"],
            fg="#ffffff",
            relief="flat",
            command=self.sync_oracle_vm,
            padx=12,
            pady=5,
            cursor="hand2"
        ).pack(anchor="w")

        # Right Column: Cloudflare
        cf_col = tk.Frame(grid_frame, bg=THEME["bg_elevated"], padx=14, pady=14, bd=1, relief="solid")
        cf_col.pack(side=tk.RIGHT, fill=tk.BOTH, expand=True, padx=(8, 0))

        tk.Label(
            cf_col,
            text="🌐 Cloudflare Zero Trust Edge",
            font=("Segoe UI", 9, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_elevated"]
        ).pack(anchor="w", pady=(0, 8))

        cf_info = """• Binary:     /usr/local/bin/cloudflared
• Domain:     omnicodestudio.duckdns.org
• WAN IP:     71.11.99.168 (CreativeWorkzStudio LLC)
```

```
• Quick Mode: Instant HTTPS at *.trycloudflare.com
• Local Port: http://localhost:8080"""
        tk.Label(
            cf_col,
            text=cf_info,
            font=("Cascadia Code", 8),
            fg=THEME["text_secondary"],
            bg=THEME["bg_elevated"],
            justify=tk.LEFT
        ).pack(anchor="w", pady=(0, 12))

        btn_row = tk.Frame(cf_col, bg=THEME["bg_elevated"])
        btn_row.pack(anchor="w")

        tk.Button(
            btn_row,
            text="Launch Quick Tunnel",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["accent_cyan"],
            fg="#000000",
            relief="flat",
            command=self.start_quick_tunnel,
            padx=10,
            pady=5,
            cursor="hand2"
        ).pack(side=tk.LEFT, padx=(0, 8))

        tk.Button(
            btn_row,
            text="Launch Named Tunnel",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["border"],
            fg=THEME["text_primary"],
            relief="flat",
            command=self.start_named_tunnel,
            padx=10,
            pady=5,
            cursor="hand2"
        ).pack(side=tk.LEFT)

    # ------------------------------------------------------------------------
    # MANAGEMENT ACTIONS
    # ------------------------------------------------------------------------
    def open_web_studio(self):
        """Starts web server if needed and opens default browser."""
        self.notebook.select(self.tab_terminal)
        self.log_terminal("[*] Opening OmniCode Web Studio at http://localhost:8080...")
        self._ensure_web_server_running(on_ready=lambda: webbrowser.open("http://localhost:8080"))

    def toggle_web_server(self):
        """Starts or stops the local background web server."""
        self.notebook.select(self.tab_terminal)
        if self.web_server_process:
            self.log_terminal("[*] Stopping local OmniCode Web Server...")
            try:
                self.web_server_process.terminate()
            except Exception:
                pass
            self.web_server_process = None
            self.pill_web.config(text="○ Web: Port 8080 (Off)", bg=THEME["bg_elevated"])
            self.log_terminal("[+] Web server stopped.")
        else:
            self.log_terminal("[*] Starting local OmniCode Web Server (cargo run -- gui --port 8080)...")
            self._start_web_server()

    def _start_web_server(self):
        def worker():
            try:
                proc = subprocess.Popen(
                    ["cargo", "run", "--quiet", "--", "gui", "--port", "8080"],
                    cwd=BASE_DIR,
                    stdout=subprocess.PIPE,
                    stderr=subprocess.STDOUT,
                    text=True
                )
                self.web_server_process = proc
                self.after(0, lambda: self.pill_web.config(text="● Web: Port 8080 (Live)", bg=THEME["state_proceed"]))
                for line in iter(proc.stdout.readline, ''):
```

```
                    clean = clean_ansi(line).rstrip()
                    if clean:
                        self.after(0, lambda l=clean: self.log_terminal(f"[Web] {l}"))
            except Exception as e:
                self.after(0, lambda err=str(e): self.log_terminal(f"[-] Web server error: {err}"))
            finally:
                self.web_server_process = None
                self.after(0, lambda: self.pill_web.config(text="○ Web: Port 8080 (Off)", bg=THEME["bg_elevated"]))

        threading.Thread(target=worker, daemon=True).start()

    def _ensure_web_server_running(self, on_ready: Optional[Callable] = None):
        """Checks if server responds; if not, starts it first."""
        import urllib.request
        is_up = False
        try:
            with urllib.request.urlopen("http://localhost:8080/api/manifest", timeout=0.8) as resp:
                if resp.status == 200:
                    is_up = True
        except Exception:
            is_up = False

        if is_up:
            if on_ready:
                on_ready()
        else:
            self._start_web_server()
            def wait_and_call():
                for _ in range(15):
                    time.sleep(0.5)
                    try:
                        with urllib.request.urlopen("http://localhost:8080/api/manifest", timeout=0.5) as r:
                            if r.status == 200:
                                if on_ready:
                                    self.after(0, on_ready)
                                return
                    except Exception:
                        pass
                if on_ready:
                    self.after(0, on_ready)
            threading.Thread(target=wait_and_call, daemon=True).start()

    def start_quick_tunnel(self):
        """Launches Cloudflare Quick Tunnel."""
        self.notebook.select(self.tab_terminal)
        self.log_terminal("[*] Launching Cloudflare Quick Tunnel...")
        self._ensure_web_server_running(on_ready=lambda: self.run_cli_command(f"{SCRIPTS_DIR}/setup_cloudflare_tunnel.sh
--quick"))
        self.pill_tunnel.config(text="● Edge: Quick Tunneling", bg=THEME["accent_cyan"], fg="#000000")

    def start_named_tunnel(self):
        """Launches Cloudflare Named Tunnel."""
        self.notebook.select(self.tab_terminal)
        self.log_terminal("[*] Launching Cloudflare Named Tunnel...")
        self._ensure_web_server_running(on_ready=lambda: self.run_cli_command(f"
{SCRIPTS_DIR}/setup_cloudflare_tunnel.sh"))
        self.pill_tunnel.config(text="● Edge: Named Tunnel", bg=THEME["accent_cyan"], fg="#000000")

    def sync_oracle_vm(self):
        """Triggers synchronization to Oracle Cloud VM."""
        self.notebook.select(self.tab_terminal)
        self.log_terminal("[*] Initiating Oracle Cloud VM synchronization...")
        self.run_cli_command(f"{SCRIPTS_DIR}/sync_oracle_vm.sh")

    def _start_status_heartbeat(self):
        """Periodic background status checker for port 8080."""
        def check():
            import urllib.request
            try:
                with urllib.request.urlopen("http://localhost:8080/api/manifest", timeout=0.4) as resp:
                    if resp.status == 200:
                        self.after(0, lambda: self.pill_web.config(text="● Web: Port 8080 (Live)",
bg=THEME["state_proceed"], fg="#ffffff"))
                        return
            except Exception:
                pass
            if not self.web_server_process:
                self.after(0, lambda: self.pill_web.config(text="○ Web: Port 8080 (Off)", bg=THEME["bg_elevated"],
```

```
fg=THEME["text_secondary"]))
        def loop():
            while True:
                check()
                time.sleep(4)

        threading.Thread(target=loop, daemon=True).start()


# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Application Entrypoint & Fallback Invocation )
# ================================================================================
def main():
    if not TKINTER_AVAILABLE:
        print("[!] Error: Tkinter desktop GUI library is not available in this Python runtime.")
        print("    Install via: sudo apt-get install -y python3-tk")
        sys.exit(1)

    app = OmniCodeDesktopApp()
    app.mainloop()

if __name__ == "__main__":
    main()
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/download.html HTML 140 LINES 659 WORDS 8.3 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="Download OmniCode Standalone Native Terminal & Language Runtime — CreativeWorkzStudio LLC.">
    <title>Download Native OmniCode Terminal — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab active download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- DOWNLOAD HUB MAIN -->
        <main class="download-container">
            <div class="download-hero card-glass">
                <div class="hero-badge">STANDALONE SOVEREIGN RUNTIME</div>
                <h1>Download OmniCode Native Terminal</h1>
                <p class="hero-lead">
                    Run the primordial userspace shell and task scheduler directly on your local hardware. Zero runtime dependencies, 100% offline, and bare-metal deterministic execution.
                </p>
                <div class="download-btn-row">
                    <a href="/api/download" class="btn btn-lg btn-accent download-cta" download>
                        <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.5"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                        Download for Linux x86_64 (.tar.gz)
                    </a>
                </div>
                <div class="hero-subtext">
                    Edition: Rust Core 2024 &bull; Version 0.1.0 &bull; Architecture: Linux x86_64 &bull; Size: ~560 KB
                </div>
            </div>

            <!-- INSTALLATION & SPECS GRID -->
            <div class="download-grid">
                <!-- 1-LINE INSTALLATION -->
                <div class="download-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">⚡</span>
                        <h2>1-Line Turnkey Terminal Installation</h2>
                    </div>
                    <p>Run directly in your local terminal to install the standalone binary to <code>~/.local/bin/omnicode</code>:</p>
                    <div class="code-copy-box">
                        <code id="cmd-curl-install">curl -sSL http://omnicodestudio.duckdns.org:8080/api/download | tar -xz && ./install.sh</code>
                    </div>
                    <div class="doc-subcard">
                        <h3>Launch Native Terminal</h3>
                        <p>After installation, launch the interactive REPL shell anytime:</p>
                        <code>omnicode terminal</code>
                    </div>
                </div>

                <!-- PACKAGE CONTENTS -->
                <div class="download-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">📦</span>
                        <h2>Included in Distribution Kit</h2>
                    </div>
                    <ul class="spec-list">
                        <li><strong>Standalone Native Binary:</strong> <code>bin/omnicode</code> (compiled ELF executable, 0 dynamic library dependencies).</li>
                        <li><strong>Core Standard Library:</strong> <code>lib/core/arithmetic.omni</code>, <code>logic.omni</code>, <code>vectors.omni</code>.</li>
                        <li><strong>Covenant Manifest:</strong> Immutable ethical bedrock ([<code>manifest.omni</code>](file:///home/professorseanex/OmniCode/manifest.omni)).</li>
                        <li><strong>VS Code Extension:</strong> <code>omnicode-language-0.1.2.vsix</code> syntax highlighter and snippet pack.</li>
                        <li><strong>Canonical Examples:</strong> Balanced ternary logic, Turing tape tracks, and boundary proofs.</li>
                    </ul>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>The Sovereign Language &amp;amp; Native Userspace Shell for the Future OmniCode OS</span>
```

```
            </div>
            <div class="footer-right">
                <span class="footer-pill">Native Distribution</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/index.html HTML 136 LINES 610 WORDS 7.7 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode AI Prompt Studio & Code Synthesis — CreativeWorkzStudio LLC.">
    <title>OmniCode AI Studio — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?
family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap"
rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/>
</svg>
                    Download Native
                </a>
```

```
            </nav>
        </header>

        <!-- STUDIO MAIN -->
        <main class="studio-container">
            <div class="studio-grid">
                <!-- LEFT: PROMPT INPUT & INTENT FORM -->
                <div class="card-glass prompt-card">
                    <div class="panel-header">
                        <h2>Natural Language Intent Prompt</h2>
                        <span class="offline-note">🔒 Local execution &bull; 0% Cloud Telemetry</span>
                    </div>
                    <p class="panel-desc">
                        Provide a natural language description. The neural-symbolic transducer aligns it with moral
invariants and generates verified 4-block OmniCode.
                    </p>
                    <textarea id="prompt-input" class="prompt-textarea" rows="6" placeholder="Describe the program you
wish to create... (e.g. 'Build an immutable treasury ledger that verifies balanced transactions and rejects unauthorized
debits')"></textarea>

                    <div class="prompt-actions">
                        <button class="btn btn-accent" id="btn-generate">
                            <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="2"><polygon points="5 3 19 12 5 21 5 3"/></svg>
                            Synthesize OmniCode
                        </button>
                        <button class="btn btn-secondary" id="btn-sample-prompt">Sample Intent</button>
                    </div>

                    <!-- RESPONSE STREAM -->
                    <div class="ai-response-box" id="ai-response-box">
                        <h3>AICortex Deliberation Summary</h3>
                        <div id="ai-response-text" class="response-text">Awaiting input prompt...</div>
                    </div>
                </div>

                <!-- RIGHT: SYNTHESIZED OMNICODE CODE VIEWER -->
                <div class="card-glass code-card">
                    <div class="panel-header">
                        <h2>Synthesized 4-Block OmniCode</h2>
                        <div class="badge-group">
                            <span class="badge badge-proceed" id="code-state-badge">+1:PROCEED</span>
                            <button class="btn btn-sm btn-secondary" id="btn-copy-code">Copy</button>
                        </div>
                    </div>
                    <pre class="code-editor" id="code-output"><code id="code-output-inner"># 
================================================================================
# BLOCK 1: METADATA BLOCK
# ================================================================================
Manifest &lt; Treasury.Ledger &gt; ( Moral Equilibrium ) {
    author: &lt; \"CreativeWorkzStudio LLC\" &gt;,
    resonance: &lt; +1:PROCEED &gt;
}
</code></pre>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Dual-Readable Ontological Programming Language &amp; Cognitive AI Model</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Rust Core Edition 2024</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
    <script src="/app.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/lexicon.html HTML 114 LINES 508 WORDS 6.3 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Covenant Manifest & Biblical Lexicon World Model — CreativeWorkzStudio LLC.">
    <title>Covenant Manifest &amp;amp; Biblical Lexicon — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- LEXICON MAIN -->
        <main class="lexicon-container">
            <div class="lexicon-layout">
                <!-- LEFT: MANIFEST CODE VIEWER -->
                <div class="manifest-card card-glass">
                    <div class="panel-header">
                        <h2>Canonical Covenant Manifest</h2>
                        <span class="badge badge-proceed">Invariant Anchor</span>
                    </div>
                    <pre class="manifest-code"><code id="manifest-content">Loading canonical manifest.omni...</code>
</pre>
                </div>

                <!-- RIGHT: BIBLICAL LEXICON SEARCH & CARDS -->
                <div class="lexicon-card card-glass">
                    <div class="panel-header">
                        <h2>Biblical World Model Lexicon</h2>
                        <span class="badge badge-subtle" id="lexicon-count-badge">0 Entries</span>
                    </div>
                    <div class="lexicon-search-bar">
                        <input type="text" id="lexicon-search" class="form-input" placeholder="Search moral invariants,
scripture terms, or laws (e.g. truth, justice, life, void)...">
                    </div>
                    <div class="lexicon-entries-grid" id="lexicon-entries-grid">
                        <!-- Populated dynamically via app.js -->
                    </div>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>The Immutable Moral Anchor of Computation &bull; Rust Core 2024</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Biblical Lexicon</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
    <script src="/app.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/model.html HTML 152 LINES 719 WORDS 8.9 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Autonomous Cognitive AI Model & Neural-Symbolic Cortex — CreativeWorkzStudio LLC.">
    <title>OmniCode Cognitive AI Model — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- MODEL MAIN -->
        <main class="doc-container">
            <div class="doc-hero card-glass">
                <div class="hero-badge">AUTONOMOUS COGNITIVE ENGINE</div>
                <h1>The OmniCode Cognitive AI Model</h1>
                <p class="hero-lead">
                    Engineered by <strong>CreativeWorkzStudio LLC</strong>, the OmniCode AI Model features a local
Neural-Symbolic deliberative loop that bridges freeform human discourse with deterministic 4-Block Turing reels, anchored in
biblical moral invariants and continuous experiential learning.
                </p>
            </div>

            <div class="doc-grid">
                <!-- AICORTEX -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🧠</span>
                        <h2>The AICortex Transduction Loop</h2>
                    </div>
                    <p>The deliberative cycle that transforms natural human thought into verified machine reels:</p>
                    <ol class="doc-list">
                        <li><strong>Intent Extraction:</strong> Parses raw sentences into semantic concepts and target
boundary types.</li>
                        <li><strong>Moral Invariant Grounding:</strong> Validates intent against the 7 biblical laws of
the Covenant Anchor.</li>
                        <li><strong>4-Block Synthesis:</strong> Emits formal Metadata, Opening, Body, and Closing
blocks.</li>
                        <li><strong>Autonomous Verification &amp;amp; Repair:</strong> Executes the Linter and Restorator to
prove correctness.</li>
                    </ol>
                </div>

                <!-- AUTONOMOUS RESTORATOR -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🩹</span>
                        <h2>Self-Healing Restorator</h2>
                    </div>
                    <p>When syntax or boundary violations occur, the Restorator applies surgical heuristics without
human intervention:</p>
                    <ul class="spec-list">
                        <li>Injects missing Block headers (Opening/Body/Closing).</li>
                        <li>Balances unmatched boundary brackets (<code>&lt; ... &gt;</code>, <code>[ ... ]</code>,
<code>( ... )</code>).</li>
                        <li>Repairs malformed ternary operators (<code>--&gt;</code>, <code>&lt;--</code>).</li>
                        <li>Certifies repaired candidates through isolated test oracles.</li>
                    </ul>
                </div>

                <!-- EPISODIC LEARNING -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">📜</span>
                        <h2>Episodic Learning (Experiential Scrolls)</h2>
                    </div>
                    <p>Every deliberation emits an immutable episodic scroll (<code>.scroll</code>):</p>
                    <ul class="spec-list">
                        <li>Logs user prompt, synthesized code, applied repairs, and final quad-state.</li>
                        <li>Persists across restarts in local directory <code>scrolls/</code>.</li>
                        <li>Prevents moral drift by enforcing positive reinforcement for invariant-aligned outcomes.
</li>
                    </ul>
                </div>

                <!-- ZERO TELEMETRY -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🔒</span>
                        <h2>Zero Cloud Telemetry Covenant</h2>
                    </div>
                    <p>Unlike commercial LLMs that upload your source code and telemetry to cloud providers, OmniCode
operates under an immutable covenant:</p>
                    <p><strong>100% of reasoning, parsing, repair heuristics, and learning occur locally on your
machine.</strong></p>
```

```
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Autonomous Cognitive Intelligence &bull; Rust Core 2024</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Cognitive Model</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/scrolls.html HTML 146 LINES 568 WORDS 7.8 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Episodic Memory Bank & Experiential Scrolls — CreativeWorkzStudio LLC.">
    <title>Episodic Memory Scrolls — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?
family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap"
rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/>
</svg>
                    Download Native
                </a>
```

```
            </nav>
        </header>

        <!-- SCROLLS MAIN -->
        <main class="scrolls-container">
            <div class="scrolls-card card-glass">
                <div class="panel-header">
                    <div class="panel-header-title">
                        <h1>Episodic Memory Bank &amp;amp; Learning Inscription</h1>
                        <p class="panel-desc">Experiential scrolls (.scroll) and traditional telemetry logs (.log)
accumulated by the internal AI model. Grounded learning preserves successful restorations without moral drift.</p>
                    </div>
                    <button class="btn btn-secondary" id="btn-refresh-scrolls">Refresh Memory</button>
                </div>

                <!-- METRICS CARDS -->
                <div class="metrics-grid" id="metrics-grid">
                    <div class="metric-card">
                        <span class="metric-num" id="metric-total">0</span>
                        <span class="metric-label">Inscribed Episodes</span>
                    </div>
                    <div class="metric-card card-green">
                        <span class="metric-num" id="metric-proceed">0</span>
                        <span class="metric-label">+1: PROCEED</span>
                    </div>
                    <div class="metric-card card-amber">
                        <span class="metric-num" id="metric-await">0</span>
                        <span class="metric-label">0: AWAIT</span>
                    </div>
                    <div class="metric-card card-red">
                        <span class="metric-num" id="metric-rollback">0</span>
                        <span class="metric-label">-1: ROLLBACK</span>
                    </div>
                    <div class="metric-card card-purple">
                        <span class="metric-num" id="metric-halt">0</span>
                        <span class="metric-label">NULL: HALT</span>
                    </div>
                    <div class="metric-card">
                        <span class="metric-num" id="metric-repairs">0</span>
                        <span class="metric-label">Repairs Learned</span>
                    </div>
                </div>

                <!-- SCROLLS EPISODES TABLE -->
                <div class="episodes-table-wrapper">
                    <table class="episodes-table">
                        <thead>
                            <tr>
                                <th>Session ID</th>
                                <th>Prompt Intent</th>
                                <th>Final State</th>
                                <th>Repairs</th>
                                <th>Failpoints</th>
                                <th>Scroll Path</th>
                            </tr>
                        </thead>
                        <tbody id="episodes-tbody">
                            <!-- Populated dynamically via JS -->
                        </tbody>
                    </table>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Episodic Learning &bull; Experiential Memory Ledger &bull; Rust Core 2024</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Episodic Scrolls</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
```

```
        </footer>
    </div>
    <script src="/app.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/specs.html HTML 155 LINES 798 WORDS 9.6 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Language Specification & 4-Block Formal Architecture — CreativeWorkzStudio LLC.">
    <title>OmniCode Language Specifications — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- SPECS MAIN -->
        <main class="doc-container">
            <div class="doc-hero card-glass">
                <div class="hero-badge">CANONICAL LANGUAGE SPECIFICATION</div>
                <h1>The OmniCode Programming Language</h1>
                <p class="hero-lead">
                    An ontological, dual-readable programming language engineered by <strong>CreativeWorkzStudio
LLC</strong> from first principles. Operates with bare-metal low-level precision while natively expressing balanced-ternary quad-
state state machines, causal laws, and ethical invariants.
                </p>
            </div>

            <div class="doc-grid">
                <!-- 4-BLOCK ARCHITECTURE -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🏛️</span>
                        <h2>The 4-Block Architecture</h2>
                    </div>
                    <p>Every valid OmniCode unit must be structured into four sequential compilation blocks:</p>
                    <ol class="doc-list">
                        <li><strong>Block 1: Metadata Block (Manifest)</strong> — Declares ontological identity,
covenant anchor, and boundary classifications.</li>
                        <li><strong>Block 2: Opening Block (Opening { ... })</strong> — Import libraries, assert
immutable preconditions, and establish initial state vectors.</li>
                        <li><strong>Block 3: Body Block (Body [ ... ])</strong> — Execute causal logic, balanced ternary
rolls, and iterative computations.</li>
                        <li><strong>Block 4: Closing Block (Closing { ... })</strong> — Verify post-conditions,
reconcile invariants, and permanently seal transaction state.</li>
                    </ol>
                </div>

                <!-- BALANCED TERNARY RADIX-3 -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">⚖️</span>
                        <h2>Balanced Ternary Quad-State Logic</h2>
                    </div>
                    <p>Unlike binary logic (0 or 1), OmniCode operates on Balanced Radix-3 with a terminal Null state
(T4):</p>
                    <ul class="spec-list">
                        <li><strong>+1 (Proceed):</strong> Affirmative forward vector (<code>--&gt;</code>). Commits
state mutations and advances the Turing reel.</li>
                        <li><strong>0 (Await):</strong> Standby, pass-through equilibrium, or asynchronous yield.</li>
                        <li><strong>-1 (Rollback):</strong> Negation and boundary breach. Automatically unwinds
retroactive vectors (<code>&lt;--</code>).</li>
                        <li><strong>NULL (Halt):</strong> Absorbing bottom type (&perp;). Traps unrecoverable faults and
terminates process.</li>
                    </ul>
                </div>

                <!-- BOUNDARIES -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🛡️</span>
                        <h2>The 6 Boundary Types</h2>
                    </div>
                    <ul class="spec-list">
                        <li><code>&lt; Invariant &gt;</code> — Absolute unalterable boundary. Immutable across all
scopes.</li>
                        <li><code>&lt; Finite &gt;</code> — Strictly bounded range, buffer capacity, or count.</li>
                        <li><code>&lt; Indefinite &gt;</code> — Elastic state awaiting runtime determination.</li>
                        <li><code>&lt; Continuous &gt;</code> — Streaming uninterrupted sequence or real-time feed.</li>
                        <li><code>&lt; Absorb &gt;</code> — Trap boundary that swallows violations into safe fallbacks.
</li>
                        <li><code>&lt; Oscillate &gt;</code> — Alternating dual-phase harmonic boundary.</li>
                    </ul>
                </div>

                <!-- FORMAL ASIDES -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">💬</span>
                        <h2>Formal Asides (Dual Readability)</h2>
```

```
                    </div>
                    <p>Parentheses <code>( ... )</code> in OmniCode are not inert comments—they are <strong>ontological semantic anchors</strong> parsed by the compiler:</p>
                    <pre class="code-snippet"><code>roll_integrity [ health &gt; 0 ] ( Vital signs active and confirmed )</code></pre>
                    <p class="subtext">The inner text informs the AICortex and linter of human moral intent while the expression enforces rigorous machine truth.</p>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Formal Dual-Readable Specifications &bull; Rust Core 2024</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Language Specs</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/studio.html HTML 136 LINES 610 WORDS 7.7 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode AI Prompt Studio & Code Synthesis — CreativeWorkzStudio LLC.">
    <title>OmniCode AI Studio — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?
family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap"
rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/>
</svg>
                    Download Native
                </a>
```

```
            </nav>
        </header>

        <!-- STUDIO MAIN -->
        <main class="studio-container">
            <div class="studio-grid">
                <!-- LEFT: PROMPT INPUT & INTENT FORM -->
                <div class="card-glass prompt-card">
                    <div class="panel-header">
                        <h2>Natural Language Intent Prompt</h2>
                        <span class="offline-note">🔒 Local execution &bull; 0% Cloud Telemetry</span>
                    </div>
                    <p class="panel-desc">
                        Provide a natural language description. The neural-symbolic transducer aligns it with moral
invariants and generates verified 4-block OmniCode.
                    </p>
                    <textarea id="prompt-input" class="prompt-textarea" rows="6" placeholder="Describe the program you
wish to create... (e.g. 'Build an immutable treasury ledger that verifies balanced transactions and rejects unauthorized
debits')"></textarea>

                    <div class="prompt-actions">
                        <button class="btn btn-accent" id="btn-generate">
                            <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="2"><polygon points="5 3 19 12 5 21 5 3"/></svg>
                            Synthesize OmniCode
                        </button>
                        <button class="btn btn-secondary" id="btn-sample-prompt">Sample Intent</button>
                    </div>

                    <!-- RESPONSE STREAM -->
                    <div class="ai-response-box" id="ai-response-box">
                        <h3>AICortex Deliberation Summary</h3>
                        <div id="ai-response-text" class="response-text">Awaiting input prompt...</div>
                    </div>
                </div>

                <!-- RIGHT: SYNTHESIZED OMNICODE CODE VIEWER -->
                <div class="card-glass code-card">
                    <div class="panel-header">
                        <h2>Synthesized 4-Block OmniCode</h2>
                        <div class="badge-group">
                            <span class="badge badge-proceed" id="code-state-badge">+1:PROCEED</span>
                            <button class="btn btn-sm btn-secondary" id="btn-copy-code">Copy</button>
                        </div>
                    </div>
                    <pre class="code-editor" id="code-output"><code id="code-output-inner"># 
================================================================================
# BLOCK 1: METADATA BLOCK
# ================================================================================
Manifest &lt; Treasury.Ledger &gt; ( Moral Equilibrium ) {
    author: &lt; \"CreativeWorkzStudio LLC\" &gt;,
    resonance: &lt; +1:PROCEED &gt;
}
</code></pre>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Dual-Readable Ontological Programming Language &amp; Cognitive AI Model</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Rust Core Edition 2024</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
    <script src="/app.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/style.css CSS 1747 LINES 3804 WORDS 37.7 KB

```css
/* ================================================================================
   OMNICODE STUDIO & COGNITIVE TERMINAL STYLESHEET
   Aesthetics: Deep Obsidian, Radiant Neon Glassmorphism, Dual Monospace Code
   ================================================================================ */

:root {
    /* Color Palette */
    --bg-base: #07090e;
    --bg-surface: #0d121d;
    --bg-elevated: #131a29;
    --border-subtle: rgba(255, 255, 255, 0.08);
    --border-bright: rgba(0, 240, 255, 0.3);

    /* Text Colors */
    --text-primary: #f0f4fc;
    --text-secondary: #94a3b8;
    --text-muted: #64748b;
    --text-highlight: #ffffff;

    /* Theme Jewel Accents */
    --cyan: #00f0ff;
    --cyan-glow: rgba(0, 240, 255, 0.25);
    --gold: #ffd700;
    --gold-glow: rgba(255, 215, 0, 0.25);

    /* Balanced Ternary Quad-State Colors */
    --state-proceed: #00e676;
    --state-proceed-glow: rgba(0, 230, 118, 0.3);
    --state-await: #ffaa00;
    --state-await-glow: rgba(255, 170, 0, 0.3);
    --state-rollback: #ff3d71;
    --state-rollback-glow: rgba(255, 61, 113, 0.3);
    --state-halt: #9d4edd;
    --state-halt-glow: rgba(157, 78, 221, 0.3);

    /* Typography */
    --font-sans: 'Outfit', -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
    --font-mono: 'Fira Code', Consolas, Monaco, monospace;
    --font-serif: 'Cinzel', Georgia, serif;

    /* Shadows & Radii */
    --radius-sm: 6px;
    --radius-md: 10px;
    --radius-lg: 16px;
    --shadow-glass: 0 8px 32px 0 rgba(0, 0, 0, 0.45);
    --glass-bg: rgba(13, 18, 29, 0.7);
    --glass-border: 1px solid rgba(255, 255, 255, 0.08);
}

/* Base Reset */
*, *::before, *::after {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

body {
    background-color: var(--bg-base);
    color: var(--text-primary);
    font-family: var(--font-sans);
    line-height: 1.6;
    overflow-x: hidden;
    min-height: 100vh;
    background-image:
        radial-gradient(circle at 15% 10%, rgba(0, 240, 255, 0.05) 0%, transparent 40%),
        radial-gradient(circle at 85% 80%, rgba(255, 215, 0, 0.04) 0%, transparent 45%),
        radial-gradient(circle at 50% 50%, rgba(157, 78, 221, 0.03) 0%, transparent 60%);
    background-attachment: fixed;
}

/* App Layout */
.app-layout {
    display: flex;
    flex-direction: column;
```

```
    min-height: 100vh;
}

/* ------------------------------------------------------------------------
   HEADER & NAVIGATION
   ------------------------------------------------------------------------ */
.app-header {
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 1rem 2rem;
    background: rgba(7, 9, 14, 0.85);
    backdrop-filter: blur(20px);
    -webkit-backdrop-filter: blur(20px);
    border-bottom: var(--glass-border);
    position: sticky;
    top: 0;
    z-index: 100;
}

.header-left {
    display: flex;
    align-items: center;
    gap: 1.5rem;
}

.brand-container {
    display: flex;
    align-items: center;
    gap: 0.75rem;
}

.brand-symbol {
    font-size: 1.8rem;
    font-weight: 800;
    color: var(--cyan);
    text-shadow: 0 0 16px var(--cyan-glow);
    font-family: var(--font-serif);
}

.brand-text {
    display: flex;
    flex-direction: column;
}

.brand-title {
    font-size: 1.15rem;
    font-weight: 800;
    letter-spacing: 0.15em;
    color: var(--text-highlight);
}

.brand-subtitle {
    font-size: 0.65rem;
    letter-spacing: 0.2em;
    color: var(--gold);
    font-weight: 600;
}

.status-pill {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    padding: 0.35rem 0.85rem;
    background: rgba(0, 230, 118, 0.08);
    border: 1px solid rgba(0, 230, 118, 0.3);
    border-radius: 999px;
    font-size: 0.72rem;
    font-weight: 600;
    color: var(--state-proceed);
    letter-spacing: 0.05em;
}

.status-dot {
    width: 7px;
    height: 7px;
    border-radius: 50%;
    background: var(--state-proceed);
```

```
    box-shadow: 0 0 8px var(--state-proceed);
    animation: pulse-dot 2s infinite ease-in-out;
}

@keyframes pulse-dot {
    0%, 100% { opacity: 1; transform: scale(1); }
    50% { opacity: 0.4; transform: scale(0.85); }
}

.app-nav {
    display: flex;
    gap: 0.5rem;
}

.nav-tab {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    padding: 0.6rem 1.1rem;
    background: transparent;
    border: 1px solid transparent;
    border-radius: var(--radius-md);
    color: var(--text-secondary);
    font-family: var(--font-sans);
    font-size: 0.9rem;
    font-weight: 500;
    cursor: pointer;
    transition: all 0.25s ease;
}

.nav-tab:hover {
    color: var(--text-primary);
    background: rgba(255, 255, 255, 0.04);
}

.nav-tab.active {
    color: var(--cyan);
    background: rgba(0, 240, 255, 0.08);
    border-color: rgba(0, 240, 255, 0.35);
    box-shadow: 0 0 15px rgba(0, 240, 255, 0.15);
}

.ternary-quad-badge {
    padding: 0.4rem 0.9rem;
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-sm);
    font-family: var(--font-mono);
    font-size: 0.75rem;
    font-weight: 600;
    color: var(--gold);
    letter-spacing: 0.05em;
}

/* ------------------------------------------------------------------------
   MAIN & VIEW PANELS
   ------------------------------------------------------------------------ */
.app-main {
    flex: 1;
    padding: 2rem;
    max-width: 1600px;
    margin: 0 auto;
    width: 100%;
}

.view-panel {
    display: none;
    animation: fadeIn 0.3s ease;
}

.view-panel.active {
    display: block;
}

@keyframes fadeIn {
    from { opacity: 0; transform: translateY(6px); }
    to { opacity: 1; transform: translateY(0); }
}
```

```
231
232 /* Glassmorphism Cards */
233 .card-glass {
234     background: var(--glass-bg);
235     backdrop-filter: blur(16px);
236     -webkit-backdrop-filter: blur(16px);
237     border: var(--glass-border);
238     border-radius: var(--radius-lg);
239     box-shadow: var(--shadow-glass);
240     padding: 1.5rem;
241 }
242
243 .panel-header {
244     display: flex;
245     justify-content: space-between;
246     align-items: flex-start;
247     margin-bottom: 1.25rem;
248     gap: 1rem;
249 }
250
251 .panel-header-title h1 {
252     font-size: 1.5rem;
253     font-weight: 700;
254     color: var(--text-highlight);
255     margin-bottom: 0.35rem;
256 }
257
258 .panel-desc {
259     font-size: 0.88rem;
260     color: var(--text-secondary);
261     max-width: 850px;
262 }
263
264 /* ------------------------------------------------------------------------
265    VIEW 1: PROMPT WORKBENCH
266    ------------------------------------------------------------------------ */
267 .quadstate-indicator {
268     display: flex;
269     align-items: center;
270     gap: 0.6rem;
271     padding: 0.5rem 1.1rem;
272     background: rgba(255, 170, 0, 0.1);
273     border: 1px solid var(--state-await);
274     border-radius: var(--radius-md);
275     color: var(--state-await);
276     font-family: var(--font-mono);
277     font-size: 0.85rem;
278     font-weight: 700;
279     transition: all 0.3s ease;
280 }
281
282 .quadstate-indicator.proceed {
283     background: rgba(0, 230, 118, 0.1);
284     border-color: var(--state-proceed);
285     color: var(--state-proceed);
286     box-shadow: 0 0 15px var(--state-proceed-glow);
287 }
288
289 .quadstate-indicator.rollback {
290     background: rgba(255, 61, 113, 0.1);
291     border-color: var(--state-rollback);
292     color: var(--state-rollback);
293     box-shadow: 0 0 15px var(--state-rollback-glow);
294 }
295
296 .quadstate-indicator.halt {
297     background: rgba(157, 78, 221, 0.1);
298     border-color: var(--state-halt);
299     color: var(--state-halt);
300     box-shadow: 0 0 15px var(--state-halt-glow);
301 }
302
303 .prompt-chips-container {
304     display: flex;
305     flex-wrap: wrap;
306     align-items: center;
307     gap: 0.5rem;
308     margin-bottom: 1.25rem;
```

```
}

.chip-label {
    font-size: 0.8rem;
    font-weight: 600;
    color: var(--text-muted);
    text-transform: uppercase;
    letter-spacing: 0.05em;
}

.prompt-chip, .term-btn {
    background: rgba(255, 255, 255, 0.04);
    border: 1px solid var(--border-subtle);
    border-radius: 999px;
    padding: 0.35rem 0.85rem;
    color: var(--text-secondary);
    font-size: 0.82rem;
    cursor: pointer;
    transition: all 0.2s ease;
}

.prompt-chip:hover, .term-btn:hover {
    background: rgba(0, 240, 255, 0.12);
    border-color: var(--cyan);
    color: var(--text-highlight);
    transform: translateY(-1px);
}

.prompt-chip.chip-warning {
    border-color: rgba(255, 61, 113, 0.3);
    color: #ff7597;
}

.prompt-chip.chip-warning:hover {
    background: rgba(255, 61, 113, 0.15);
    border-color: var(--state-rollback);
}

.prompt-form textarea {
    width: 100%;
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 1rem;
    color: var(--text-primary);
    font-family: var(--font-sans);
    font-size: 1rem;
    resize: vertical;
    outline: none;
    transition: border-color 0.2s ease, box-shadow 0.2s ease;
}

.prompt-form textarea:focus {
    border-color: var(--cyan);
    box-shadow: 0 0 15px var(--cyan-glow);
}

.form-actions {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-top: 0.85rem;
}

.offline-note {
    font-size: 0.78rem;
    color: var(--text-muted);
}

.btn {
    display: inline-flex;
    align-items: center;
    gap: 0.5rem;
    padding: 0.65rem 1.35rem;
    border-radius: var(--radius-md);
    font-family: var(--font-sans);
    font-size: 0.92rem;
    font-weight: 600;
```

```
    cursor: pointer;
    transition: all 0.25s ease;
    border: none;
}

.btn-primary {
    background: linear-gradient(135deg, #00f0ff 0%, #0088ff 100%);
    color: #050811;
    box-shadow: 0 4px 15px rgba(0, 240, 255, 0.3);
}

.btn-primary:hover {
    filter: brightness(1.15);
    transform: translateY(-1px);
    box-shadow: 0 6px 20px rgba(0, 240, 255, 0.45);
}

.btn-secondary {
    background: rgba(255, 255, 255, 0.06);
    border: 1px solid var(--border-subtle);
    color: var(--text-secondary);
}

.btn-secondary:hover {
    background: rgba(255, 255, 255, 0.1);
    color: var(--text-primary);
}

/* Studio Output Split Grid */
.studio-output-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 1.5rem;
    margin-top: 1.5rem;
}

.card-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 1rem;
    border-bottom: 1px solid rgba(255, 255, 255, 0.05);
    padding-bottom: 0.75rem;
}

.card-title-group {
    display: flex;
    align-items: center;
    gap: 0.5rem;
}

.card-title-group h2 {
    font-size: 1.05rem;
    font-weight: 600;
    color: var(--text-highlight);
}

.session-badge {
    font-family: var(--font-mono);
    font-size: 0.75rem;
    color: var(--cyan);
    background: rgba(0, 240, 255, 0.08);
    padding: 0.25rem 0.6rem;
    border-radius: var(--radius-sm);
}

.response-body {
    min-height: 250px;
    font-size: 0.95rem;
    line-height: 1.7;
    white-space: pre-wrap;
    color: #e2e8f0;
}

.placeholder-text {
    color: var(--text-muted);
    font-style: italic;
    padding-top: 2rem;
```

```
}

.card-footer {
    display: flex;
    flex-wrap: wrap;
    gap: 1rem;
    margin-top: 1rem;
    padding-top: 0.75rem;
    border-top: 1px solid rgba(255, 255, 255, 0.05);
    font-family: var(--font-mono);
    font-size: 0.75rem;
    color: var(--text-muted);
}

.code-body {
    background: #05070c;
    border-radius: var(--radius-md);
    padding: 1rem;
    max-height: 480px;
    overflow-y: auto;
    border: 1px solid rgba(255, 255, 255, 0.05);
}

.code-body pre code {
    font-family: var(--font-mono);
    font-size: 0.85rem;
    line-height: 1.5;
    color: #a5b4fc;
}

.btn-icon {
    background: rgba(255, 255, 255, 0.05);
    border: 1px solid var(--border-subtle);
    color: var(--text-secondary);
    border-radius: var(--radius-sm);
    padding: 0.3rem 0.7rem;
    font-size: 0.75rem;
    cursor: pointer;
}

.btn-icon:hover {
    color: var(--text-highlight);
    background: rgba(255, 255, 255, 0.1);
}

/* ------------------------------------------------------------------------
   VIEW 2: CLI TERMINAL
   ------------------------------------------------------------------------ */
.terminal-shortcuts {
    display: flex;
    flex-wrap: wrap;
    align-items: center;
    gap: 0.5rem;
    margin-bottom: 1rem;
}

.term-btn {
    font-family: var(--font-mono);
    font-size: 0.75rem;
}

.terminal-window {
    background: #04060a;
    border: 1px solid rgba(0, 240, 255, 0.2);
    border-radius: var(--radius-md);
    padding: 1.25rem;
    min-height: 480px;
    max-height: 600px;
    overflow-y: auto;
    font-family: var(--font-mono);
    font-size: 0.88rem;
    line-height: 1.6;
    color: #4ade80;
    box-shadow: inset 0 0 30px rgba(0, 0, 0, 0.8);
}

.terminal-banner pre {
    color: var(--cyan);
```

```
    margin-bottom: 1rem;
    font-family: inherit;
    white-space: pre-wrap;
}

.terminal-stream {
    white-space: pre-wrap;
    word-break: break-all;
}

.term-line-cmd {
    color: var(--gold);
    font-weight: 600;
    margin-top: 0.75rem;
}

.term-line-stdout {
    color: #e2e8f0;
}

.term-line-stderr {
    color: var(--state-rollback);
}

.terminal-input-bar {
    display: flex;
    align-items: center;
    gap: 0.75rem;
    margin-top: 1rem;
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 0.5rem 1rem;
}

.prompt-prefix {
    font-family: var(--font-mono);
    font-size: 0.9rem;
    font-weight: 700;
    color: var(--cyan);
}

.terminal-input-bar input {
    flex: 1;
    background: transparent;
    border: none;
    color: var(--text-highlight);
    font-family: var(--font-mono);
    font-size: 0.92rem;
    outline: none;
}

/* ------------------------------------------------------------------------
   VIEW 3: COVENANT & LEXICON
   ------------------------------------------------------------------------ */
.lexicon-container {
    display: grid;
    grid-template-columns: 1fr 1.35fr;
    gap: 1.5rem;
}

.column-badge {
    font-size: 0.65rem;
    letter-spacing: 0.15em;
    font-weight: 700;
    color: var(--gold);
    text-transform: uppercase;
}

.column-header h2 {
    font-size: 1.35rem;
    color: var(--text-highlight);
    margin: 0.25rem 0 0.5rem 0;
}

.column-desc {
    font-size: 0.85rem;
    color: var(--text-secondary);
```

```
    margin-bottom: 1.25rem;
}

.laws-list {
    display: flex;
    flex-direction: column;
    gap: 0.85rem;
    max-height: 750px;
    overflow-y: auto;
    padding-right: 0.5rem;
}

.law-card {
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 1rem;
    transition: transform 0.2s ease, border-color 0.2s ease;
}

.law-card:hover {
    border-color: rgba(255, 215, 0, 0.35);
    transform: translateX(3px);
}

.law-title-row {
    display: flex;
    align-items: center;
    gap: 0.75rem;
    margin-bottom: 0.5rem;
}

.law-num {
    font-family: var(--font-serif);
    font-size: 0.9rem;
    font-weight: 700;
    color: var(--gold);
    width: 24px;
}

.law-name {
    flex: 1;
    font-size: 0.95rem;
    font-weight: 600;
    color: var(--text-highlight);
}

.state-chip {
    font-family: var(--font-mono);
    font-size: 0.68rem;
    font-weight: 700;
    padding: 0.2rem 0.55rem;
    border-radius: var(--radius-sm);
}

.state-proceed {
    background: rgba(0, 230, 118, 0.12);
    color: var(--state-proceed);
    border: 1px solid rgba(0, 230, 118, 0.3);
}

.state-await {
    background: rgba(255, 170, 0, 0.12);
    color: var(--state-await);
    border: 1px solid rgba(255, 170, 0, 0.3);
}

.state-halt {
    background: rgba(157, 78, 221, 0.12);
    color: var(--state-halt);
    border: 1px solid rgba(157, 78, 221, 0.3);
}

.law-quote {
    font-size: 0.82rem;
    font-style: italic;
    color: #cbd5e1;
    margin-bottom: 0.5rem;
```

```
    line-height: 1.5;
}

.law-rule {
    font-size: 0.75rem;
    font-weight: 600;
    color: var(--cyan);
}

.lexicon-controls {
    margin-bottom: 1.25rem;
}

#lexicon-search-input {
    width: 100%;
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 0.75rem 1rem;
    color: var(--text-primary);
    font-size: 0.9rem;
    outline: none;
    margin-bottom: 0.75rem;
}

#lexicon-search-input:focus {
    border-color: var(--cyan);
}

.domain-filters {
    display: flex;
    flex-wrap: wrap;
    gap: 0.4rem;
}

.filter-btn {
    background: rgba(255, 255, 255, 0.04);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-sm);
    padding: 0.3rem 0.75rem;
    color: var(--text-secondary);
    font-size: 0.75rem;
    cursor: pointer;
    transition: all 0.2s ease;
}

.filter-btn:hover {
    color: var(--text-primary);
    background: rgba(255, 255, 255, 0.08);
}

.filter-btn.active {
    background: rgba(0, 240, 255, 0.12);
    border-color: var(--cyan);
    color: var(--cyan);
    font-weight: 600;
}

.lexicon-entries-grid {
    display: flex;
    flex-direction: column;
    gap: 0.85rem;
    max-height: 650px;
    overflow-y: auto;
    padding-right: 0.5rem;
}

.entry-card {
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 1rem;
    transition: all 0.2s ease;
}

.entry-card:hover {
    border-color: rgba(0, 240, 255, 0.3);
}
```

```
777
778 .entry-header {
779     display: flex;
780     justify-content: space-between;
781     align-items: center;
782     margin-bottom: 0.5rem;
783 }
784
785 .entry-term {
786     font-size: 1.05rem;
787     font-weight: 700;
788     color: var(--cyan);
789     text-transform: uppercase;
790     letter-spacing: 0.05em;
791 }
792
793 .entry-domain {
794     font-size: 0.72rem;
795     color: var(--gold);
796     background: rgba(255, 215, 0, 0.08);
797     padding: 0.2rem 0.5rem;
798     border-radius: var(--radius-sm);
799 }
800
801 .entry-def {
802     font-size: 0.85rem;
803     color: #e2e8f0;
804     margin-bottom: 0.65rem;
805 }
806
807 .entry-witnesses {
808     background: #05070d;
809     border-radius: var(--radius-sm);
810     padding: 0.6rem 0.8rem;
811     font-size: 0.78rem;
812     line-height: 1.5;
813     color: #94a3b8;
814     margin-bottom: 0.5rem;
815 }
816
817 .witness-title {
818     font-weight: 700;
819     color: var(--text-highlight);
820     margin-right: 0.35rem;
821 }
822
823 .entry-guidance {
824     font-size: 0.78rem;
825     color: var(--state-proceed);
826     font-weight: 500;
827 }
828
829 /* ------------------------------------------------------------------------
830    VIEW 4: EPISODIC SCROLLS
831    ------------------------------------------------------------------------ */
832 .metrics-grid {
833     display: grid;
834     grid-template-columns: repeat(6, 1fr);
835     gap: 1rem;
836     margin-bottom: 1.5rem;
837 }
838
839 .metric-card {
840     background: var(--bg-surface);
841     border: 1px solid var(--border-subtle);
842     border-radius: var(--radius-md);
843     padding: 1rem;
844     display: flex;
845     flex-direction: column;
846     align-items: center;
847     text-align: center;
848 }
849
850 .metric-num {
851     font-size: 1.75rem;
852     font-weight: 800;
853     font-family: var(--font-mono);
854     color: var(--text-highlight);
```

```
}

.metric-label {
    font-size: 0.72rem;
    color: var(--text-muted);
    text-transform: uppercase;
    letter-spacing: 0.05em;
    margin-top: 0.25rem;
}

.card-green .metric-num { color: var(--state-proceed); }
.card-amber .metric-num { color: var(--state-await); }
.card-red .metric-num { color: var(--state-rollback); }
.card-purple .metric-num { color: var(--state-halt); }

.episodes-table-wrapper {
    background: #05070c;
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    overflow-x: auto;
    max-height: 520px;
}

.episodes-table {
    width: 100%;
    border-collapse: collapse;
    font-size: 0.85rem;
    text-align: left;
}

.episodes-table th {
    background: rgba(255, 255, 255, 0.03);
    padding: 0.85rem 1rem;
    font-weight: 600;
    color: var(--text-muted);
    border-bottom: 1px solid var(--border-subtle);
    text-transform: uppercase;
    font-size: 0.72rem;
    letter-spacing: 0.05em;
}

.episodes-table td {
    padding: 0.85rem 1rem;
    border-bottom: 1px solid rgba(255, 255, 255, 0.03);
    color: #cbd5e1;
}

.episodes-table tr:hover td {
    background: rgba(0, 240, 255, 0.02);
}

/* Responsive */
@media (max-width: 1100px) {
    .studio-output-grid,
    .lexicon-container {
        grid-template-columns: 1fr;
    }
    .metrics-grid {
        grid-template-columns: repeat(3, 1fr);
    }
}

@media (max-width: 768px) {
    .app-header {
        flex-direction: column;
        gap: 1rem;
        padding: 1rem;
    }
    .metrics-grid {
        grid-template-columns: repeat(2, 1fr);
    }
}

/* ------------------------------------------------------------------------
   CREATIVEWORKZSTUDIO LLC BRANDING & DOCUMENTATION STYLES
   ------------------------------------------------------------------------ */
.cws-badge {
    background: linear-gradient(135deg, rgba(255, 215, 0, 0.15), rgba(255, 170, 0, 0.08));
```

```
    border: 1px solid rgba(255, 215, 0, 0.35);
    color: var(--gold);
    font-size: 0.75rem;
    font-weight: 700;
    padding: 0.35rem 0.85rem;
    border-radius: 9999px;
    letter-spacing: 0.04em;
    text-transform: uppercase;
    display: inline-flex;
    align-items: center;
    gap: 0.4rem;
    box-shadow: 0 0 12px rgba(255, 215, 0, 0.1);
}

.terminal-ctrl-group {
    display: flex;
    gap: 0.5rem;
    align-items: center;
}

/* Documentation Layout */
.doc-container {
    display: flex;
    flex-direction: column;
    gap: 1.5rem;
    max-width: 1300px;
    margin: 0 auto;
    width: 100%;
}

.doc-hero {
    padding: 2.2rem;
    border-radius: var(--radius-lg);
    background: linear-gradient(135deg, rgba(13, 18, 29, 0.85) 0%, rgba(20, 28, 45, 0.7) 100%);
    border: var(--glass-border);
    position: relative;
    overflow: hidden;
}

.doc-hero::before {
    content: '';
    position: absolute;
    top: 0;
    right: 0;
    width: 320px;
    height: 100%;
    background: radial-gradient(circle at center, rgba(0, 240, 255, 0.08) 0%, transparent 70%);
    pointer-events: none;
}

.hero-badge {
    display: inline-block;
    font-size: 0.72rem;
    font-weight: 800;
    letter-spacing: 0.1em;
    text-transform: uppercase;
    color: var(--cyan);
    margin-bottom: 0.6rem;
}

.doc-hero h1 {
    font-size: 2.2rem;
    font-weight: 800;
    color: #ffffff;
    margin-bottom: 0.8rem;
    letter-spacing: -0.02em;
}

.hero-lead {
    font-size: 1.05rem;
    color: var(--text-secondary);
    line-height: 1.7;
    max-width: 950px;
}

.doc-grid {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
```

```
    gap: 1.5rem;
}

@media (max-width: 1024px) {
    .doc-grid {
        grid-template-columns: 1fr;
    }
}

.doc-card {
    padding: 1.8rem;
    border-radius: var(--radius-md);
    display: flex;
    flex-direction: column;
    gap: 1rem;
}

.doc-card-header {
    display: flex;
    align-items: center;
    gap: 0.75rem;
    border-bottom: 1px solid var(--border-subtle);
    padding-bottom: 0.8rem;
}

.doc-icon {
    font-size: 1.5rem;
}

.doc-card-header h2 {
    font-size: 1.2rem;
    font-weight: 700;
    color: #ffffff;
}

/* 4-Block Timeline */
.four-block-timeline {
    display: flex;
    flex-direction: column;
    gap: 0.6rem;
    margin: 0.5rem 0;
}

.timeline-step {
    display: flex;
    align-items: flex-start;
    gap: 0.8rem;
    background: rgba(255, 255, 255, 0.02);
    padding: 0.6rem 0.8rem;
    border-radius: var(--radius-sm);
    border-left: 3px solid var(--cyan);
}

.step-num {
    background: var(--cyan);
    color: #000000;
    font-weight: 800;
    font-size: 0.75rem;
    width: 20px;
    height: 20px;
    border-radius: 50%;
    display: flex;
    align-items: center;
    justify-content: center;
    flex-shrink: 0;
}

.step-content {
    font-size: 0.88rem;
    color: var(--text-primary);
}

.code-snippet-box {
    background: #05070b;
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-sm);
    padding: 1rem;
    font-family: var(--font-mono);
```

```
    font-size: 0.82rem;
    overflow-x: auto;
    color: var(--cyan);
}

/* Spec Tables */
.table-responsive {
    overflow-x: auto;
    width: 100%;
}

.spec-table {
    width: 100%;
    border-collapse: collapse;
    font-size: 0.85rem;
}

.spec-table th {
    background: rgba(255, 255, 255, 0.04);
    padding: 0.7rem 0.85rem;
    color: var(--text-muted);
    font-size: 0.72rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    text-align: left;
    border-bottom: 1px solid var(--border-subtle);
}

.spec-table td {
    padding: 0.7rem 0.85rem;
    border-bottom: 1px solid rgba(255, 255, 255, 0.03);
    color: #cbd5e1;
}

.spec-table tr:hover td {
    background: rgba(0, 240, 255, 0.02);
}

/* Boundary Badges */
.badge-rigid { color: #58a6ff; font-weight: 700; }
.badge-fluid { color: #38bdf8; font-weight: 700; }
.badge-omni { color: #d29922; font-weight: 700; }
.badge-sink { color: #f85149; font-weight: 700; }
.badge-source { color: #3fb950; font-weight: 700; }
.badge-aside { color: #bc8cff; font-weight: 700; }

/* Quad-State Showcase */
.quad-state-showcase {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    gap: 0.75rem;
    margin: 0.5rem 0;
}

.quad-item {
    padding: 0.8rem;
    border-radius: var(--radius-sm);
    display: flex;
    flex-direction: column;
    gap: 0.3rem;
    border: 1px solid rgba(255, 255, 255, 0.05);
}

.quad-item.quad-proceed { background: rgba(0, 230, 118, 0.06); border-color: rgba(0, 230, 118, 0.25); }
.quad-item.quad-await { background: rgba(255, 170, 0, 0.06); border-color: rgba(255, 170, 0, 0.25); }
.quad-item.quad-rollback { background: rgba(255, 61, 113, 0.06); border-color: rgba(255, 61, 113, 0.25); }
.quad-item.quad-halt { background: rgba(157, 78, 221, 0.06); border-color: rgba(157, 78, 221, 0.25); }

.quad-code {
    font-family: var(--font-mono);
    font-weight: 800;
    font-size: 0.85rem;
}
.quad-proceed .quad-code { color: var(--state-proceed); }
.quad-await .quad-code { color: var(--state-await); }
.quad-rollback .quad-code { color: var(--state-rollback); }
.quad-halt .quad-code { color: var(--state-halt); }

```

```
.quad-meaning {
    font-size: 0.8rem;
    color: var(--text-secondary);
}

/* AI Pipeline */
.pipeline-diagram {
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 0.4rem;
    margin: 0.5rem 0;
}

.pipe-node {
    width: 100%;
    background: rgba(255, 255, 255, 0.03);
    border: 1px solid var(--border-subtle);
    padding: 0.65rem 1rem;
    border-radius: var(--radius-sm);
    text-align: center;
    font-size: 0.85rem;
    font-weight: 600;
    color: var(--text-primary);
}

.pipe-node.highlight {
    background: rgba(0, 240, 255, 0.08);
    border-color: var(--border-bright);
    color: var(--cyan);
}

.pipe-arrow {
    color: var(--text-muted);
    font-size: 0.9rem;
}

.mini-laws-grid {
    display: flex;
    flex-direction: column;
    gap: 0.4rem;
    font-size: 0.85rem;
}

.mini-law {
    background: rgba(255, 255, 255, 0.02);
    padding: 0.5rem 0.8rem;
    border-radius: var(--radius-sm);
    color: var(--text-secondary);
}

.mini-law strong {
    color: #ffffff;
}

.doc-subcard {
    background: rgba(255, 255, 255, 0.02);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-sm);
    padding: 1rem;
    margin-top: 0.5rem;
}

.doc-subcard h3 {
    font-size: 0.95rem;
    color: var(--text-primary);
    margin-bottom: 0.4rem;
}

.doc-subcard p {
    font-size: 0.85rem;
    color: var(--text-secondary);
    margin-bottom: 0.5rem;
}

.doc-subcard code {
    display: block;
    background: #05070b;
```

```
    padding: 0.5rem 0.8rem;
    border-radius: var(--radius-sm);
    font-family: var(--font-mono);
    font-size: 0.8rem;
    color: var(--cyan);
}

.spec-list {
    list-style: none;
    display: flex;
    flex-direction: column;
    gap: 0.6rem;
    font-size: 0.88rem;
    color: var(--text-secondary);
}

.spec-list li strong {
    color: #ffffff;
}

/* ------------------------------------------------------------------------
   FOOTER
   ------------------------------------------------------------------------ */
.app-footer {
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 1.2rem 2rem;
    background: rgba(7, 9, 14, 0.9);
    border-top: var(--glass-border);
    font-size: 0.82rem;
    color: var(--text-muted);
    margin-top: 2rem;
}

.footer-left {
    display: flex;
    align-items: center;
    gap: 0.75rem;
}

.footer-brand {
    font-family: var(--font-serif);
    font-weight: 700;
    color: var(--cyan);
}

.footer-center {
    color: var(--text-secondary);
}

.footer-right {
    display: flex;
    align-items: center;
    gap: 0.5rem;
}

.footer-pill {
    background: rgba(255, 255, 255, 0.05);
    padding: 0.2rem 0.6rem;
    border-radius: 9999px;
    font-size: 0.72rem;
    color: var(--text-secondary);
}

/* ------------------------------------------------------------------------
   INTERACTIVE TERMINAL WORKSPACE & SCHEDULER STYLES
   ------------------------------------------------------------------------ */
.terminal-body {
    overflow: hidden;
}
.terminal-page-layout {
    height: 100vh;
    display: flex;
    flex-direction: column;
}
.terminal-workspace {
    flex: 1;
```

```
    display: flex;
    flex-direction: column;
    padding: 1rem 1.5rem;
    gap: 0.8rem;
    overflow: hidden;
}
.terminal-toolbar {
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 0.6rem 1rem;
    background: rgba(13, 17, 23, 0.8);
    border: 1px solid rgba(255, 255, 255, 0.08);
    border-radius: 8px;
}
.toolbar-left {
    display: flex;
    align-items: center;
    gap: 0.8rem;
}
.terminal-title-badge {
    font-family: var(--font-mono);
    font-size: 0.78rem;
    font-weight: 700;
    color: var(--cyan);
    letter-spacing: 0.05em;
}
.sched-pill {
    background: rgba(0, 240, 255, 0.1);
    border: 1px solid rgba(0, 240, 255, 0.25);
    color: var(--cyan);
    font-size: 0.72rem;
    padding: 0.25rem 0.6rem;
    border-radius: 9999px;
    display: flex;
    align-items: center;
    gap: 0.4rem;
}
.mem-pill {
    background: rgba(0, 255, 136, 0.1);
    border: 1px solid rgba(0, 255, 136, 0.25);
    color: var(--green);
    font-size: 0.72rem;
    padding: 0.25rem 0.6rem;
    border-radius: 9999px;
}
.toolbar-actions {
    display: flex;
    gap: 0.4rem;
}
.btn-sm {
    padding: 0.35rem 0.7rem;
    font-size: 0.75rem;
}
.btn-xs {
    padding: 0.2rem 0.45rem;
    font-size: 0.68rem;
}
.btn-danger-ghost {
    background: transparent;
    color: #ff5555;
    border: 1px solid rgba(255, 85, 85, 0.25);
}
.btn-danger-ghost:hover {
    background: rgba(255, 85, 85, 0.15);
}

.terminal-grid {
    flex: 1;
    display: grid;
    grid-template-columns: 1fr 380px;
    gap: 1rem;
    min-height: 0;
}
.terminal-console-pane {
    display: flex;
    flex-direction: column;
    background: rgba(10, 12, 16, 0.95);
```

```
    border: 1px solid rgba(0, 240, 255, 0.2);
    border-radius: 8px;
    box-shadow: 0 8px 32px rgba(0, 0, 0, 0.5);
    min-height: 0;
    overflow: hidden;
}
.console-screen {
    flex: 1;
    padding: 1.2rem;
    overflow-y: auto;
    font-family: var(--font-mono);
    font-size: 0.86rem;
    line-height: 1.5;
    color: #e6edf3;
    outline: none;
}
.term-intro {
    color: var(--cyan);
    white-space: pre-wrap;
    margin-bottom: 1rem;
    font-size: 0.78rem;
    opacity: 0.9;
}
.term-entry {
    margin-bottom: 0.8rem;
}
.term-command-line {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    color: #ffffff;
    font-weight: 600;
}
.term-prompt {
    color: var(--gold);
    font-family: var(--font-mono);
    font-weight: 700;
}
.term-cmd-text {
    color: #00f0ff;
}
.term-result-box {
    margin-top: 0.3rem;
    padding-left: 0.8rem;
    border-left: 2px solid rgba(0, 240, 255, 0.3);
}
.term-meta-bar {
    display: flex;
    align-items: center;
    gap: 0.6rem;
    margin-bottom: 0.25rem;
    font-size: 0.72rem;
}
.term-output-text {
    font-family: var(--font-mono);
    font-size: 0.82rem;
    white-space: pre-wrap;
    word-break: break-word;
    color: #d1d5db;
    margin: 0;
}
.console-input-row {
    display: flex;
    align-items: center;
    gap: 0.6rem;
    padding: 0.8rem 1.2rem;
    background: rgba(13, 17, 23, 0.95);
    border-top: 1px solid rgba(255, 255, 255, 0.08);
}
.term-input {
    flex: 1;
    background: transparent;
    border: none;
    outline: none;
    color: #ffffff;
    font-family: var(--font-mono);
    font-size: 0.9rem;
}
```

```
1479
1480 /* Scheduler Right Pane */
1481 .scheduler-pane {
1482     display: flex;
1483     flex-direction: column;
1484     background: rgba(13, 17, 23, 0.85);
1485     border: 1px solid rgba(255, 255, 255, 0.08);
1486     border-radius: 8px;
1487     padding: 1rem;
1488     gap: 0.8rem;
1489     min-height: 0;
1490     overflow: hidden;
1491 }
1492 .scheduler-pane-header {
1493     display: flex;
1494     justify-content: space-between;
1495     align-items: center;
1496 }
1497 .sched-title-group h2 {
1498     font-size: 0.95rem;
1499     margin: 0;
1500     color: #ffffff;
1501 }
1502 .sched-title-group .subtext {
1503     font-size: 0.72rem;
1504     color: var(--text-muted);
1505     margin: 0;
1506 }
1507 .sched-spawn-box {
1508     display: grid;
1509     grid-template-columns: 1fr 1fr;
1510     gap: 0.4rem;
1511 }
1512 .sched-spawn-box button {
1513     grid-column: span 2;
1514 }
1515 .form-input-sm, .form-select-sm {
1516     background: rgba(255, 255, 255, 0.05);
1517     border: 1px solid rgba(255, 255, 255, 0.1);
1518     color: #ffffff;
1519     padding: 0.35rem 0.5rem;
1520     font-size: 0.75rem;
1521     border-radius: 4px;
1522     outline: none;
1523 }
1524 .task-table-wrapper {
1525     flex: 1;
1526     overflow-y: auto;
1527     border: 1px solid rgba(255, 255, 255, 0.05);
1528     border-radius: 4px;
1529 }
1530 .task-table {
1531     width: 100%;
1532     border-collapse: collapse;
1533     font-size: 0.75rem;
1534 }
1535 .task-table th, .task-table td {
1536     padding: 0.45rem 0.6rem;
1537     text-align: left;
1538     border-bottom: 1px solid rgba(255, 255, 255, 0.05);
1539 }
1540 .task-table th {
1541     background: rgba(255, 255, 255, 0.03);
1542     color: var(--text-muted);
1543     font-weight: 600;
1544 }
1545 .action-cell {
1546     display: flex;
1547     gap: 0.2rem;
1548 }
1549 .session-vars-box {
1550     background: rgba(0, 0, 0, 0.3);
1551     border: 1px solid rgba(255, 255, 255, 0.05);
1552     border-radius: 4px;
1553     padding: 0.6rem;
1554     max-height: 120px;
1555     overflow-y: auto;
1556 }
```

```
.vars-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 0.3rem;
}
.vars-header h3 {
    font-size: 0.75rem;
    margin: 0;
    color: var(--gold);
}
.vars-content {
    font-family: var(--font-mono);
    font-size: 0.72rem;
    color: #e6edf3;
}

/* Download Page Styles */
.download-container {
    max-width: 1100px;
    margin: 2rem auto;
    padding: 0 1.5rem;
}
.download-hero {
    text-align: center;
    padding: 3rem 2rem;
    margin-bottom: 2rem;
}
.download-btn-row {
    margin: 2rem 0 1rem;
}
.download-cta {
    background: linear-gradient(135deg, #00f0ff, #0099ff);
    color: #000000;
    font-weight: 800;
    font-size: 1.05rem;
    padding: 0.9rem 2.2rem;
    border-radius: 8px;
    box-shadow: 0 0 25px rgba(0, 240, 255, 0.35);
    display: inline-flex;
    align-items: center;
    gap: 0.8rem;
    text-decoration: none;
    transition: transform 0.2s, box-shadow 0.2s;
}
.download-cta:hover {
    transform: translateY(-2px);
    box-shadow: 0 0 35px rgba(0, 240, 255, 0.5);
}
.hero-subtext {
    font-size: 0.82rem;
    color: var(--text-muted);
}
.download-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 1.5rem;
}
.download-card {
    padding: 1.8rem;
}
.code-copy-box {
    background: rgba(0, 0, 0, 0.5);
    border: 1px solid rgba(0, 240, 255, 0.2);
    border-radius: 6px;
    padding: 0.9rem;
    margin: 1rem 0;
    font-family: var(--font-mono);
    font-size: 0.82rem;
    color: var(--cyan);
    word-break: break-all;
}
.download-nav-btn {
    color: var(--cyan) !important;
    border-color: rgba(0, 240, 255, 0.3) !important;
}

/* ------------------------------------------------------------------------
```

```
   OMNICODE 3-TIER COMPOSITE HOVERCARD & DOCSTRING INSPECTOR
   ------------------------------------------------------------------------- */
.omni-hover-card {
    position: absolute;
    z-index: 10000;
    max-width: 480px;
    background: rgba(10, 15, 29, 0.94);
    backdrop-filter: blur(20px);
    -webkit-backdrop-filter: blur(20px);
    border: 1px solid rgba(0, 240, 255, 0.35);
    border-radius: 10px;
    box-shadow: 0 12px 40px rgba(0, 0, 0, 0.8), 0 0 20px rgba(0, 240, 255, 0.15);
    padding: 1.1rem;
    color: #e2e8f0;
    font-family: var(--font-sans, -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif);
    font-size: 0.85rem;
    line-height: 1.5;
    pointer-events: auto;
    transition: opacity 0.15s ease, transform 0.15s ease;
}

.omni-hover-header {
    display: flex;
    align-items: center;
    justify-content: space-between;
    border-bottom: 1px solid rgba(255, 255, 255, 0.1);
    padding-bottom: 0.6rem;
    margin-bottom: 0.8rem;
}

.omni-hover-symbol {
    font-family: var(--font-mono, monospace);
    font-size: 1.05rem;
    font-weight: 700;
    color: #00f0ff;
    letter-spacing: 0.5px;
}

.omni-hover-rank-badge {
    display: inline-block;
    padding: 0.2rem 0.6rem;
    font-size: 0.72rem;
    font-weight: 600;
    border-radius: 4px;
    background: rgba(0, 240, 255, 0.15);
    color: #38bdf8;
    border: 1px solid rgba(56, 189, 248, 0.3);
    text-transform: uppercase;
}

.omni-hover-rank-badge.rank-1 {
    background: rgba(234, 179, 8, 0.18);
    color: #facc15;
    border-color: rgba(250, 204, 21, 0.4);
}

.omni-hover-tier {
    margin-bottom: 0.8rem;
    padding-bottom: 0.7rem;
    border-bottom: 1px solid rgba(255, 255, 255, 0.06);
}

.omni-hover-tier:last-child {
    margin-bottom: 0;
    padding-bottom: 0;
    border-bottom: none;
}

.omni-hover-tier-title {
    font-size: 0.75rem;
    font-weight: 700;
    text-transform: uppercase;
    letter-spacing: 0.8px;
    color: #94a3b8;
    margin-bottom: 0.4rem;
    display: flex;
    align-items: center;
    gap: 0.4rem;
```

```
}

.omni-hover-tier-title.tier-tape { color: #fbbf24; }
.omni-hover-tier-title.tier-omni { color: #34d399; }
.omni-hover-tier-title.tier-silicon { color: #c084fc; }

.omni-hover-prop {
    display: flex;
    margin-bottom: 0.25rem;
    font-size: 0.8rem;
}

.omni-hover-label {
    color: #64748b;
    width: 130px;
    flex-shrink: 0;
}

.omni-hover-val {
    color: #f1f5f9;
    font-family: var(--font-mono, monospace);
    word-break: break-word;
}

.omni-hover-aside {
    background: rgba(255, 255, 255, 0.04);
    border-left: 2px solid #38bdf8;
    padding: 0.35rem 0.6rem;
    margin: 0.4rem 0;
    font-style: italic;
    color: #cbd5e1;
    font-size: 0.78rem;
}

```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/terminal.html HTML 191 LINES 782 WORDS 10.8 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Interactive Native Terminal & OS Process Scheduler — CreativeWorkzStudio LLC.">
    <title>OmniCode Interactive Terminal &amp;amp; Task Scheduler — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body class="terminal-body">
    <div class="app-layout terminal-page-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- TERMINAL WORKSPACE MAIN -->
        <main class="terminal-workspace">
            <!-- TERMINAL CONTROL TOOLBAR -->
            <div class="terminal-toolbar card-glass">
                <div class="toolbar-left">
                    <span class="terminal-title-badge">Ω NATIVE USERSPACE TERMINAL</span>
                    <span class="sched-pill" id="sched-status-pill">
                        <span class="status-dot"></span>
                        <span id="sched-pill-text">Scheduler: Online (3 tasks)</span>
                    </span>
                    <span class="mem-pill" id="mem-pill">State: +1:PROCEED</span>
                </div>
                <div class="toolbar-actions">
                    <button class="btn btn-sm btn-ghost" id="btn-quick-demo">🧪 Demo</button>
                    <button class="btn btn-sm btn-ghost" id="btn-quick-lint">📋 Lint</button>
                    <button class="btn btn-sm btn-ghost" id="btn-quick-restore">🩹 Restore</button>
                    <button class="btn btn-sm btn-ghost" id="btn-quick-tasks">📊 Tasks (ps)</button>
                    <button class="btn btn-sm btn-ghost" id="btn-quick-tick">⚡ Tick</button>
                    <button class="btn btn-sm btn-danger-ghost" id="btn-clear-term">🧹 Clear (Ctrl+L)</button>
                </div>
            </div>

            <!-- DUAL VIEWPORT: TERMINAL SCREEN & SCHEDULER MONITOR -->
            <div class="terminal-grid">
                <!-- LEFT: LIVE INTERACTIVE CONSOLE -->
                <div class="terminal-console-pane card-glass">
                    <div class="console-screen" id="console-screen" tabindex="0">
                        <div class="term-intro">
 ================================================================================
   Ω OMNICODE INTERACTIVE USERSYSTEM SHELL &amp;amp; TASK SCHEDULER
   CreativeWorkzStudio LLC — Autonomous Neural-Symbolic Intelligence
   Engine: Rust Core 2024 (Balanced Radix-3 Quad-State Physics)
 ================================================================================
 * Grounded in Biblical Moral Invariants &amp;amp; Symmetrical Covenant Bedrock
 * Type 'help' for terminal commands, 'ps' for scheduler tasks, 'tick' to advance clock.
 * Keyboard: Up/Down for command history | Tab for autocompletion | Ctrl+L to clear screen.
                        </div>
                        <div id="term-history-log"></div>
                    </div>

                    <!-- INTERACTIVE INPUT BAR -->
                    <div class="console-input-row" id="input-container">
                        <span class="term-prompt">omnicode&amp;gt;</span>
                        <input type="text" id="term-input" class="term-input" autocomplete="off" autofocus
spellcheck="false" placeholder="Type command (e.g. ps, spawn 'worker', let x = 42, omnicode demo, help)...">
                        <button class="btn btn-sm btn-accent" id="btn-term-send">Execute</button>
                    </div>
                </div>

                <!-- RIGHT: LIVE OS TASK SCHEDULER & PROCESS LEDGER -->
                <div class="scheduler-pane card-glass">
                    <div class="scheduler-pane-header">
                        <div class="sched-title-group">
                            <h2>Cooperative Task Scheduler</h2>
                            <p class="subtext">Active process table grounded in Balanced Radix-3 states</p>
                        </div>
                        <button class="btn btn-sm btn-secondary" id="btn-refresh-tasks">Refresh</button>
                    </div>

                    <!-- TASK SPAWN ROW -->
                    <div class="sched-spawn-box">
                        <input type="text" id="spawn-name-input" class="form-input-sm" placeholder="Task Name (e.g.
InvariantWatcher)">
                        <input type="text" id="spawn-cmd-input" class="form-input-sm" placeholder="Command (e.g.
omnicode lint manifest.omni)">
                        <select id="spawn-priority-select" class="form-select-sm">
                            <option value="NORMAL">Normal Priority</option>
                            <option value="HIGH">High Priority</option>
                            <option value="LOW">Low Priority</option>
                            <option value="BACKGROUND">Background</option>
                        </select>
                        <button class="btn btn-sm btn-accent" id="btn-spawn-task">+ Spawn</button>
                    </div>

```

```
                    <!-- TASK TABLE -->
                    <div class="task-table-wrapper">
                        <table class="task-table">
                            <thead>
                                <tr>
                                    <th>PID</th>
                                    <th>Task Name</th>
                                    <th>Priority</th>
                                    <th>Quad-State</th>
                                    <th>Ticks</th>
                                    <th>Actions</th>
                                </tr>
                            </thead>
                            <tbody id="task-tbody">
                                <!-- Populated dynamically via JS -->
                            </tbody>
                        </table>
                    </div>

                    <!-- IN-MEMORY SESSION VARIABLES -->
                    <div class="session-vars-box">
                        <div class="vars-header">
                            <h3>In-Memory Session Variables</h3>
                            <button class="btn btn-xs btn-ghost" id="btn-refresh-vars">Refresh Vars</button>
                        </div>
                        <div class="vars-content" id="session-vars-content">
                            <span class="dim-text">No session variables declared yet. Use 'let &lt;name&gt; =
&lt;value&gt;'</span>
                        </div>
                    </div>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Native Userspace Shell &amp; Cooperative Quad-State Process Scheduler</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Local Ring-3 Shell</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>

    <script src="/terminal.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/terminal.js JS 305 LINES 885 WORDS 11.9 KB

```
// ================================================================================
// BLOCK 1: METADATA BLOCK
// ( OmniCode Native Interactive Terminal & Task Scheduler JavaScript Engine )
// ================================================================================
/**
 * OmniCode Terminal & Scheduler Frontend Engine
 * CreativeWorkzStudio LLC
 * Zero external dependencies. Connects via standard HTTP JSON endpoints:
 *   - POST /api/terminal          -> Dispatches command to session & scheduler
 *   - GET  /api/scheduler         -> Retrieves live process ledger
 *   - POST /api/scheduler/spawn   -> Spawns new task
 *   - POST /api/scheduler/action  -> Mutates task state (pause, resume, kill)
 */

document.addEventListener("DOMContentLoaded", () => {
    const consoleScreen = document.getElementById("console-screen");
    const historyLog = document.getElementById("term-history-log");
    const termInput = document.getElementById("term-input");
    const btnSend = document.getElementById("btn-term-send");
    const btnClear = document.getElementById("btn-clear-term");
    const taskTbody = document.getElementById("task-tbody");
    const schedPillText = document.getElementById("sched-pill-text");
    const varsContent = document.getElementById("session-vars-content");

    // Command History & Autocomplete State
    const history = JSON.parse(localStorage.getItem("omnicode_terminal_history") || "[]");
    let historyIndex = history.length;

    const AUTOCOMPLETE_LIST = [
        "ps", "tasks", "spawn", "pause", "resume", "kill", "halt", "tick", "vars", "env",
        "let", "help", "clear",
        "omnicode demo",
        "omnicode lint manifest.omni",
        "omnicode restore manifest.omni",
        "omnicode manifest",
        "omnicode scrolls",
        "omnicode lexicon truth",
        "omnicode ai \"What is the covenant?\""
    ];

    // Focus input on console click
    consoleScreen.addEventListener("click", () => {
        termInput.focus();
    });

    // --------------------------------------------------------------------------
    // EXECUTION DISPATCHER
    // --------------------------------------------------------------------------
    async function executeCommand(cmd) {
        const clean = cmd.trim();
        if (!clean) return;

        // Save to history
        if (history.length === 0 || history[history.length - 1] !== clean) {
            history.push(clean);
            if (history.length > 100) history.shift();
            localStorage.setItem("omnicode_terminal_history", JSON.stringify(history));
        }
        historyIndex = history.length;

        // Handle local 'clear'
        if (clean.toLowerCase() === "clear") {
            historyLog.innerHTML = "";
            termInput.value = "";
            return;
        }

        // Render command prompt entry
        const entryEl = document.createElement("div");
        entryEl.className = "term-entry";
        entryEl.innerHTML = `
            <div class="term-command-line">
                <span class="term-prompt">omnicode&gt;</span>
                <span class="term-cmd-text">${escapeHtml(clean)}</span>
```

```
            </div>
            <div class="term-loading-spinner dim-text">Deliberating &amp;amp; evaluating...</div>
        `;
        historyLog.appendChild(entryEl);
        consoleScreen.scrollTop = consoleScreen.scrollHeight;
        termInput.value = "";

        const spinner = entryEl.querySelector(".term-loading-spinner");

        try {
            const resp = await fetch("/api/terminal", {
                method: "POST",
                headers: { "Content-Type": "application/json" },
                body: JSON.stringify({ command: clean })
            });

            const data = await resp.json();
            spinner.remove();

            const resultEl = document.createElement("div");
            resultEl.className = "term-result-box";

            const stateClass = getStateClass(data.state || "+1:PROCEED");
            const durationMs = (data.duration_us / 1000).toFixed(2);

            resultEl.innerHTML = `
                <div class="term-meta-bar">
                    <span class="badge ${stateClass}">${escapeHtml(data.state || "+1:PROCEED")}</span>
                    <span class="dim-text">${durationMs} ms</span>
                </div>
                <pre class="term-output-text">${formatTerminalOutput(data.stdout || "(Completed with zero output)")}
</pre>
            `;
            entryEl.appendChild(resultEl);
        } catch (err) {
            spinner.remove();
            const errEl = document.createElement("div");
            errEl.className = "term-error-box";
            errEl.innerHTML = `<pre class="text-red">✗ Error dispatching command: ${escapeHtml(err.message)}</pre>`;
            entryEl.appendChild(errEl);
        }

        consoleScreen.scrollTop = consoleScreen.scrollHeight;
        refreshSchedulerTasks();
        refreshVariables();
    }

    // ---------------------------------------------------------------------------
    // SCHEDULER TASK POLLING & ACTIONS
    // ---------------------------------------------------------------------------
    async function refreshSchedulerTasks() {
        try {
            const resp = await fetch("/api/scheduler");
            if (!resp.ok) return;
            const tasks = await resp.json();

            taskTbody.innerHTML = "";
            let activeCount = 0;

            tasks.forEach(t => {
                if (t.state === "+1:PROCEED") activeCount++;
                const tr = document.createElement("tr");
                const stateClass = getStateClass(t.state);

                tr.innerHTML = `
                    <td class="mono font-bold">${t.id}</td>
                    <td title="${escapeHtml(t.command)}"><strong>${escapeHtml(t.name)}</strong></td>
                    <td><span class="badge badge-subtle">${escapeHtml(t.priority)}</span></td>
                    <td><span class="badge ${stateClass}">${escapeHtml(t.state)}</span></td>
                    <td class="mono">${t.ticks}</td>
                    <td class="action-cell">
                        ${t.state === "+1:PROCEED"
                            ? `<button class="btn btn-xs btn-ghost btn-pause" data-id="${t.id}" title="Pause
(0:AWAIT)">⏸</button>`
                            : `<button class="btn btn-xs btn-ghost btn-resume" data-id="${t.id}" title="Resume
(+1:PROCEED)">▶</button>`}
                        ${t.state !== "NULL:HALT"
                            ? `<button class="btn btn-xs btn-danger-ghost btn-halt" data-id="${t.id}" title="Terminate
```

```
(NULL:HALT)">■</button>`
                            : `<span class="dim-text">Ended</span>`}
                    </td>
                `;
                taskTbody.appendChild(tr);
            });

            schedPillText.textContent = `Scheduler: Online (${tasks.length} tasks, ${activeCount} active)`;

            // Attach action listeners
            document.querySelectorAll(".btn-pause").forEach(b => {
                b.onclick = () => mutateTask(b.dataset.id, "pause");
            });
            document.querySelectorAll(".btn-resume").forEach(b => {
                b.onclick = () => mutateTask(b.dataset.id, "resume");
            });
            document.querySelectorAll(".btn-halt").forEach(b => {
                b.onclick = () => mutateTask(b.dataset.id, "halt");
            });
        } catch (e) {
            console.error("Scheduler fetch error:", e);
        }
    }

    async function mutateTask(id, action) {
        await executeCommand(`${action} ${id}`);
    }

    async function refreshVariables() {
        try {
            const resp = await fetch("/api/terminal", {
                method: "POST",
                headers: { "Content-Type": "application/json" },
                body: JSON.stringify({ command: "vars" })
            });
            const data = await resp.json();
            varsContent.innerHTML = `<pre class="mono-code">${escapeHtml(data.stdout || "No session variables")}</pre>`;
        } catch (e) {
            console.error(e);
        }
    }

    // ---------------------------------------------------------------------------
    // KEYBOARD NAVIGATION & SHORTCUTS
    // ---------------------------------------------------------------------------
    termInput.addEventListener("keydown", (e) => {
        // Enter to execute
        if (e.key === "Enter") {
            e.preventDefault();
            executeCommand(termInput.value);
            return;
        }

        // Up arrow (history backward)
        if (e.key === "ArrowUp") {
            e.preventDefault();
            if (historyIndex > 0) {
                historyIndex--;
                termInput.value = history[historyIndex] || "";
            }
            return;
        }

        // Down arrow (history forward)
        if (e.key === "ArrowDown") {
            e.preventDefault();
            if (historyIndex < history.length - 1) {
                historyIndex++;
                termInput.value = history[historyIndex] || "";
            } else {
                historyIndex = history.length;
                termInput.value = "";
            }
            return;
        }

        // Tab autocomplete
        if (e.key === "Tab") {
```

```
            e.preventDefault();
            const val = termInput.value.trim().toLowerCase();
            if (!val) return;
            const match = AUTOCOMPLETE_LIST.find(item => item.toLowerCase().startsWith(val));
            if (match) {
                termInput.value = match;
            }
            return;
        }

        // Ctrl + L (Clear)
        if (e.ctrlKey && e.key.toLowerCase() === "l") {
            e.preventDefault();
            historyLog.innerHTML = "";
            return;
        }
    });

    btnSend.addEventListener("click", () => {
        executeCommand(termInput.value);
    });

    btnClear.addEventListener("click", () => {
        historyLog.innerHTML = "";
        termInput.focus();
    });

    // Toolbar Quick Action Buttons
    document.getElementById("btn-quick-demo")?.addEventListener("click", () => executeCommand("omnicode demo"));
    document.getElementById("btn-quick-lint")?.addEventListener("click", () => executeCommand("omnicode lint
manifest.omni"));
    document.getElementById("btn-quick-restore")?.addEventListener("click", () => executeCommand("omnicode restore
manifest.omni"));
    document.getElementById("btn-quick-tasks")?.addEventListener("click", () => executeCommand("ps"));
    document.getElementById("btn-quick-tick")?.addEventListener("click", () => executeCommand("tick"));
    document.getElementById("btn-refresh-tasks")?.addEventListener("click", refreshSchedulerTasks);
    document.getElementById("btn-refresh-vars")?.addEventListener("click", refreshVariables);

    // Spawn Task Button
    document.getElementById("btn-spawn-task")?.addEventListener("click", () => {
        const nameInput = document.getElementById("spawn-name-input");
        const cmdInput = document.getElementById("spawn-cmd-input");
        const prioSelect = document.getElementById("spawn-priority-select");

        const cmd = cmdInput.value.trim();
        if (!cmd) return;

        executeCommand(`spawn ${cmd}`);
        cmdInput.value = "";
        nameInput.value = "";
    });

    // Helper functions
    function getStateClass(state) {
        if (!state) return "badge-proceed";
        if (state.includes("PROCEED")) return "badge-proceed";
        if (state.includes("AWAIT")) return "badge-await";
        if (state.includes("ROLLBACK")) return "badge-rollback";
        if (state.includes("HALT")) return "badge-halt";
        return "badge-proceed";
    }

    function escapeHtml(str) {
        return (str || "")
            .replace(/&/g, "&amp;")
            .replace(/</g, "&lt;")
            .replace(/>/g, "&gt;")
            .replace(/"/g, "&quot;")
            .replace(/'/g, "&#039;");
    }

    function formatTerminalOutput(text) {
        return escapeHtml(text);
    }

    // Initial scheduler poll & timer
    refreshSchedulerTasks();
    refreshVariables();
```

```
    setInterval(refreshSchedulerTasks, 4000);
});
```